

CMS Draft Analysis Note

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Measurements of properties of the Higgs boson in the four-lepton final state at $\sqrt{s} = 13$ TeV with full Run II data

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Abstract

Properties of the Higgs boson are measured in the $H \rightarrow ZZ \rightarrow 4\ell$ ($\ell = e, \mu$) decay channel. A data sample of proton-proton collisions at a center-of-mass energy of 13 TeV is used, corresponding to an integrated luminosity of 137.1 fb^{-1} recorded in Run II by the CMS detector at the LHC. The signal-strength modifier μ , defined as the ratio of the observed Higgs boson rate in the $H \rightarrow ZZ \rightarrow 4\ell$ decay channel to the standard model expectation, is measured to be $\mu = 0.94^{+0.11}_{-0.10}$ with the Higgs boson mass profiled in the fit. The signal-strength modifiers for the main Higgs boson production modes are also constrained. Measurements of the simplified template cross sections, designed to quantify the different Higgs boson production processes in specific regions of phase space, are reported. The cross section for the production of the $H \rightarrow ZZ \rightarrow 4\ell$ in a fiducial region closely matching the experimental selection of the leptons is measured to be $\sigma_{\text{fid.}} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{sys.}) \text{ fb}$, in agreement with the Standard Model prediction of $2.76 \pm 0.14 \text{ fb}$. Differential cross sections as a function of the p_T and rapidity of the Higgs boson, the number of associated jets, and the p_T of the leading associated jet are determined. All results are found to be compatible with the Standard Model predictions, within the measurements precision.

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1 Introduction

The ATLAS and CMS collaborations first reported the discovery of a new boson in 2012 [1, 2] consistent with the standard model (SM) Higgs boson [3–8] based on proton-proton (pp) collisions delivered by the CERN LHC at a center-of-mass energy of $\sqrt{s} = 7$ TeV in 2011 and 8 TeV in 2012. Subsequent studies by CMS using the full LHC Run 1 data set in various decay channels and production modes and combined measurements from ATLAS and CMS [9–12] showed that the properties of the new boson are so far consistent with expectations for the SM Higgs boson.

The $H \rightarrow ZZ \rightarrow 4\ell$ decay channel ($\ell = e, \mu$) has a large signal-to-background ratio due to the complete reconstruction of the final state decay products and excellent lepton momentum resolution and is one of the most important channels for studies of the Higgs boson's properties. Measurements performed using this decay channel and the Run 1 data set include the determination of the mass and spin-parity of the new boson [13–15], its width [16, 17] and fiducial cross sections [18], as well as tests for anomalous HVV couplings [15, 17].

This analysis note presents measurements of properties of the Higgs boson in the $H \rightarrow ZZ \rightarrow 4\ell$ decay channel at $\sqrt{s} = 13$ TeV using 137.1 fb^{-1} of pp collision data collected with the CMS experiment at the LHC in 2016, 2017 and 2018. The data collected in 2018 corresponding to 59.7 fb^{-1} is analyzed carefully and combined with 41.4 fb^{-1} of the data from 2017 [19] and with 36.8 fb^{-1} of data from 2016 [20] that were previously analyzed. Therefore, the main focus of this analysis note is to present the results of the analysis of 2018 data that is later on combined on a datacard level with the previously analyzed 2016 and 2017 data.

2 Summary of major changes and improvements in the analysis

In this section a concise and brief summary of major analysis changes and improvements with respect to the previous analysis of 2016 data [20] and of 2017 data [19] are given:

- Move to electron BDT ID+Iso v2 that exploits xgboost training instead of TMVA used in the analysis of 2017 data. Results in reduction of fakes for same signal efficiency,
- Introduction of the Stage 1.1. STXS measurement.,
- New binning and observable introduced for the differential cross section measurement when compared to the results on 2016 data.

DRAFT

3 Datasets

3.1 Data

3.1.1 Triggers and Datasets

This analysis uses a data sample recorded by the CMS experiment during 2016, 2017 and 2018, corresponding to 137.1 fb^{-1} of data.

The datasets used are listed in Table 1. The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. To avoid duplicate events from different primary datasets, events are taken:

- from EGamma if they pass the diEle or triEle or singleElectron triggers,
- from DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,
- from MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,
- from SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

The HLT paths used for 2018 collision data is listed in Table 2.

Run-range	Dataset	Integrated luminosity
xxx-xxx	/DoubleMuon/Run2018A-17Sep2018-v2/MINIAOD /MuonEG/Run2018A-17Sep2018-v1/MINIAOD /EGamma/Run2018A-17Sep2018-v2/MINIAOD /SingleMuon/Run2018A-17Sep2018-v2/MINIAOD	X.XXX fb^{-1}
xxx-xxx	/DoubleMuon/Run2018B-17Sep2018-v1/MINIAOD /MuonEG/Run2018B-17Sep2018-v1/MINIAOD /EGamma/Run2018B-17Sep2018-v1/MINIAOD /SingleMuon/Run2018B-17Sep2018-v1/MINIAOD	X.XXX fb^{-1}
xxx-xxx	/DoubleMuon/Run2018C-17Sep2018-v1/MINIAOD /MuonEG/Run2018C-17Sep2018-v1/MINIAOD /EGamma/Run2018C-17Sep2018-v1/MINIAOD /SingleMuon/Run2018C-17Sep2018-v1/MINIAOD	X.XXX fb^{-1}
xxx-xxx	/DoubleMuon/Run2018D-PromptReco-v2/MINIAOD /MuonEG/Run2018D-PromptReco-v2/MINIAOD /EGamma/Run2018D-PromptReco-v2/MINIAOD /SingleMuon/Run2018D-PromptReco-v2/MINIAOD	X.XXX fb^{-1}

Table 1: Datasets used in the 2018 analysis.

HLT path	prescale	primary dataset
HLT_Ele23_Ele12_CaloIdL_TrackIdL_IsoVL_v*	1	DoubleEG
HLT_DoubleEle25_CaloIdL_MW_v*	1	DoubleEG
HLT_Mu17_TrkIsoVVL_Mu8_TrkIsoVVL_DZ_Mass3p8_v*	1	DoubleMuon
HLT_Mu23_TrkIsoVVL_Ele12_CaloIdL_TrackIdL_IsoVL_v*	1	MuonEG
HLT_Mu8_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ_v*	1	MuonEG
HLT_Mu12_TrkIsoVVL_Ele23_CaloIdL_TrackIdL_IsoVL_DZ_v*	1	MuonEG
HLT_DiMu9_Ele9_CaloIdL_TrackIdL_DZ_v*	1	MuonEG
HLT_Ele32_WPTight_Gsf_v*	1	SingleElectron
HLT_IsoMu24_v*	1	SingleMuon

Table 2: Trigger paths used in 2018 collision data.

3.1.2 Trigger Efficiency

The efficiency in data of the combination of triggers used in the analysis with respect to the offline reconstruction and selection is measured by considering 4ℓ events triggered by single lepton triggers. One of the four reconstructed leptons (the “tag”) is geometrically matched to

a trigger object passing the final filter of one of the single muon or single electron triggers. The other three leptons are used as “probes”. In each 4ℓ event there are up to 4 possible tag-probe combinations, and all possible combinations are counted in the denominator of the efficiency. For each of the three probe leptons all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the analysis. If any of the analysis triggers can be formed using the probe leptons, the set of probes is also counted in the numerator of the efficiency.

This method does not have a perfect closure in MC events due to the fact that the presence of a fourth lepton increases the trigger efficiency, and this effect is not accounted for. Also, in the $2e2\mu$ final state, the three probe leptons cannot be combined to form all possible triggers which can collect events with two electrons and two muons (e.g. if the tag lepton is an electron, the three remaining leptons cannot pass a double electron trigger). Therefore the method is also exercised on MC and the difference between data and MC is used to determine the reliability of the simulation. The efficiency plotted as a function of the minimum p_T of the three probe leptons in data and MC using this method can be seen in Fig. 1 that shows the trigger efficiency in 2018 data for different final states. The MC efficiency describes well the data within the statistical uncertainties.

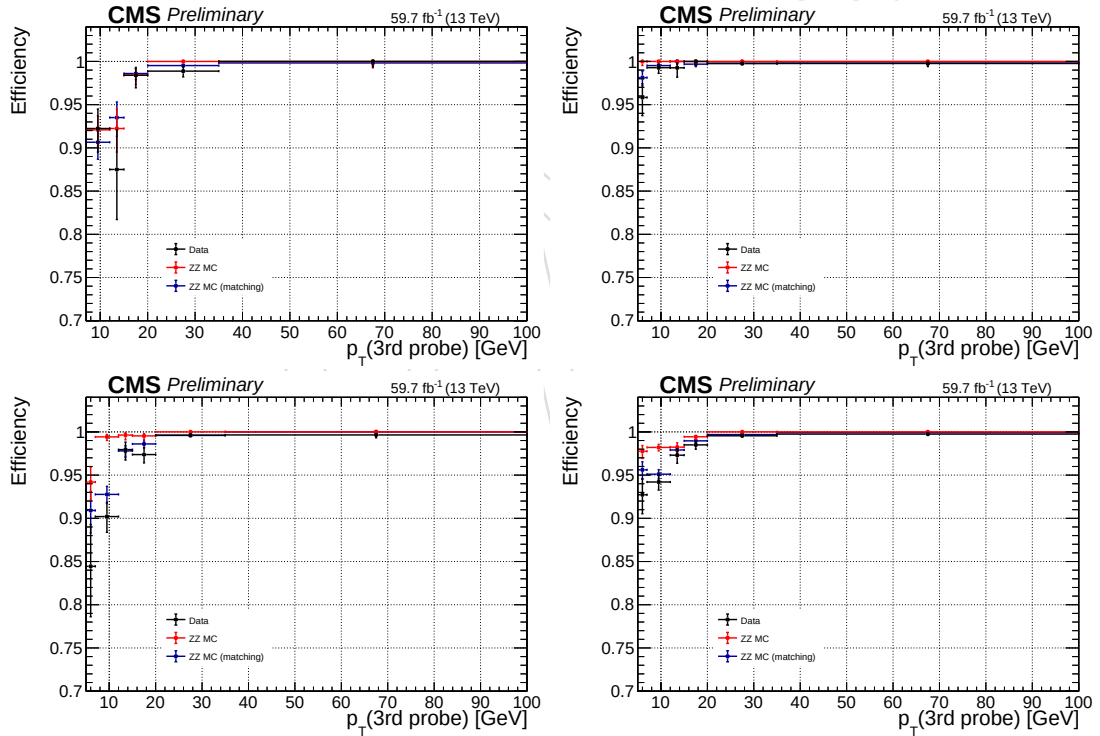


Figure 1: Trigger efficiency measured in 2018 data using 4ℓ events collected by single lepton triggers for the $4e$ (top left), 4μ (top right), $2e2\mu$ (bottom left) and 4ℓ (bottom right) final states.

A summary of the trigger efficiencies in MC truth, and in MC and data using the tag and probe method are summarized in table 3. The trigger efficiency in simulation is found to be $> 99\%$ in each final state.

Final State	gg $\rightarrow$ H MC	gg $\rightarrow$ H MC (matching)	Data (matching)
$4e$	$0.991^{+0.002}_{-0.002}$	$0.948^{+0.004}_{-0.004}$	$0.982^{+0.005}_{-0.007}$
4μ	$0.997^{+0.001}_{-0.001}$	$0.997^{+0.001}_{-0.001}$	$1.000^{+0.000}_{-0.001}$
$2e2\mu$	$0.995^{+0.001}_{-0.001}$	$0.964^{+0.002}_{-0.002}$	$0.983^{+0.003}_{-0.004}$

Table 3: **FIXME: TO BE UPDATED, CURRENTLY COPY PASTE OF 2017 NUMBERS.** Trigger efficiencies measured using 4ℓ events in 2018 data.

Process	Dataset Name	$\sigma \times BR(\times \epsilon_{\text{filter}})$
gg $\rightarrow$ H $\rightarrow$ ZZ $\rightarrow$ 4ℓ	/GluGluHToZZTo4L_M125_13TeV_powheg2_JHUGenV709_pythia8/[1]	12.18 fb
qq $\rightarrow$ Hqq $\rightarrow$ ZZqq $\rightarrow$ 4ℓ qq	/VBF_HToZZTo4L_M125_13TeV_powheg2_JHUGenV709_pythia8/[1]	1.044 fb
q $\bar{q}$ $\rightarrow$ W ⁺ H $\rightarrow$ W ⁺ ZZ $\rightarrow$ 4ℓ + X	/WplusH_HToZZTo4L_M125_13TeV_powheg2-minlo-HWJ_JHUGenV709_pythia8/[1]	0.232 fb
q $\bar{q}$ $\rightarrow$ W ⁻ H $\rightarrow$ W ⁻ ZZ $\rightarrow$ 4ℓ + X	/WminusH_HToZZTo4L_M125_13TeV_powheg2-minlo-HWJ_JHUGenV709_pythia8/[1]	0.147 fb
q $\bar{q}$ $\rightarrow$ ZH $\rightarrow$ ZZ $\rightarrow$ 4ℓ + X	/ZH_HToZZ_4LFilter_M125_13TeV_powheg2-minlo-HZJ_JHUGenV709_pythia8/[1]	0.668 fb
gg $\rightarrow$ ttH $\rightarrow$ ttZZ $\rightarrow$ 4ℓ + X	/ttH_HToZZ_4LFilter_M125_13TeV_powheg2_JHUGenV709_pythia8/[1]	0.393 fb
gg $\rightarrow$ bbH $\rightarrow$ bbZZ $\rightarrow$ 4ℓ + X	/bbH_HToZZTo4L_M125_13TeV_JHUGenV7011_pythia8/[1]	0.135 fb
q $\bar{q}$ /qg $\rightarrow$ tHq $\rightarrow$ tqZZ $\rightarrow$ 4ℓ + X	/tqH_HToZZTo4L_M125_13TeV_JHUGenV7011_pythia8/[1]	0.0213 fb

[1] RunIIAutumn18MiniAOD-102X_upgrade2018_realistic_v15

Table 4: Signal Monte Carlo samples and cross sections.

3.2 Simulation

3.2.1 Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 [21–23] generator for the five main production modes: gluon fusion (gg $\rightarrow$ H) including quark mass effects [24], vector boson fusion (VBF) [25], and associated production (WH, ZH and t $\bar{t}$ H [26]). In the case of WH and ZH the MINLO HVJ extension of POWHEG is used [27]. The description of the decay of the Higgs boson to four leptons is obtained using the JHUGEN generator [28]. In the case of WH, ZH and t $\bar{t}$ H, the Higgs boson is allowed to decay to H $\rightarrow$ ZZ $\rightarrow$ $2\ell 2X$ such that 4-lepton events where two leptons originate from the decay of associated Z, W bosons or top quarks are also taken into account in the simulation. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the NNPDF 3.1 NLO parton distribution functions (PDFs) [29]. The list of signal samples and their cross sections are shown in Table 4.

3.2.2 Background Samples

Production of ZZ via quark-antiquark annihilation is generated at next-to-leading order (NLO) using POWHEG V2 [30] and PYTHIA8, with the same settings as for the Higgs signal. As this simulation covers a large range of ZZ invariant masses, dynamical QCD factorization and renormalization scales have been chosen, equal to m_{ZZ} .

The gg $\rightarrow$ ZZ process is simulated at leading order (LO) with MCFM [31, 32]. In order to match the gg $\rightarrow$ H $\rightarrow$ ZZ transverse momentum spectra predicted by POWHEG at NLO, the showering for MCFM samples is performed with different PYTHIA8 settings, allowing only emissions up to the parton-level scale (“wimpy” shower).

Although not directly used to model data observations, additional MC samples of WZ, Drell-Yan+jets, t $\bar{t}$, and tribosons are generated using MADGRAPH5_AMCATNLO [33] either inclusively or merging several jet multiplicities, as detailed in the table. Table 5 summarizes the MC simulation datasets used for this analysis.

Process	Dataset Name	$\sigma \cdot BR$
$qq \rightarrow ZZ \rightarrow 4\ell$	/ZZTo4L_13TeV_powheg_pythia8/[1]	1.256pb
$qq \rightarrow ZZ \rightarrow 4\ell$	/ZZTo4L_13TeV-amcatnloFXFX-pythia8/[1]	1.212pb
$gg \rightarrow ZZ \rightarrow 4e$	/GluGluToContinToZZTo4e_13TeV_MCFM701/[1]	0.00159pb
$gg \rightarrow ZZ \rightarrow 4\mu$	/GluGluToContinToZZTo4mu_13TeV_MCFM701/[1]	0.00159pb
$gg \rightarrow ZZ \rightarrow 4\tau$	/GluGluToContinToZZTo4tau_13TeV_MCFM701/[1]	0.00159pb
$gg \rightarrow ZZ \rightarrow 2e2\mu$	/GluGluToContinToZZTo2e2mu_13TeV_MCFM701/[1]	0.00319pb
$gg \rightarrow ZZ \rightarrow 2e2\tau$	/GluGluToContinToZZTo2e2tau_13TeV_MCFM701/[1]	0.00319pb
$gg \rightarrow ZZ \rightarrow 2\mu2\tau$	/GluGluToContinToZZTo2mu2tau_13TeV_MCFM701/[1]	0.00319pb
$Z \rightarrow \ell\ell + \text{jets}$	/DYJetsToLL_M-50_TuneCUETP8M1_13TeV-amcatnloFXFX-pythia8/[1]	6104pb
$Z \rightarrow \ell\ell + \text{jets}$	/DYJetsToLL_M-10to50_TuneCUETP8M1_13TeV-amcatnloFXFX-pythia8/[1]	18610pb
$WZ \rightarrow 3\ell\nu$	/WZTo3LNU_TuneCUETP8M1_13TeV_powheg_pythia8/[1]	4.430pb
$t\bar{t}$	/TTJets_TuneCUETP8M1_13TeV-amcatnloFXFX-pythia8/[1]	815.96pb
$t\bar{t} \rightarrow 2\ell 2\nu 2b$	/TTTo2L2Nu_13TeV_powheg/[1]	87.31pb

[1] RunIIAutumn18MiniAOD-102X_upgrade2018_realistic.v15

Table 5: Background Monte Carlo samples and cross sections.

3.2.3 Pileup Reweighting

For each year, corresponding simulation samples are reweighted to match the pileup distribution in data. An example of reweighting procedure for 2018 data is shown in Fig. 2.

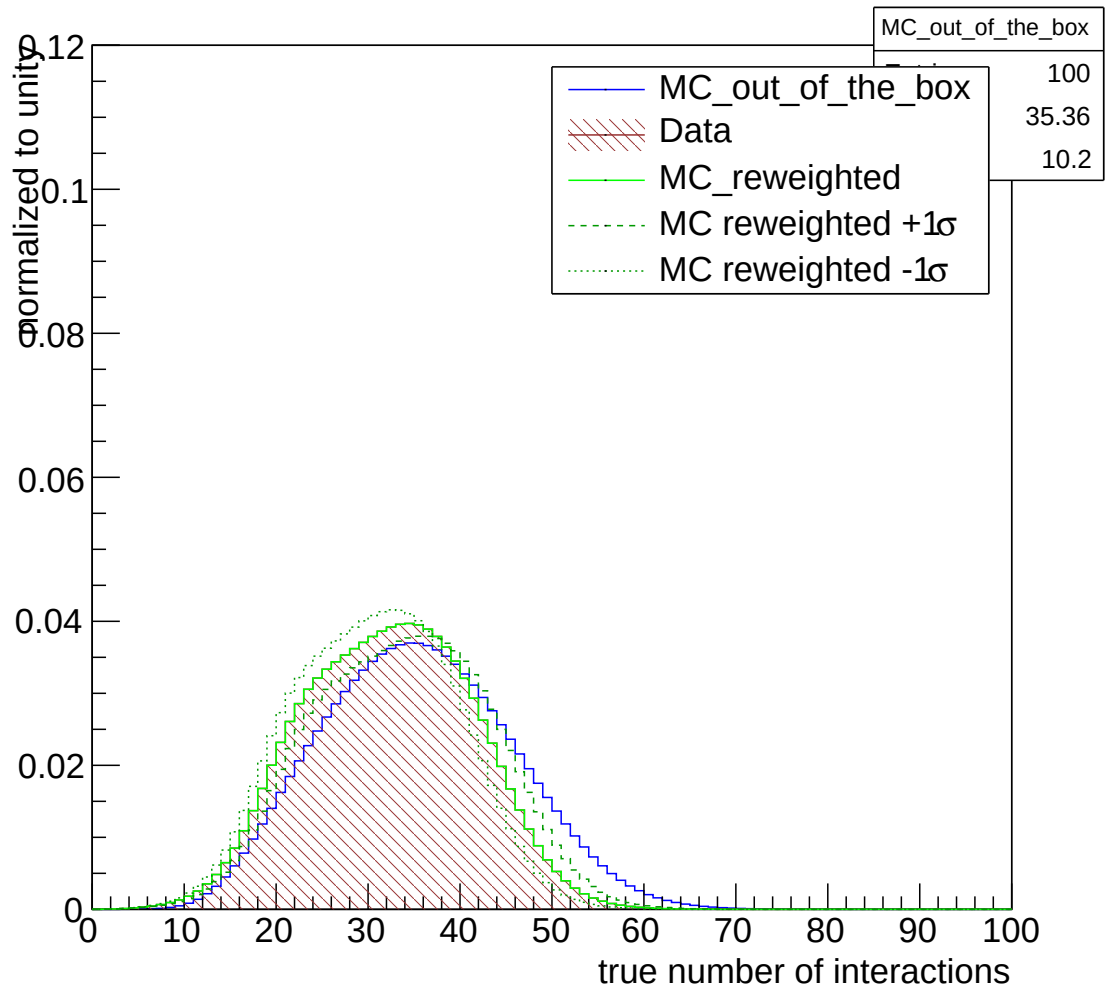


Figure 2: Distribution of pileup in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weights are also shown.

4 Objects

4.1 Electrons

4.1.1 Electron Reconstruction

More details on electron reconstruction can be found in Ref. [34].

Electron candidates are preselected using loose cuts on track-cluster matching observables, so as to preserve the highest possible efficiency while rejecting part of the QCD background. To be considered for the analysis, electrons are required to have a transverse momentum $p_T^e > 7$ GeV, a reconstructed $|\eta^e| < 2.5$, and to satisfy a loose primary vertex constraint defined as $d_{xy} < 0.5$ cm and $d_z < 1$ cm. Such electrons are called **loose electrons**.

The data-MC discrepancy is corrected using scale factors as is done for the electron selection with data efficiencies measured using the same tag-and-probe technique outlined later (see Section 4.1.5). These studies for reconstructions are carried out by the EGM POG and the results are only summarised here.

The electron reconstruction scale factors are shown Fig. 3 and are applied as a function of the super cluster η and electron p_T .

4.1.2 Electron Identification and Isolation

One of the main improvements brought in the analysis is the usage of a new multivariate discriminant for electron selection in all data taking periods.

Reconstructed electrons are now identified and isolated by means of a Gradient Boosted Decision Tree (GBDT) multivariate classifier algorithm, which exploits observables from the electromagnetic cluster, the matching between the cluster and the electron track, observables based exclusively on tracking measurements as well as particle flow isolation sums. The full list of observables used can be found in the Table 6.

observable type	observable name
cluster shape	RMS of the energy-crystal number spectrum along η and ϕ ; $\sigma_{i\eta i\eta}, \sigma_{i\phi i\phi}$
	super cluster width along η and ϕ
	'ratio of the hadronic energy behind the electron supercluster to the supercluster energy, H/E
	circularity $(E_{5 \times 5} - E_{5 \times 1})/E_{5 \times 5}$
	sum of the seed and adjacent crystal over the super cluster energy R_9
track-cluster matching	for endcap traing bins: energy fraction in pre-shower E_{PS}/E_{raw}
	energy-momentum agreement $E_{tot}/p_{in}, E_{ele}/p_{out}, 1/E_{tot} - 1/p_{in}$
tracking	position matching $\Delta\eta_{in}, \Delta\phi_{in}, \Delta\eta_{seed}$
	fractional momentum loss $f_{brem} = 1 - p_{out}/p_{in}$
	number of hits of the KF and GSF track $N_{KF}, N_{GSF} (\cdot)$
	reduced χ^2 of the KF and GSF track $\chi_{KF}^2, \chi_{GSF}^2 (\cdot)$
	number of expected but missing inner hits $(\cdot)$
isolation	probability transform of conversion vertex fit $\chi^2 (\cdot)$
	Particle Flow photon isolation sum $(\cdot)$
	Particle Flow charged hadrons isolation sum $(\cdot)$
For PU-resilience	Particle Flow neutral hadrons isolation sum $(\cdot)$
	Mean energy density in the event: $\rho (\cdot)$

Table 6: Overview of input variables to the identification classifier. Variables not used in the run I MVA are marked with $(\cdot)$.

The classifier is trained on 2017 Drell-Yan plus jets MC sample for both signal and background:

/DYJetsToLL M=50 TuneCUETP8M1 13TeV-madgraphMLM-pythia8/RunIISummer17MiniAOD-92X upgrade2017 realistic v10 ext1-v1/MINIAODSIM.

Fig. 4 shows the output of the multiclassifier discriminant for prompt electrons from DY events and fakes from jets in DY events.

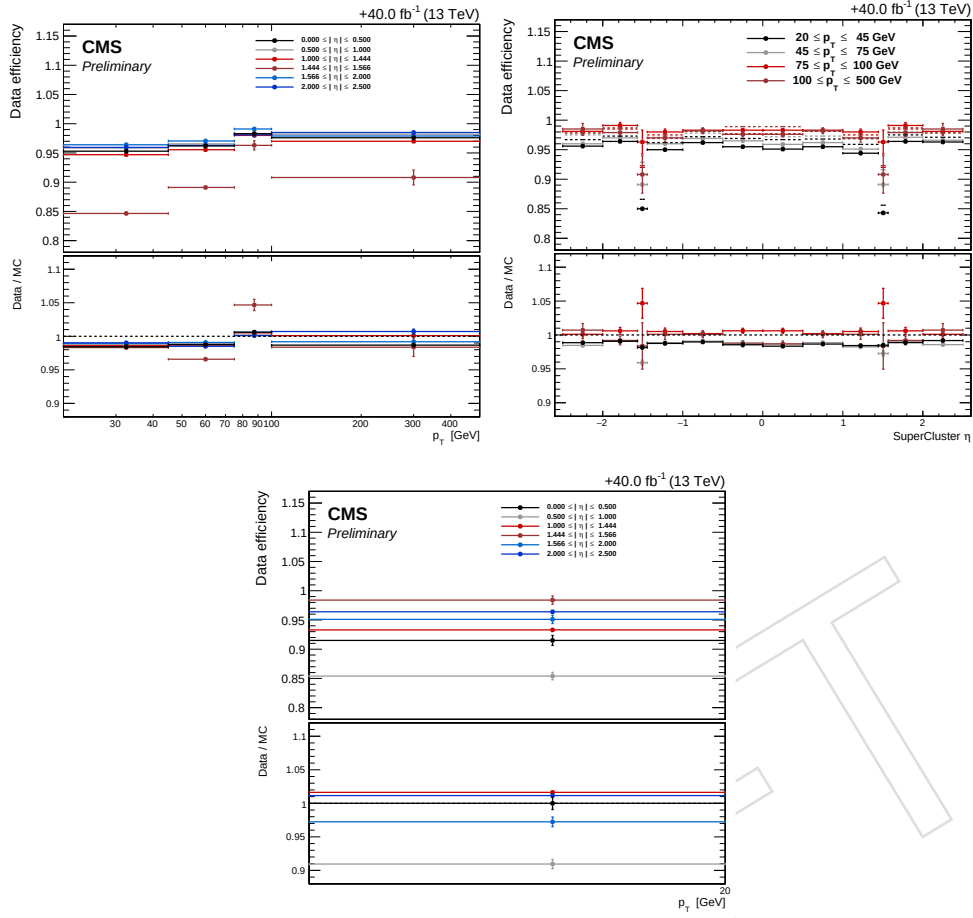


Figure 3: Electron reconstruction efficiencies efficiency in data versus p_T (left) and η (right) for electrons with $p_T > 20$ GeV (top) and $p_T < 20$ GeV (bottom) with corresponding data/MC scale factors as provided by the EGM POG.

The impact of the transition from the TMVA(v1) to the xgboost(v2) training library is shown in Fig. 5. The working points shown are chosen so as to get the same signal efficiency as a cut on MVA ID and a cut on the PF isolation, in each p_T and η bin.

Table 7 lists the cut values applied to the BDT score for the chosen working point. For the analysis, loose electrons have to pass this MVA identification and isolation working point.

		$ \eta < 0.8$	
		Cut on BDT score	Signal eff. Background eff.
$5 < p_T < 10$ GeV	$p_T > 10$ GeV	1.264	81.04% 4.4%
		2.364	97.1% 2.9%
		$0.8 < \eta < 1.479$	
		Cut on BDT score	Signal eff. Background eff.
$5 < p_T < 10$ GeV	$p_T > 10$ GeV	1.178	79.3% 4.6%
		2.078	96.3% 3.6%
		$ \eta > 1.479$	
		Cut on BDT score	Signal eff. Background eff.
$5 < p_T < 10$ GeV	$p_T > 10$ GeV	1.330	72.97% 3.6%
		1.080	95.7% 6.7%

Table 7: Minimum BDT score required for passing the electron identification, together with the corresponding signal and background efficiencies.

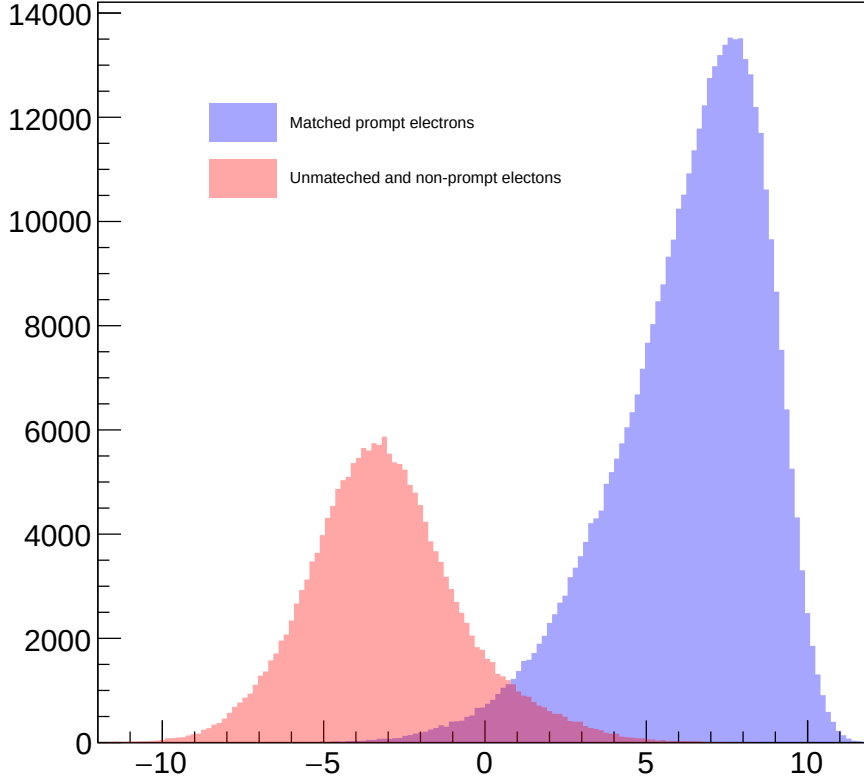


Figure 4: Output of the multiclassifier discriminant for prompt electrons matched to truth electrons from Z decay (blue) and for fakes (red). Events are all taken from DY events MC sample.

4.1.3 Electron Impact Parameter Selection

In order to ensure that the leptons are consistent with a common primary vertex we require that they have an associated track with a small impact parameter with respect to the event primary vertex. We use the significance of the impact parameter to the event vertex, $|\text{SIP}_{3\text{D}}| = \frac{|\text{IP}|}{\sigma_{\text{IP}}}$, where IP is the lepton impact parameter in three dimensions at the point of closest approach with respect to the primary interaction vertex, and σ_{IP} the associated uncertainty. Hereafter, a “primary lepton” is a lepton satisfying $|\text{SIP}_{3\text{D}}| < 4$.

4.1.4 Electron Energy Calibrations

Electrons in data are corrected for features in ECAL energy scale in bins of p_{T} and $|\eta|$. Corrections are calculated on a $\text{Z} \rightarrow \text{ee}$ sample to align the dielectron mass spectrum in the data to that in the MC, and to minimize its width.

The $\text{Z} \rightarrow \text{ee}$ mass resolution in Monte Carlo is made to match data by applying a pseudorandom Gaussian smearing to electron energies, with Gaussian parameters varying in bins of p_{T} and $|\eta|$. This has the effect of convoluting the electron energy spectrum with a Gaussian.

The electron energy scale is measured in data by fitting a Crystall-ball function to the di-

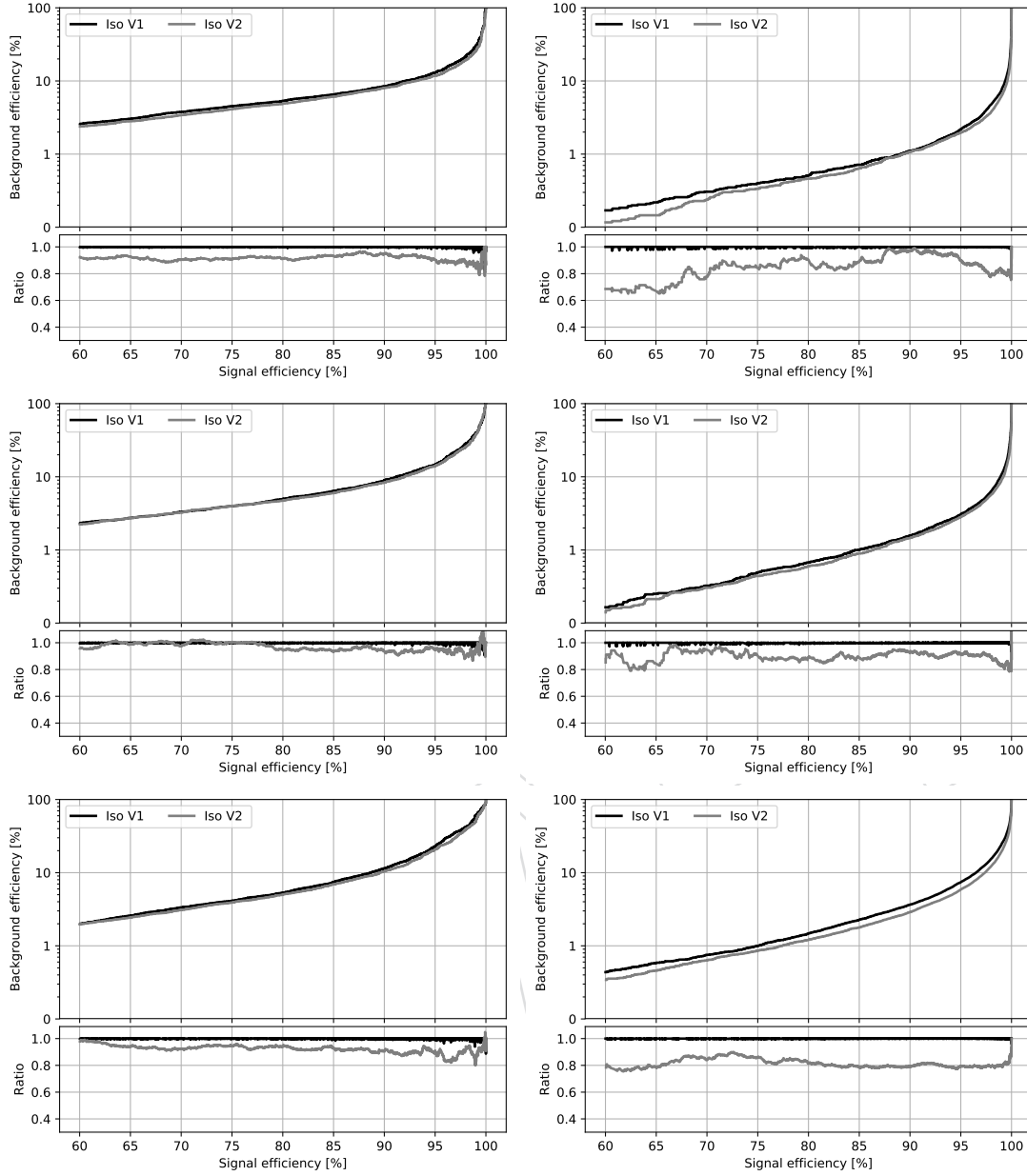


Figure 5: The receiver operating characteristic curves (background efficiency vs signal efficiency) of the MVA's trained on the 2017 simulation and applied on the 2018 simulation. The training combines identification and isolation variables. Performance are shown for electrons with $5 < p_T < 10$ GeV (left), $p_T > 10$ GeV (right) and $|\eta| < 0.8$ (top), $0.8 < |\eta| < 1.479$ (middle) and $|\eta| > 1.479$ (bottom). V1 and V2 versions of training are compared, exploiting TMVA and xgboost training libraries respectively.

electron mass spectrum around the Z peak in the $Z \rightarrow ee$ control region. The energy scale for the full 2018 dataset is shown in Fig. 6(a) and agrees decently with the MC with the preliminary corrections released by EGAMMA POG for Moriond.

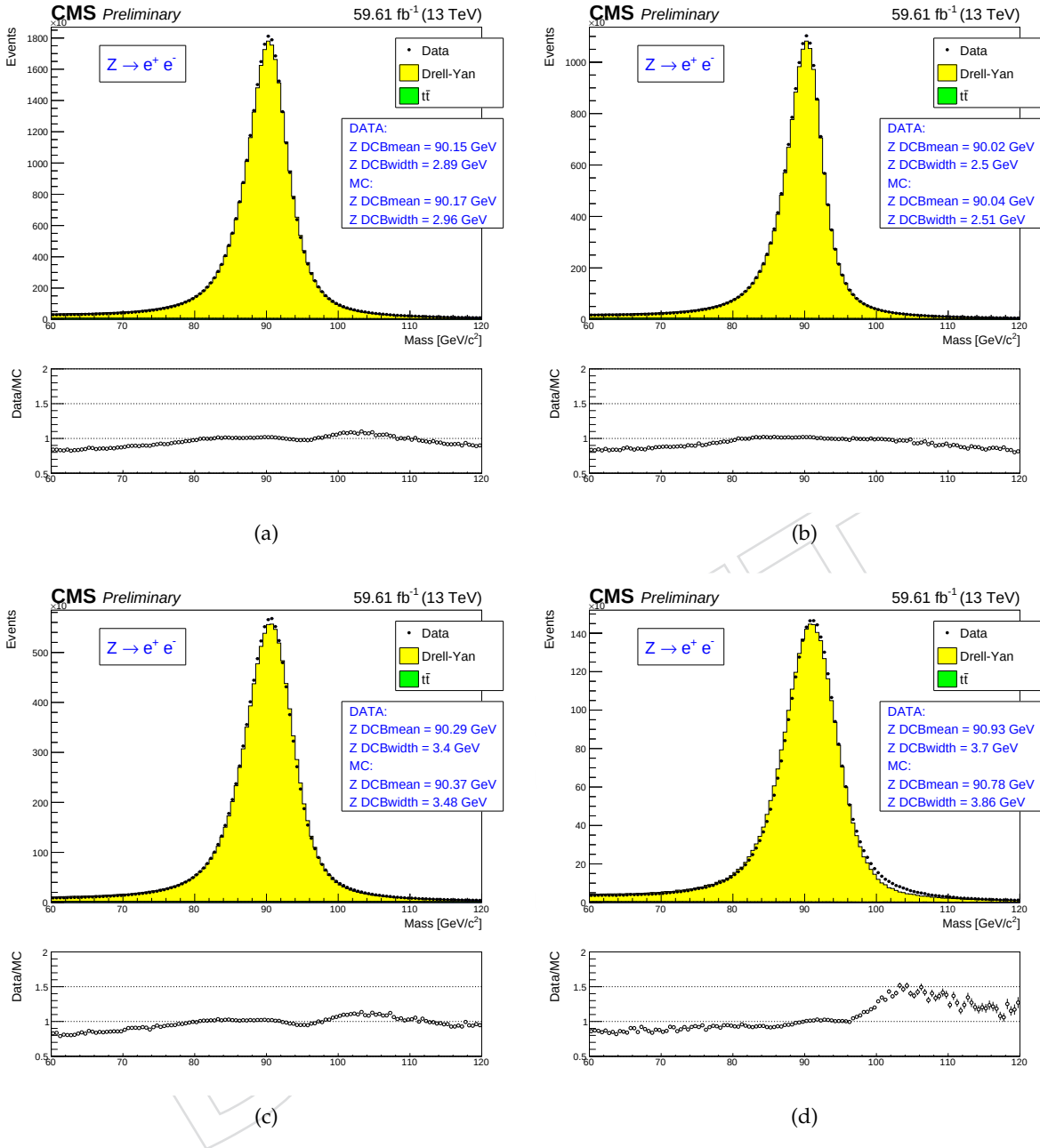


Figure 6: (a): electron energy scale measured in the $Z \rightarrow ee$ control region for all electrons, for both electrons in the barrel (b), for one electron in the barrel, one in the endcaps (c) and for both electrons in the endcaps (d). The results of the Crystall-ball fit are reported in the figures.

4.1.5 Electron Efficiency Measurements

The Tag-and-Probe study was performed on the EGM dataset using the golden JSON of 58.83 fb⁻¹. More details on the Tag-and-Probe method can be found in Ref. [35].

Tag electrons need to satisfy the following quality requirements:

- trigger matched to HLT_Ele32_WPTight_Gsf_L1DoubleEG_v*

- $p_T > 30$ GeV, super cluster (SC) $\eta < 2.17$

For the bin between 7 and 20 GeV, additional criteria are required: the tag has to pass a cut on the MVA ID > 0.92 , $\sqrt{(2 * PFMET * p_T(tag) * (1 - \cos(\phi_{MET} - \phi_{tag})))} < 45 \text{ GeV}$.

Probe electrons only need to be reconstructed as GsfElectron while the FSR recovery algorithm is not applied in efficiency measurement.

The nominal MC efficiencies are evaluated from the LO MadGraph Drell-Yan, while the NLO systematics use the MadGraph_AMCatNLO sample.

For the efficiency measurements a template fit is used. The m_{ee} signal shape of the passing and failing probes is taken from MC and convoluted with a Gaussian. The data is then fitted with the convoluted MC template and a CMSShape (an Error-function with a one-sided exponential tail). This change follows from the usage of the T&P tool developed by the EGM POG.

The electron selection efficiency is measured as a function of the probe electron p_T and its SC η , and separately for electrons falling in the ECAL gaps. Figure 7 and ?? shows the p_T and η turn-on curves measured in data.

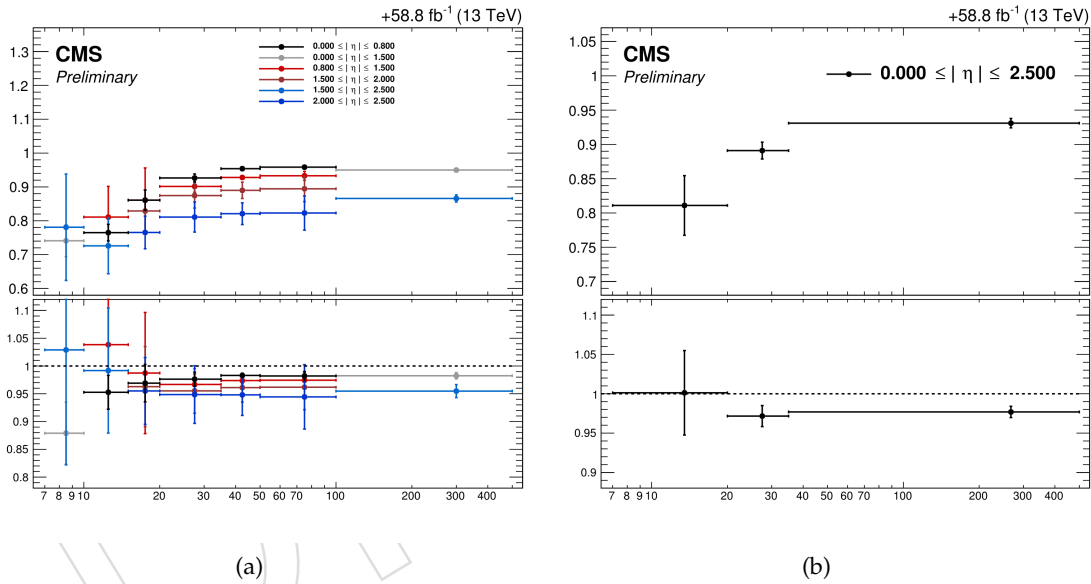


Figure 7: Electron selection efficiencies vs p_T measured using the Tag-and-Probe technique described in the text, non-gap electrons (left) and gap electrons (right), together with the corresponding data/MC ratio (bottom).

The EGM recommendations on the evaluation of Tag-and-Probe uncertainties for efficiency measurements are followed. Specifically, we consider

- Variation of the signal shape from a MC shape to an analytic shape (Crystal Ball) fitted to the MC
- Variation of the background shape from a CMS-shape to a simple exponential in fits to data
- Using an NLO MC sample for the signal templates

The total uncertainty for the measurement of the scale factors is the quadratic sum of the statistical uncertainties returned from the fit and the aforementioned systematic uncertainties.

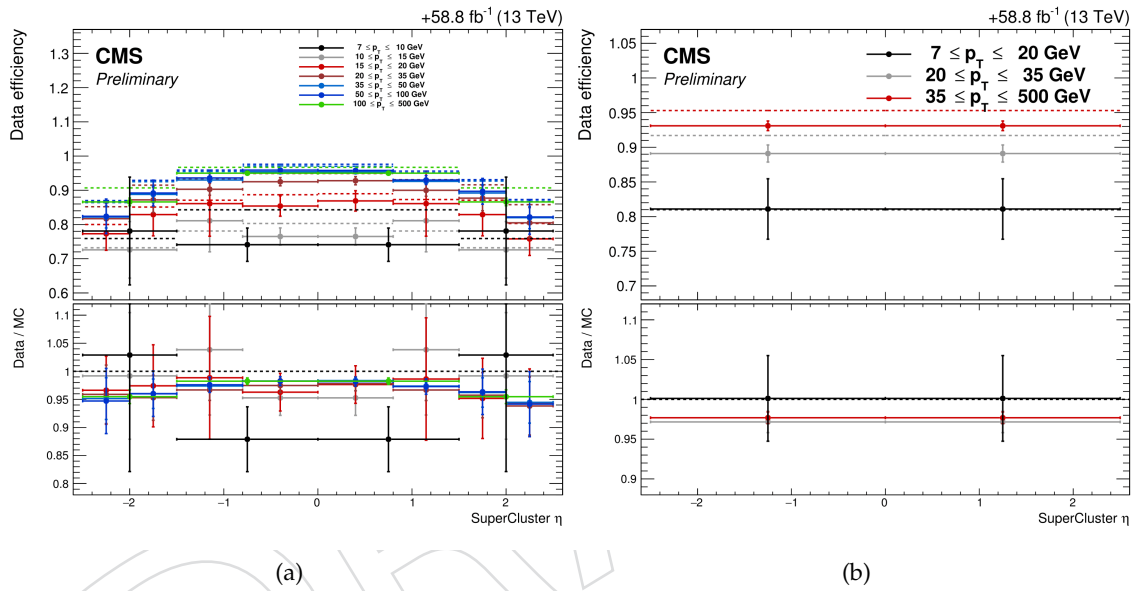


Figure 8: Electron selection efficiencies vs η measured using the Tag-and-Probe technique described in the text, non-gap electrons (left) and gap electrons (right), together with the corresponding data/MC ratio at the bottom of each plot.

4.2 Muons

4.2.1 Muon Reconstruction and Identification

More details on muon reconstruction can be found in Ref. [35]. We define **loose muons** as the muons that satisfy $p_T > 5$, $|\eta| < 2.4$, $d_{xy} < 0.5$ cm, $dz < 1$ cm, where d_{xy} and dz are defined w.r.t. the PV and using the 'muonBestTrack'. Muons have to be reconstructed by either the Global Muon or Tracker Muon algorithm. Standalone Muon tracks that are only reconstructed in the muon system are rejected. Muons with `muonBestTrackType==2` (standalone) are discarded even if they are marked as global or tracker muons.

Loose muons with p_T below 200 GeV are considered identified muons if they also pass the PF muon ID (note that the naming convention used for these IDs differs from the muon POG naming scheme, in which the "tight ID" used here is called the "loose ID"). Loose muons with p_T above 200 GeV are considered identified muons if they pass the PF ID or the Tracker High- p_T ID, the definition of which is shown in Table 8. This relaxed definition is used to increase signal efficiency for the high-mass search. When a very heavy resonance decays to two Z bosons, both bosons will be very boosted. In the lab frame, the leptons coming from the decay of a highly boosted Z will be nearly collinear, and the PF ID loses efficiency for muons separated by approximately $\Delta R < 0.4$, which roughly corresponds to muons originating from Z bosons with $p_T > 500$ GeV.

Table 8: The requirements for a muon to pass the Tracker High- p_T ID. Note that these are equivalent to the Muon POG High- p_T ID with the global track requirements removed.

Plain-text description	Technical description
Muon station matching	Muon is matched to segments in at least two muon stations NB: this implies the muon is an arbitrated tracker muon.
Good p_T measurement	$\frac{p_T}{\sigma_{p_T}} < 0.3$
Vertex compatibility ($x - y$)	$d_{xy} < 2$ mm
Vertex compatibility (z)	$d_z < 5$ mm
Pixel hits	At least one pixel hit
Tracker hits	Hits in at least six tracker layers

An additional "ghost-cleaning" step is performed to deal with situations when a single muon can be incorrectly reconstructed as two or more muons:

- Tracker Muons that are not Global Muons are required to be arbitrated.
- If two muons are sharing 50% or more of their segments then the muon with lower quality is removed.

4.2.2 Muon Isolation

Particle-Flow based isolation is used for the muons. The so-called $\Delta\beta$ correction is applied in order to subtract the pileup contribution for the muons, whereby $\Delta\beta = \frac{1}{2} \sum_{\text{PU}}^{\text{charged had.}} p_T$ gives an estimate of the energy deposit of neutral particles (hadrons and photons) from pile-up vertices. The relative isolation for muons is then defined as:

$$\text{RelPFiso} = \frac{\sum^{\text{charged had.}} p_T + \max(\sum^{\text{neutral had.}} E_T + \sum^{\text{photon}} E_T - \Delta\beta, 0)}{p_T^{\text{lepton}}} \quad (1)$$

The isolation working point for muons was optimized in Ref. [35] and the working point was chosen to be the same as electrons, namely $\text{RelPFiso}(\Delta R = 0.3) < 0.35$.

4.2.3 Muon Impact Parameter Selection

We apply the same selection on the muon significance of impact parameter as for the electrons, as described in Sec. 4.1.3:

- $|\text{SIP}_{3D} = \frac{\text{IP}}{\sigma_{\text{IP}}}| < 4$

4.2.4 Muon Energy Calibrations

Similar to electrons the muon momentum scale is measured in data by fitting a Crystall-ball function to the di-muon mass spectrum around the Z peak in the $Z \rightarrow \mu\mu$ control region. At the moment no muon scale and resolution corrections are available but from Fig. 9 shows a very good agreement between data and simulation.

4.2.5 Muon Efficiency Measurements

Muon efficiencies are measured with the Tag and Probe (T&P) method performed on $Z \rightarrow \mu\mu$ and $J/\psi \rightarrow \mu\mu$ events in bins of p_T and η . More details on the methodology can be found in Ref. [36]. Measurements are extracted using 2018 RunA,B,C,D data while the measurements corresponding to 2016 and 2017 datasets have already been reported in Ref. [37] and Ref. [38] respectively. The Z sample is used to measure the muon reconstruction and identification efficiency at high p_T , and the efficiency of the isolation and impact parameter requirements at all p_T . The J/ψ sample is used to measure the reconstruction efficiency at low p_T , as it benefits from a better purity in that kinematic regime. In this case, events are collected using `HLT_Mu7p5_Track2_Jpsi_v*` when probing the reconstruction and identification efficiency in the muon system, and using the `HLT_Mu7p5_L2Mu2_Jpsi_v*` when probing the tracking efficiency.

Reconstruction and identification Results for the muon reconstruction and identification efficiency for $p_T > 20 \text{ GeV}$ have been derived by the Muon POG. The probe in this measurement are tracks reconstructed in the inner tracker, and the passing probes are those that are also reconstructed as a global or tracker muon and passing the Muon POG Loose muon identification. Results for low p_T muons were derived using J/ψ events, with the same definitions of probe and passing probes. The systematic uncertainties are estimated by varying the analytical signal and background shape models used to fit the dimuon invariant mass. Details on the procedure can be found in Ref. [35]. The efficiency and scale factors used for low p_T muons are the ones derived using single muon prompt-reco dataset.

The efficiency in data and simulation is shown in Fig. 10.

Impact parameter requirements The measurement is performed using Z events. Events are selected with `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers. For this measurement, the probe is a muon passing the POG Loose identification criteria, and it is considered a passing probe if satisfies the SIP_{3D} , d_{xy} , d_z cuts of this analysis. The results are shown in Fig. 11.

Isolation requirements The isolation efficiency is measured using events from the Z decay for any p_T . The events are selected with either of `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers. The isolation of the muons are calculated after recovery of the FSR photons and subtracting their contribution to the isolation cone of the muons. More detailed description of the method can be found in Ref. [39].

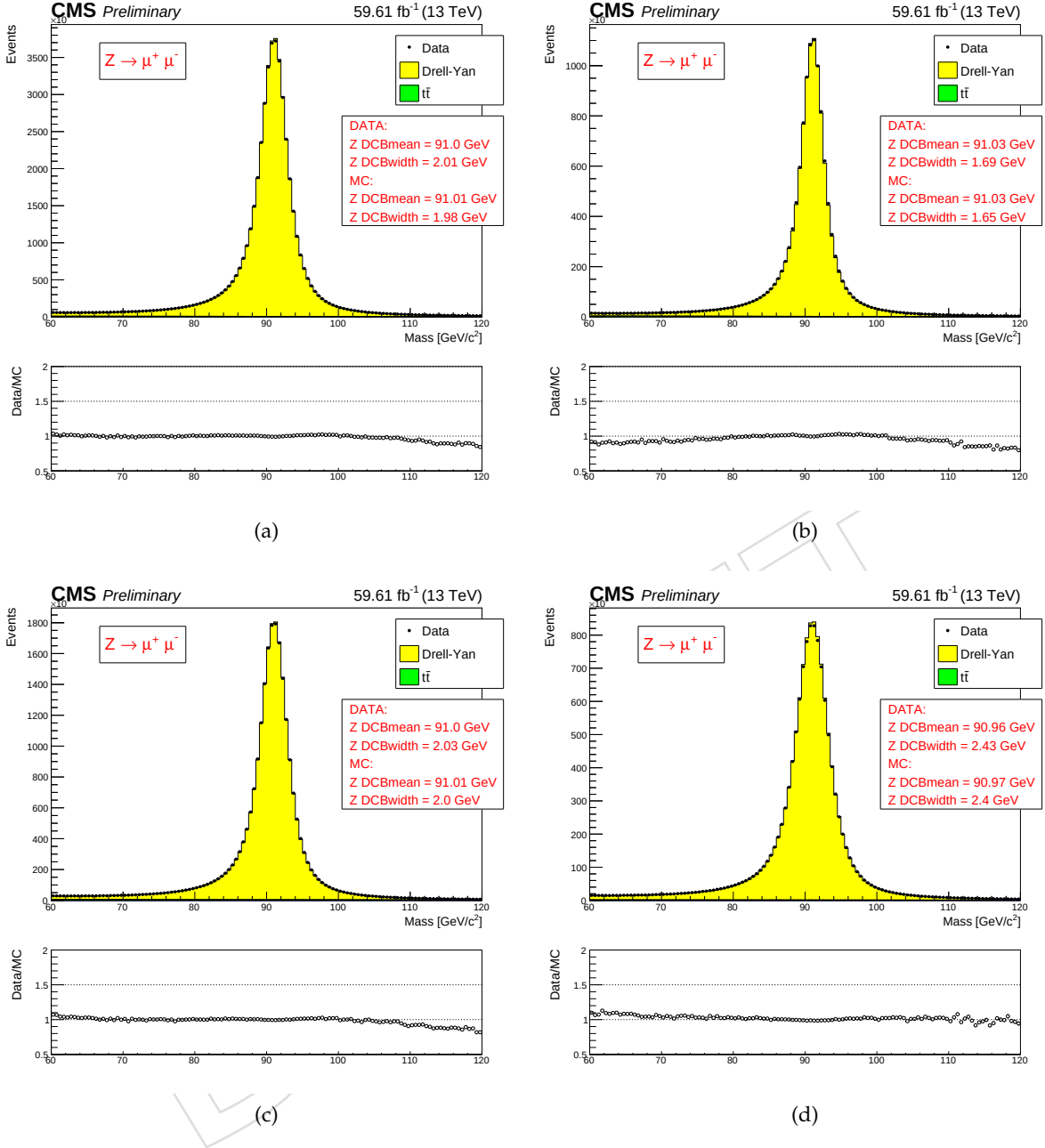


Figure 9: (a): muon energy scale measured in the $Z \rightarrow \mu\mu$ control region for all muons, for both muons in the barrel (b), for one muon in the barrel, one in the endcaps (c) and for both muons in the endcaps (d). The results of the Crystall-ball fit are reported in the figures.

281 The results are shown in Fig. 12.

282 **Tracking** The efficiency to reconstruct a muon track in the inner detector is measured using
 283 as probes tracks reconstructed in the muon system alone. The method for measuring the
 284 tracking efficiency is the same as in Ref. [40], and the results on 2018 data are briefly discussed
 285 here. The efficiency and data to mc scale factors are measured from Z events as a function of
 286 η for $p_T > 10$ GeV and $p_T < 10$ GeV. The values of data to mc scale factors used are from the

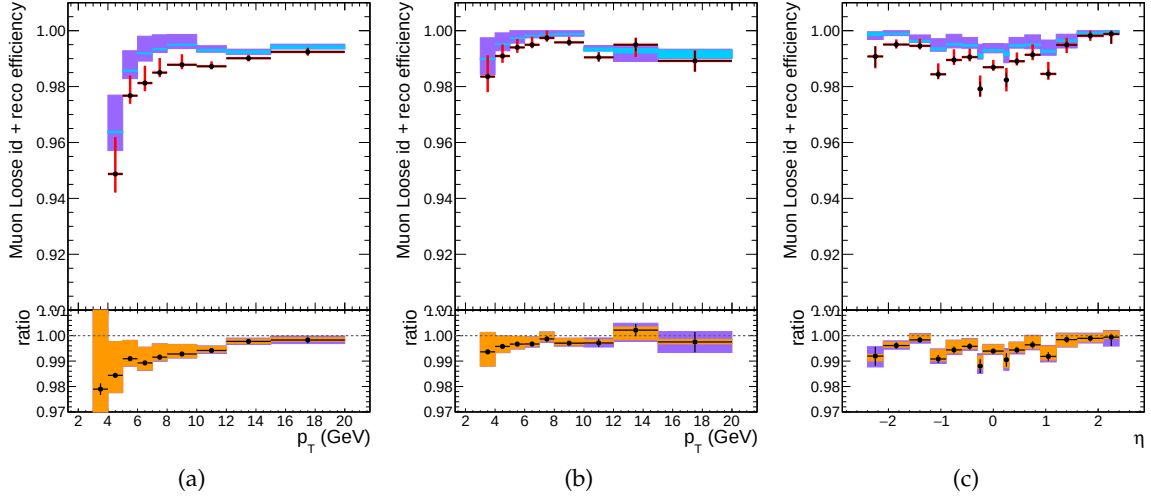


Figure 10: Muon reconstruction and identification efficiency at low p_T , measured with the tag&probe method on J/ψ events, as function of p_T in the barrel (left) and endcaps (center), and as function of η for $p_T > 7$ GeV (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

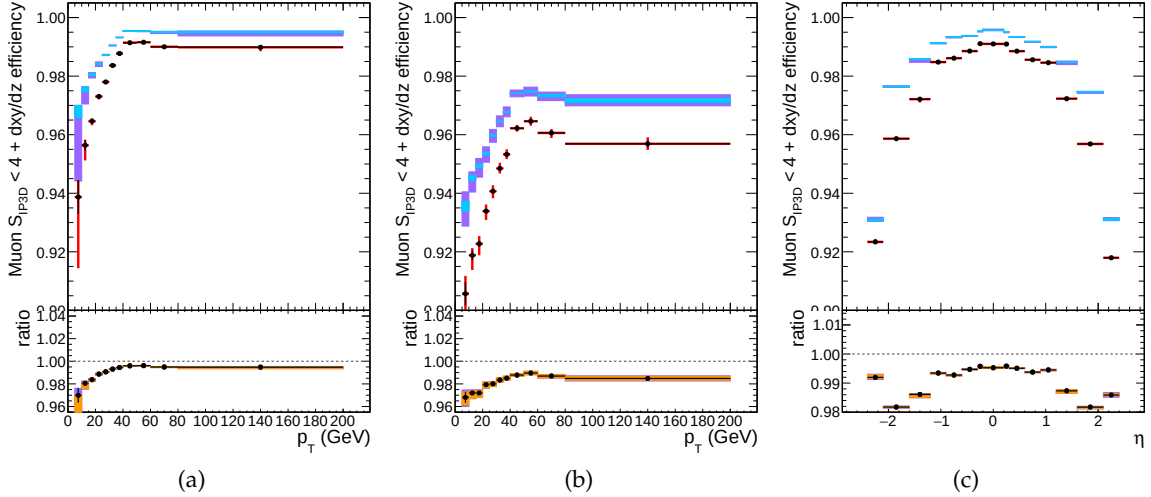


Figure 11: Efficiency of the muon impact parameter requirements, measured with the tag&probe method on Z events, as function of p_T in the barrel (left) and endcaps (center), and as function of η for $p_T > 20$ GeV (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

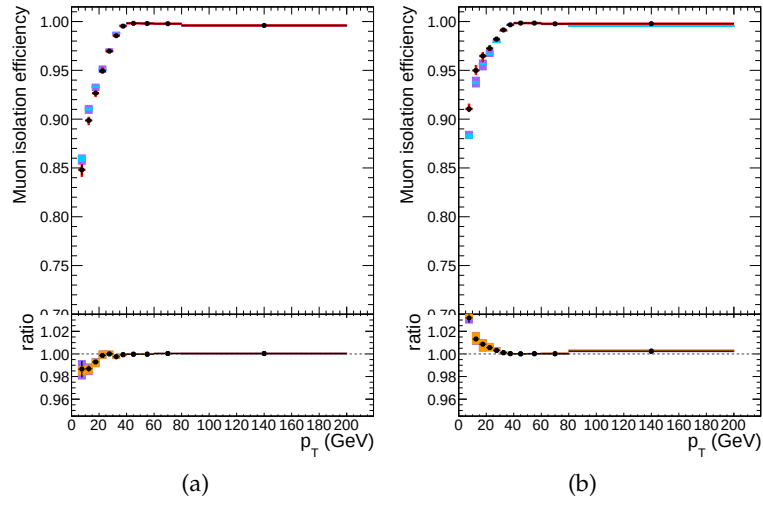


Figure 12: Efficiency of the muon isolation requirement, measured with the tag&probe method on Z events, as function of p_T in the barrel (left) and endcaps (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

287 ReReco version of the full dataset collected in 2018.

288 **Overall results** The product of all the data to simulation scale factors for muon tracking,
 289 reconstruction, identification, impact parameter and isolation requirements is shown in Fig. 13.

290

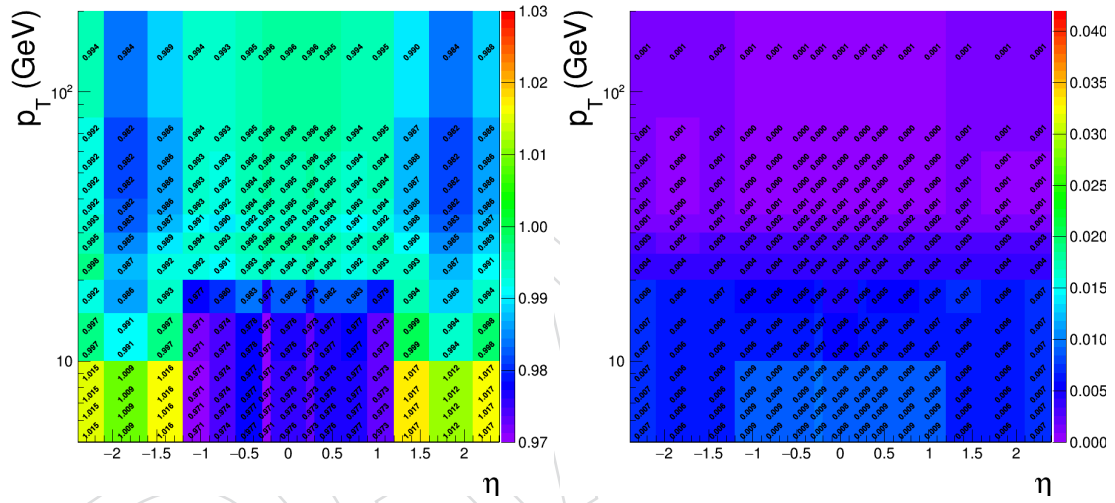


Figure 13: Left: Overall data to simulation scale factors for muons, as function of p_T and η . Right: Uncertainties on data to simulation scale factors for muons, as function of p_T and η .

4.3 Photons for FSR recovery

The FSR recovery algorithm was considerably simplified with respect to what was done in Run I, while maintaining similar performance. The selection of FSR photons is now only done per-lepton and no longer depends on any Z mass criterion, thus much simplifying the subsequent ZZ candidate building and selection. As regards the association of photons with leptons, the rectangular cuts on $\Delta R(\gamma, l)$ and $E_{T,\gamma}$ have been replaced by a cut on $\Delta R(\gamma, l)/E_{T,\gamma}^2$.

Starting from the collection of ‘PF photons’ provided by the particle-flow algorithm, the selection of photons and their association to a lepton proceeds as follows:

1. The preselection of PF photons is done by requiring $p_{T,\gamma} > 2$ GeV, $|\eta^\gamma| < 2.4$, and a relative Particle-flow isolation smaller than 1.8. The latter variable is computed using a cone of radius $R = 0.3$, a threshold of 0.2 GeV on charged hadrons with a veto cone of 0.0001, and 0.5 GeV on neutral hadrons and photons with a veto cone of 0.01, also including the contribution from pileup vertices (with the same radius and threshold as per charged isolation).
2. Supercluster veto: we remove all PF photons that match with any electron passing both the loose ID and SIP cuts. The matching is performed by directly associating the two PF candidates.
3. Photons are associated to the closest lepton in the event among all those pass both the loose ID and SIP cuts.
4. We discard photons that do not satisfy the cuts $\Delta R(\gamma, l)/E_{T,\gamma}^2 < 0.012$, and $\Delta R(\gamma, l) < 0.5$.
5. If more than one photon is associated to the same lepton, the lowest- $\Delta R(\gamma, l)/E_{T,\gamma}^2$ is selected.
6. For each FSR photon that was selected, we exclude that photon from the isolation sum of all the leptons in the event that pass both the loose ID and SIP cuts. This concerns the photons that are in the isolation cone and outside the isolation veto of said leptons ($\Delta R < 0.4$ AND $\Delta R > 0.01$ for muons and $\Delta R < 0.4$ AND ($\eta^{\text{SC}} < 1.479$ OR $\Delta R > 0.08$) for electrons).

More details on the optimization of the FSR photon selection can be found in Ref. [35, 39].

4.4 Jets

Vector Boson Fusion (VBF) and other production mechanisms of Higgs Boson normally differ as regards the jet kinematics. In this analysis, jets are thus used for the event categorization, which will be introduced in Section 7.

4.4.1 Jet Identification

Jets are reconstructed by using the anti- k_T clustering algorithm out of particle flow candidates, with a distance parameter $R = 0.4$, after rejecting the charged hadrons that are associated to a pileup primary vertex.

To reduce instrumental background, the tight working point jet ID suggested by the JetMET Physics Object Group is applied [41]. In addition, jets from Pile-Up are rejected using the PileUp jet ID criteria suggested by the JetMET POG [42]. In this analysis, the jets are required to be within $|\eta| < 4.7$ area and have a transverse momentum above 30 GeV. In addition, the

jets are cleaned from any of the tight leptons (passing the SIP and isolation cut computed after FSR correction) and FSR photons by a separation criterion: $\Delta R(\text{jet}, \text{lepton}/\text{photon}) > 0.4$.

4.4.2 Jet Energy Corrections

The calorimeter response to particles is not linear and it is not straightforward to translate the measured jet energy to the true particle or parton energy, therefore we need Jet Energy Corrections. In this analysis, standard jet energy corrections are applied to the reconstructed jets, which consist of L1 Pileup, L2 Relative Jet Correction, L3 Absolute Jet Correction for both Monte Carlo samples and data, and also residual calibration for data [43]. Jet Energy resolution corrections are also applied to MC.

4.4.3 B-tagging

For categorization purpose, we need to distinguish whether a jet is b-jet or not. The *DeepCSV* algorithm is used as our b-tagging algorithm. It combines the same information as the previous tagger *CSVv2*, impact parameter significance, secondary vertex and jet kinematics but uses information of more tracks. Also, the b-tag output discriminator is computed with a Deep Neural Network. In this analysis, a jet is considered to be b-tagged if it passes the *medium* working point, i.e. if its `|pfDeepCSVJetTags:probb+ pfDeepCSVJetTags:probbb|` discriminator is greater than 0.4184 [44].

Data to simulation scale factors for b-tagging efficiency are provided for this working point for the full dataset as a function of jet p_T , η and flavour. They are applied to simulated jets by downgrading (upgrading) the b-tagging status of a fraction of the b-tagged (untagged) jets that have a scale factor smaller (larger) than one. **They are not yet available but will be applied as soon as they arrive.**

4.4.4 Validation on data

Similarly to what was done with electrons and muons, we performed some validation studies to probe the agreement between data and simulated samples for jets. Events containing $Z \rightarrow e^+e^-$ or $Z \rightarrow \mu^+\mu^-$ were selecting, following the same trigger and lepton selection criteria of the main analysis, but increasing the p_T threshold on leptons to 30 GeV and asking the invariant mass of the two leptons to be within the 70-110 GeV range. Basic properties of additional jets in the event, with $p_T > 30$ GeV and $\eta < 4.7$, were then compared. Figure 14 shows the pseudo-rapidity of the leading jet in data and MC. Simulated samples of Drell-Yan and $t\bar{t}$ are used. Uncertainties from the jet energy corrections are also displayed on the data/MC ratio plots.

As it has been shown, the Jet Energy Resolution corrections are creating the “horns” one can see in the MC for η of about 2.8. Although not satisfactory, the disagreement we see between DATA and MC in this region is covered by the very large uncertainty. After discussion with jet experts in the JetMET POG, it has been decided that this is the configuration we should use for this preliminary result. We also verified that it was not inducing significant migration of events in the signal MC between the categories.

The disagreement between data and MC for the MET distribution prevents us at this stage to use this variable in the analysis, in particular to better target VH events where, for instance, a Z would decay into neutrinos.

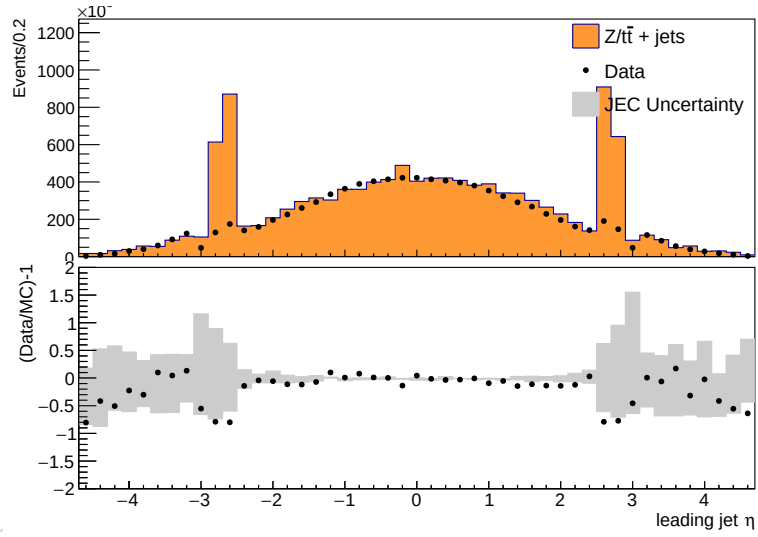


Figure 14: Comparison between data and MC for the leading jet η in $Z \rightarrow \ell\ell + \text{jets}$ events. MC is normalized to data. Jet ID and Jet PUID are applied. MC samples include DY and $t\bar{t}$. Data/MC ratio plots are shown in the bottom of each plots, together with the uncertainties (shaded histograms) from Jet Energy Corrections.

4.5 Object Summary

The requirements on all objects used for the analysis with 2018 data are summarized in the Table 9. In addition, a “ghost-cleaning” procedure is applied to the muons, as described in Sec. 4.2.1.

A lepton is declared **loose** if it passes the reconstruction, kinematics and d_{xy}/d_z cuts and declared **tight** if it passes in addition the identification, isolation and SIP3D cut.

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Table 9: Summary of physics object selection for the $H \rightarrow ZZ \rightarrow 4\ell$ analysis performed on the 2018 data.

Electrons
$p_T^e > 7 \text{ GeV}$ $ \eta^e < 2.5$ $d_{xy} < 0.5 \text{ cm}$ $d_z < 1 \text{ cm}$ $ \text{SIP}_{3D} < 4$ BDT ID with isolation (Fall17v2 training) with cuts from Table 7
Muons
Global or Tracker Muon Discard Standalone Muon tracks if reconstructed in muon system only Discard muons with muonBestTrackType==2 even if they are global or tracker muons $p_T^\mu > 5 \text{ GeV}$ $ \eta^\mu < 2.4$ $d_{xy} < 0.5 \text{ cm}$ $d_z < 1 \text{ cm}$ $ \text{SIP}_{3D} < 4$ PF muon ID if $p_T < 200 \text{ GeV}$, PF muon ID or High- p_T muon ID (Table 8) if $p_T > 200 \text{ GeV}$ $\mathcal{I}_{\text{PF}}^\mu < 0.35$
FSR photons
$p_T^\gamma > 2 \text{ GeV}$ $ \eta^\gamma < 2.4$ $\mathcal{I}_{\text{PF}}^\gamma < 1.8$ $\Delta R(\ell, \gamma) < 0.5$ $\frac{\Delta R(\ell, \gamma)}{(p_T^\gamma)^2} < 0.012 \text{ GeV}^{-2}$
Jets
$p_T^{\text{jet}} > 30 \text{ GeV}$ $ \eta^{\text{jet}} < 4.7$ $\Delta R(\ell/\gamma, \text{jet}) > 0.4$ Cut-based jet ID (tight WP) Jet pileup ID (tight WP) Deep CSV b-tagging (medium WP)

5 Event Selection

5.1 Trigger Selection

The events are required to have fired the High-Level Trigger paths described in section 3.1.1. Unlike in the Run I analysis, the trigger requirement does not depend on the selected final state: it is always the OR of all HLT paths. The reason is in Run II we will be targeting associated production modes that can come with additional leptons, thus improving trigger efficiency further.

5.2 Vertex Selection

The events are required to have at least one good primary vertex (PV) fulfilling the following criteria: high number of degree of freedom ($N_{PV} > 4$), collisions restricted along the z -axis ($z_{PV} < 24$ cm) and small radius of the PV ($r_{PV} < 2$ cm).

5.3 ZZ Candidate Selection

The four-lepton candidates are built from what we call **selected leptons**, which are the tight leptons (defined in sections 4.1.2 and 4.2.1) that pass the $SIP_{3D} < 4$ vertex constraint and the isolation cuts (defined in sections 4.1.2 and 4.2.2), where FSR photons are subtracted as described in Section 4.3. A lepton cross cleaning is applied by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are defined as pairs of selected leptons of opposite charge and matching flavour (e^+e^- , $\mu^+\mu^-$) that satisfy $12 < m_{\ell\ell(\gamma)} < 120$ GeV/ c^2 , where the Z candidate mass includes the selected FSR photons if any.
2. **ZZ candidates** are defined as pairs of non-overlapping Z candidates. The Z candidate with reconstructed mass $m_{\ell\ell}$ closest to the nominal Z boson mass is denoted as Z_1 , and the second one is denoted as Z_2 . ZZ candidates are required to satisfy the following list of requirements:
 - **Ghost removal** : $\Delta R(\eta, \phi) > 0.02$ between each of the four leptons.
 - **lepton p_T** : Two of the four selected leptons should pass $p_{T,i} > 20$ GeV/ c and $p_{T,j} > 10$ GeV/ c .
 - **QCD suppression**: all four opposite-sign pairs that can be built with the four leptons (regardless of lepton flavor) must satisfy $m_{\ell\ell} > 4$ GeV/ c^2 . Here, selected FSR photons are not used in computing $m_{\ell\ell}$, since a QCD-induced low mass dilepton (eg. J/Ψ) may have photons nearby (e.g. from π_0).
 - **Z_1 mass**: $m_{Z1} > 40$ GeV/ c^2
 - **'smart cut'**: defining Z_a and Z_b as the mass-sorted alternative pairing Z candidates (Z_a being the one closest to the nominal Z boson mass), require NOT($|m_{Za} - m_Z| < |m_{Z1} - m_Z|$ AND $m_{Zb} < 12$). Selected FSR photons are included in m_Z 's computations. This cut discards 4μ and $4e$ candidates where the alternative pairing looks like an on-shell Z + low-mass $\ell^+\ell^-$. (NB. In Run I, such a situation was avoided by choosing the best ZZ candidate before applying kinematic cuts to it, most precisely before the $m_{ZZ} > 12$ GeV/ c^2 cut. The present smart cut allows to choose the best ZZ candidate after all kinematic cuts.)

- **four-lepton invariant mass:** $m_{4\ell} > 70 \text{ GeV}/c^2$

3. Events containing at least one selected ZZ candidate form the **signal region**.

5.4 Choice of the best ZZ Candidate

Unlike in the Run I analysis, the best ZZ candidate is now chosen after all kinematic cuts, a change that allows to test other selection strategies for this candidate choice. This is especially relevant for events with more than four selected leptons, having in mind the search for the Higgs boson production modes where associated particles can decay to leptons, such as VH and ttH.

For the current analysis, we adopt a different approach compared to Run 1: if more than one ZZ candidate survives the above selection, we choose the one with the highest value of $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ (defined in Section 6, except when two candidates consist of the same four leptons in which case the candidate with Z_1 closest in mass to nominal Z boson mass is retained).

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6 Observables and Kinematic Discriminants

Several observables, such as the four-lepton invariant mass, $m_{4\ell}$, or the kinematic discriminants, are used either as input to likelihood fits or to categorize events in this analysis.

6.1 Description of discriminant setup and algorithms

The full kinematic information from each event using either the Higgs boson decay or associated particles in its production is extracted using the matrix element calculations. These discriminants use a complete set of mass and angular input observables $\vec{\Omega}$ [28, 45, 46] to describe kinematics at LO in QCD, see Fig. 15. The p_T of either the combined H boson and 2 jets system for the production discriminant (e.g. $\mathcal{D}_{2\text{jet}/1\text{jet}/\text{VH}}$) [47, 48] or the H boson itself for the decay discriminants (e.g. $\mathcal{D}^{\text{kin}}$) [1, 49] is not included in the input observables in order to reduce associated QCD uncertainties.

The kinematic discriminants used in this study are computed using the MELA package [1, 28, 45, 50], which provides the full set of processes studied in this paper and uses JHUGEN matrix elements for the signal and MCFM matrix elements for the background. The signal includes both the four-lepton decay kinematics in the processes $g\bar{g}$ or $q\bar{q} \rightarrow X \rightarrow ZZ / Z\gamma^* / \gamma^*\gamma^* \rightarrow 4\ell$, and kinematics of associated particles in production $H+\text{jet}$, $H+2\text{jets}$, VBF, ZH, WH, $t\bar{t}H$, tqH , or $b\bar{b}H$. The background includes $g\bar{g}$ or $q\bar{q} \rightarrow ZZ / Z\gamma^* / \gamma^*\gamma^* / Z \rightarrow 4\ell$ processes. Additional algorithms are available for the cross-checks of the four-lepton kinematics in decay. Within the MELA framework, an analytic parameterization of the matrix elements for signal [28, 50] and background [51] was adopted in the previous CMS analyses, reported in Refs. [1, 52, 53], and is used for cross-checks in this analysis. The four-lepton matrix element calculations are also tested with the MEKD package [54], which is based on MADGRAPH and FEYNRULES [55], for a subset of four-lepton decay processes implemented in common with MELA.

6.2 Description of kinematic discriminants

6.2.1 Kinematic discriminants for final fit

The discriminant sensitive to the $g\bar{g}/q\bar{q} \rightarrow 4\ell$ kinematics is calculated as [1, 49]

$$\mathcal{D}_{\text{bkg}}^{\text{kin}} = \left[1 + \frac{\mathcal{P}_{\text{bkg}}^{q\bar{q}}(\vec{\Omega}^{H \rightarrow 4\ell} | m_{4\ell})}{\mathcal{P}_{\text{sig}}^{g\bar{g}}(\vec{\Omega}^{H \rightarrow 4\ell} | m_{4\ell})} \right]^{-1}, \quad (2)$$

where the denominator contains the probability for the signal and the numerator includes the probability for the dominant $q\bar{q} \rightarrow 4\ell$ background process, all calculated either with the JHUGEN or MCFM matrix elements within the MELA framework. Figure 16 shows conditional distribution of $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ for signal and background as a function of $m_{4\ell}$, in the Untagged category.

One of the main improvements brought by this analysis is the usage of a new kinematic discriminant. In the final statistical analysis, the signal strength is computed via a 2D fit using the information on the invariant mass of the four leptons and $\mathcal{D}_{\text{bkg}}^{\text{kin}}$. However, in VBF and VH-hadronic categories, the contamination from $g\bar{g}H$ process is still important. In order to better separate $g\bar{g}H$ from the other production modes and from the backgrounds, the idea is thus to add the jet information (i.e., information about the production mode) to the already present information on decay.

The new dedicated production-dependent $\mathcal{D}_{\text{bkg}}$ discriminants used in the VBF-2jet tagged and

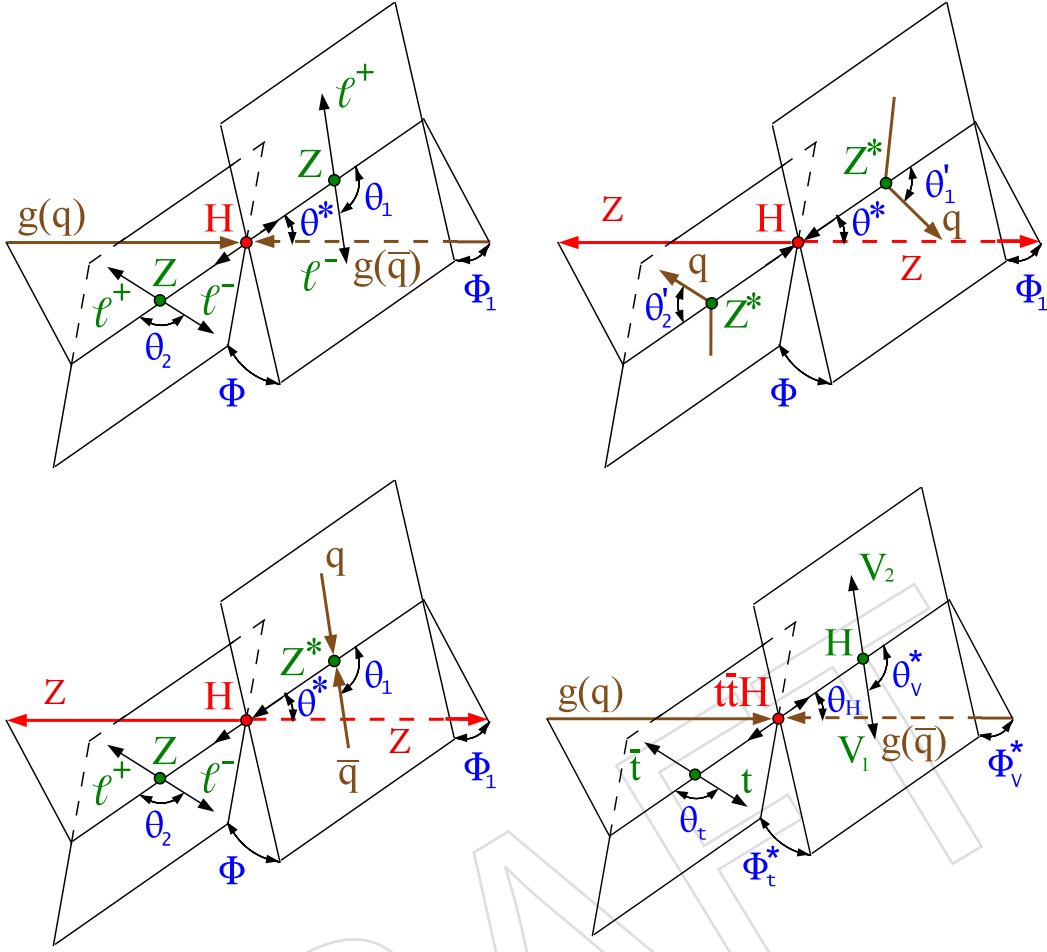


Figure 15: Illustrations of H particle production and decay $gg/q\bar{q} \rightarrow H \rightarrow ZZ \rightarrow 4\ell^\pm$ (top-left), $VBF qq' \rightarrow qq'H$ (top-right), $q\bar{q} \rightarrow V^* \rightarrow VH$ (bottom-left), and $gg/q\bar{q} \rightarrow t\bar{t}H$ (bottom-right). Angles and invariant masses fully characterize the orientation of the production and decay chain and are defined in the suitable rest frames [28, 45, 46].

harmonic VH tagged categories are then defined as:

$$\mathcal{D}_{\text{bkg}}^{\text{VBF+dec}} = \frac{\mathcal{P}_{\text{sig}}^{\text{VBF+VH+dec}}(\vec{\Omega})}{\mathcal{P}_{\text{sig}}^{\text{VBF+VH+dec}}(\vec{\Omega}) + c^{\text{VBF2jet}}(m_{4\ell}) \times (\mathcal{P}_{\text{bkg}}^{\text{VBS+}}(\vec{\Omega}) + \mathcal{P}_{\text{bkg}}^{\text{QCD+dec}}(\vec{\Omega}))}$$

$$\mathcal{D}_{\text{bkg}}^{\text{VH+dec}} = \frac{\mathcal{P}_{\text{sig}}^{\text{VBF+VH+dec}}(\vec{\Omega})}{\mathcal{P}_{\text{sig}}^{\text{VBF+VH+dec}}(\vec{\Omega}) + c^{\text{had.VH}}(m_{4\ell}) \times (\mathcal{P}_{\text{bkg}}^{\text{VBS+}}(\vec{\Omega}) + \mathcal{P}_{\text{bkg}}^{\text{QCD+dec}}(\vec{\Omega}))}, \quad (3)$$

where $c^p(m_{4\ell})$ for category p is the $m_{4\ell}$ -dependent constant to calibrate the distribution. The performance of the new background discriminants in these categories are compared to $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ on Fig. 17 as ROC curves.

A test of this new discriminant was performed, running the full statistical analysis. The likelihood scans for the expected signal strength modifiers corresponding to the five main Higgs production modes were computed for the nominal configuration and for the one using kinematic discriminant with jet information:

- μ_{VBF} went from $+0.999_{-0.937}^{+1.224}$ to $+0.999_{-0.783}^{+1.152}$

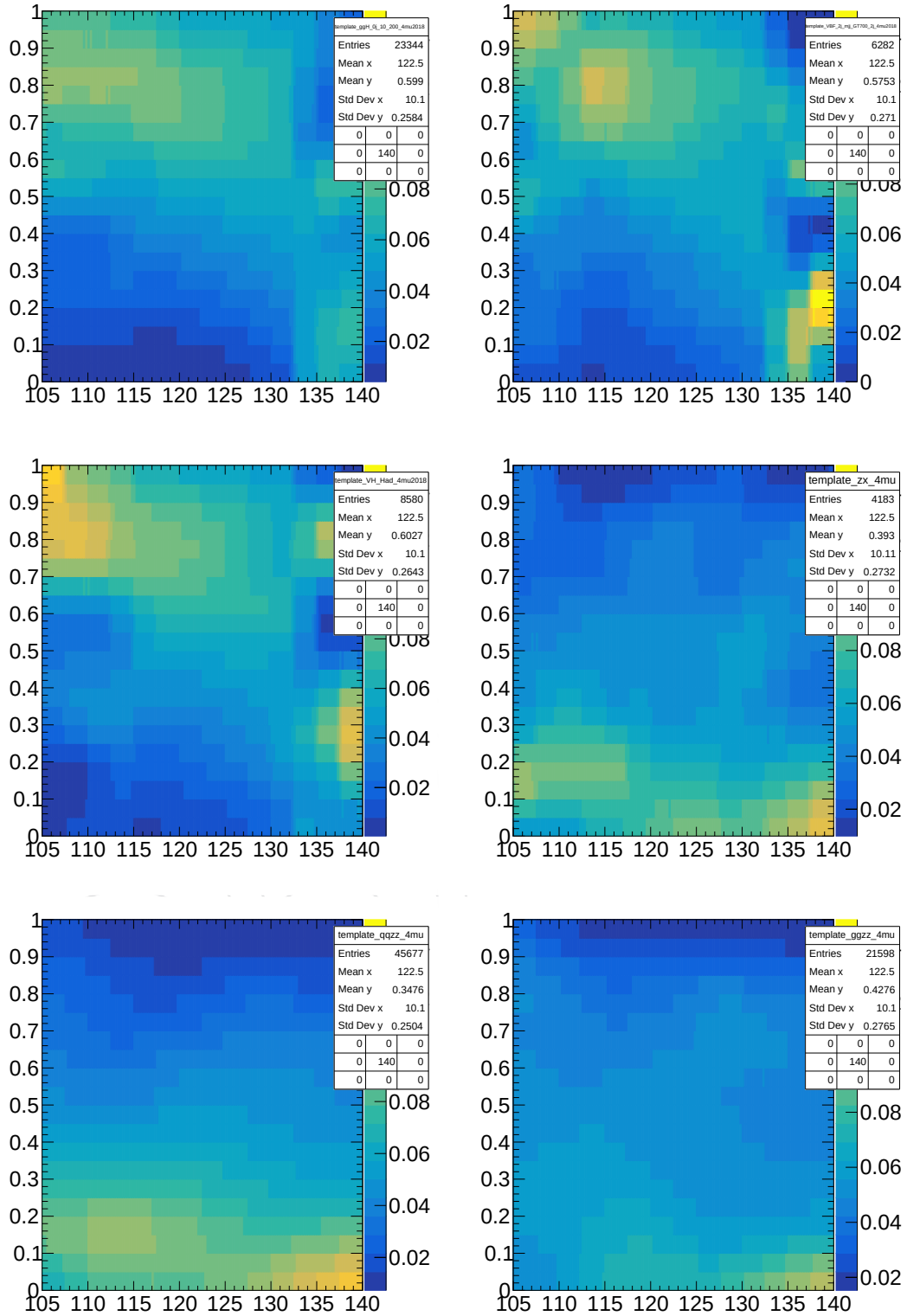


Figure 16: Conditional distribution of $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ in Untagged category for ggH signal (top left), VBF signal (top right), ZH signal (middle left), Z+X background (middle right), $q\bar{q} \rightarrow 4\ell$ (bottom left) and $gg \rightarrow ZZ$ (bottom right) as a function of $m_{4\ell}$.

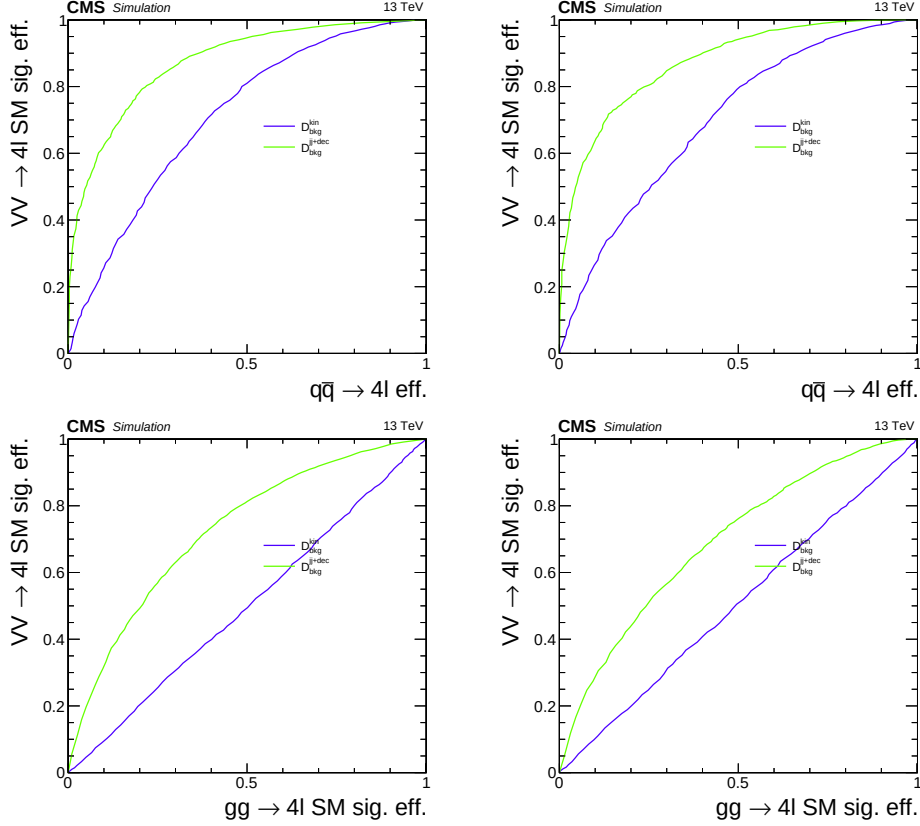


Figure 17: Top panels: Comparison of $\mathcal{D}_{\text{bkg}}^{\text{VBF}+\text{dec}}$ and $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ in the VBF-2jet tagged category (left) and $\mathcal{D}_{\text{bkg}}^{\text{VH}+\text{dec}}$ and $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ in the had. VH tagged category (right) between the VBF (left) or VH (right) processes and the $q\bar{q}$ background. Bottom panels: Comparison of $\mathcal{D}_{\text{bkg}}^{\text{VBF}+\text{dec}}$ and $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ in the VBF-2jet tagged category (left) and $\mathcal{D}_{\text{bkg}}^{\text{VH}+\text{dec}}$ and $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ in the had. VH tagged category (right) between the VBF (left) or VH (right) processes and the gluon fusion signal process.

• $\mu_{\text{VH-had}}$ went from $+0.998^{+3.851}_{-0.998}$ to $+0.999^{+3.286}_{-0.999}$

leading thus to an improvement of about 10 to 15 percent.

Figure ?? (resp. Figure ?? shows conditional distribution of $\mathcal{D}_{\text{bkg}}^{\text{VBF}+\text{dec}}$ (resp. $\mathcal{D}_{\text{bkg}}^{\text{VH}+\text{dec}}$) for signal and backgrounds, as a function of $m_{4\ell}$, in the VBF-2 jets (resp. VH-hadronic) category.

6.2.2 Kinematic discriminants for categorization

The discriminant sensitive to the VBF signal topology with two associated jets is calculated as [47, 48]

$$\mathcal{D}_{2\text{jet}} = \left[1 + \frac{\mathcal{P}_{\text{HJJ}}(\vec{\Omega}^{\text{H+JJ}}|m_{4\ell})}{\mathcal{P}_{\text{VBF}}(\vec{\Omega}^{\text{H+JJ}}|m_{4\ell})} \right]^{-1}, \quad (4)$$

where $\mathcal{P}_{\text{VBF}}$ and $\mathcal{P}_{\text{HJJ}}$ are probabilities obtained from the JHUGEN matrix elements for the VBF process and gluon fusion (technically combination of $gg/qg/qq'$ parton collisions) in association with two jets (H + 2 jets) within the MELA framework. This discriminant is equally efficient in separating VBF from either $gg \rightarrow \text{H} + 2\text{jets}$ signal or gg or $q\bar{q} \rightarrow 4\ell + 2\text{jets}$ background because jet correlations in these processes are distinct from the VBF process.

491 The discriminant sensitive to the VBF signal topology with one associated jet is calculated
 492 as [47, 48]

$$\mathcal{D}_{1\text{jet}} = \left[1 + \frac{\mathcal{P}_{\text{HJ}}(\vec{\Omega}^{\text{H+J}}|m_{4\ell})}{\int d\eta_J \mathcal{P}_{\text{VBF}}(\vec{\Omega}^{\text{H+J}}|m_{4\ell})} \right]^{-1}, \quad (5)$$

493 where and $\mathcal{P}_{\text{HJ}}$ is the probability obtained from the JHUGEN matrix element for gluon fusion
 494 in association with one jet (H + 1 jet) within the MELA framework. The $\int d\eta_J \mathcal{P}_{\text{VBF}}$ is the in-
 495 tegral of the two-jet VBF matrix element probability discussed above over the η_J values of the
 496 unobserved jet with the constraint that the total transverse momentum of the HJJ system is
 497 zero.

498 The discriminant sensitive to the VH signal topology with two associated jets is calculated
 499 as [47, 48]

$$\mathcal{D}_{\text{WH}} = \left[1 + \frac{\mathcal{P}_{\text{HJJ}}(\vec{\Omega}^{\text{H+JJ}}|m_{4\ell})}{\mathcal{P}_{\text{ZH}}(\vec{\Omega}^{\text{H+JJ}}|m_{4\ell})} \right]^{-1}, \quad (6)$$

$$\mathcal{D}_{\text{ZH}} = \left[1 + \frac{\mathcal{P}_{\text{HJJ}}(\vec{\Omega}^{\text{H+JJ}}|m_{4\ell})}{\mathcal{P}_{\text{WH}}(\vec{\Omega}^{\text{H+JJ}}|m_{4\ell})} \right]^{-1}, \quad (7)$$

500 where $\mathcal{P}_{\text{VH}}$ and $\mathcal{P}_{\text{HJJ}}$ are probabilities obtained from the JHUGEN matrix elements for the VH
 501 (either ZH or WH) process and gluon fusion in association with two jets (H + 2 jets) within the
 502 MELA framework.

503 Figure 18 shows distributions of the four production discriminants discussed above. In ad-
 504 dition, it is possible to calculate ttH, bbH, and tqH discriminants to isolate these production
 505 mechanisms, using kinematics shown in the last graph of Fig. 15, but expected yields in these
 506 categories are very small. Therefore full reconstruction needed for the matrix element calcula-
 507 tion and it will be inefficient with the presently expected statistics. However, for details of such
 508 matrix element calculations and possible future applications, see Refs. [46, 56].

509 These discriminants are used to enhance the purity of the categories described in Section 7.
 510 Operating points then are defined, both for the 4 MELA-only discriminants and the 4 com-
 511 bined discriminants, so as to obtain a good compromise between expected category yields and
 512 purities.

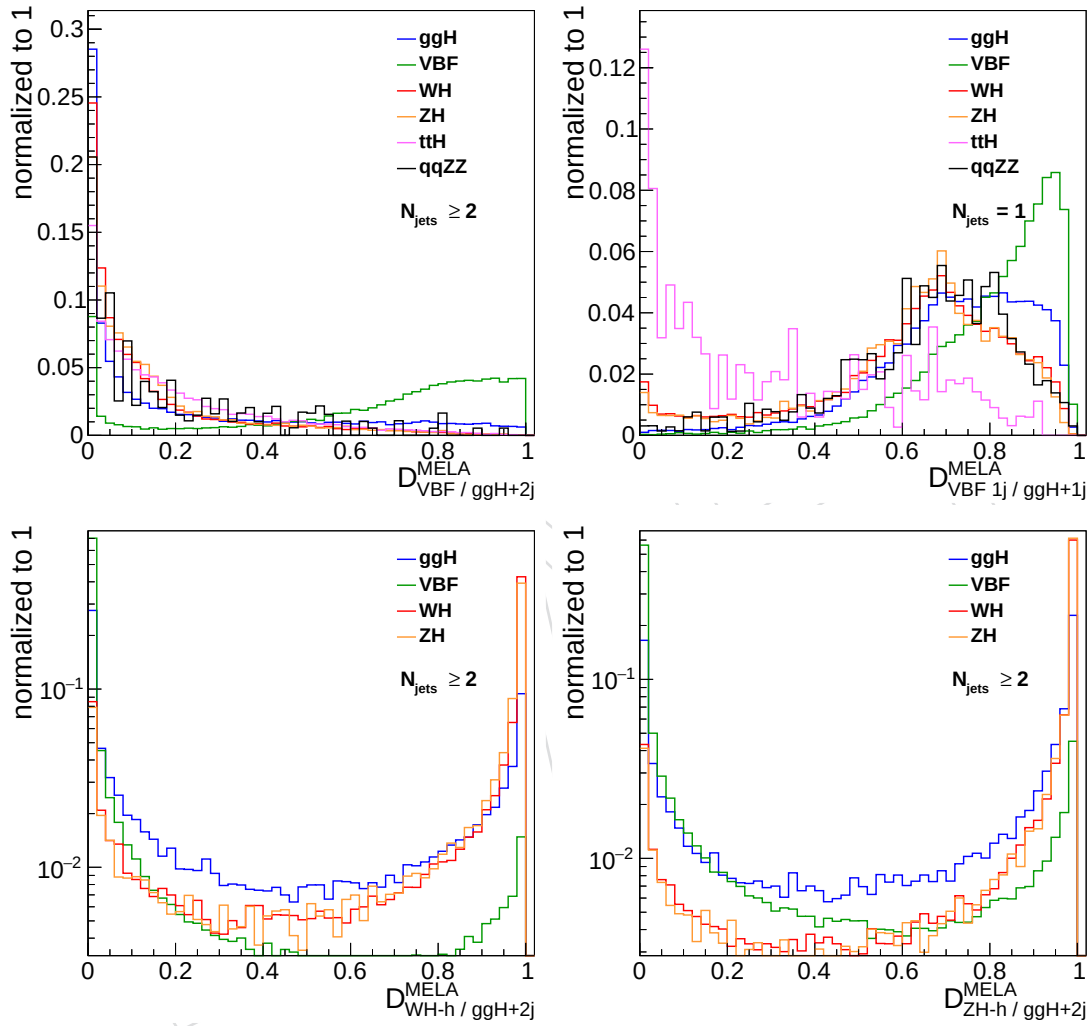


Figure 18: Distribution of the $\mathcal{D}_{2\text{jet}}$ (top left), $\mathcal{D}_{1\text{jet}}$ (top right), $\mathcal{D}_{\text{WH}}$ (bottom left), $\mathcal{D}_{\text{ZH}}$ (bottom right) discriminant for various production mechanisms of the H(125) signal and background in the $H \rightarrow 4\ell$ analysis. MC simulation at 13 TeV is shown.

7 Event Categorization

7.1 Event Categories targeting production mechanisms

In order to probe different Higgs boson production mechanisms, the selected events are classified into mutually exclusive categories. Seven categories are defined in the same way as previous Run2 analyses [19], except that in the VBF-1jet-tagged category, $\mathcal{D}_{1\text{jet}}$ requirement is shifted from 0.5 to 0.7 to improve the purity. The criteria is applied in this exact order as follows (i.e. an event is considered for the subsequent category only if it does not satisfy the requirements of the previous category):

- **VBF-2jet-tagged category** requires exactly 4 leptons. In addition there must be either 2 or 3 jets of which at most 1 is b-tagged, or at least 4 jets and no b-tagged jets. Finally, $\mathcal{D}_{2\text{jet}} > 0.5$ is required.
- **VH-hadronic-tagged category** requires exactly 4 leptons. In addition there must be 2 or 3 jets, or at least 4 jets and no b-tagged jets. Finally, $\mathcal{D}_{\text{VH}} \equiv \max(\mathcal{D}_{\text{ZH}}, \mathcal{D}_{\text{WH}}) > 0.5$ is required.
- **VH-leptonic-tagged category** requires no more than 3 jets and no b-tagged jets in the event, and exactly 1 additional lepton or 1 additional pair of opposite sign same flavor leptons. This category also includes events with no jets and at least 1 additional lepton.
- **$\text{t}\bar{\text{t}}\text{H}$ -hadronic-tagged category** requires at least 4 jets of which at least 1 is b-tagged and no additional leptons.
- **$\text{t}\bar{\text{t}}\text{H}$ -leptonic-tagged category** requires at least 1 additional lepton in the event.
- **VBF-1jet-tagged category** requires exactly 4 leptons, exactly 1 jet and $\mathcal{D}_{1\text{jet}} > 0.5$.
- **Untagged category** consists of the remaining events.

Figure 19 shows the event yields in the seven categories calculated using 2018 MC and scaled to full Run II luminosity.

7.2 Finer Event Categories targeting simplified template cross sections

Simplified template cross sections (STXS) [57] were developed to provide finely-grained measurements than signal strength and multiplicative coupling modifiers. The primary goals of the STXS framework are to maximize the sensitivity of the measurements while at the same time to minimize their theory dependence. The measured exclusive regions of phase space, called bins for simplicity, are specific to the different production modes. To account for the evolving experimental sensitivity, the STXS bins are defined in stages (corresponding to increasingly fine granularity). The stage 0 bins correspond to the Higgs boson production mechanisms. The previous Run2 analysis has reported the measured stage 0 results [19]. With full Run2 data, this analysis targets the finer stage 1.1 bins. The bin definitions are shown in Figure 20, 21 and 22 for ggF, VBF and VH productions, respectively.

Figure 23 shows the expected signal event yields at 136.2 fb^{-1} of all stage 1.1 bins reconstructed. Some of the stage 1.1 bins expect less than 1 event, thus the analysis is not sensitive to them. These bins are merged and the measurement is provided for the merged bins. More specifically, ggF 2jets and 3 jets bins above $m_{jj} > 350 \text{ GeV}$ are merged to one; two bins of VBF 3 jets ($350 < m_{jj} < 700 \text{ GeV}$ and $m_{jj} > 700 \text{ GeV}$) are merged to one; WH and ZH bins are merged as one VH bin, and only splitted to $p_T^V < 150 \text{ GeV}$ and $p_T^V > 150 \text{ GeV}$ bins.

To maximize the sensitivity to the stage 1.1 bins, the above 7 reconstructed categories are further

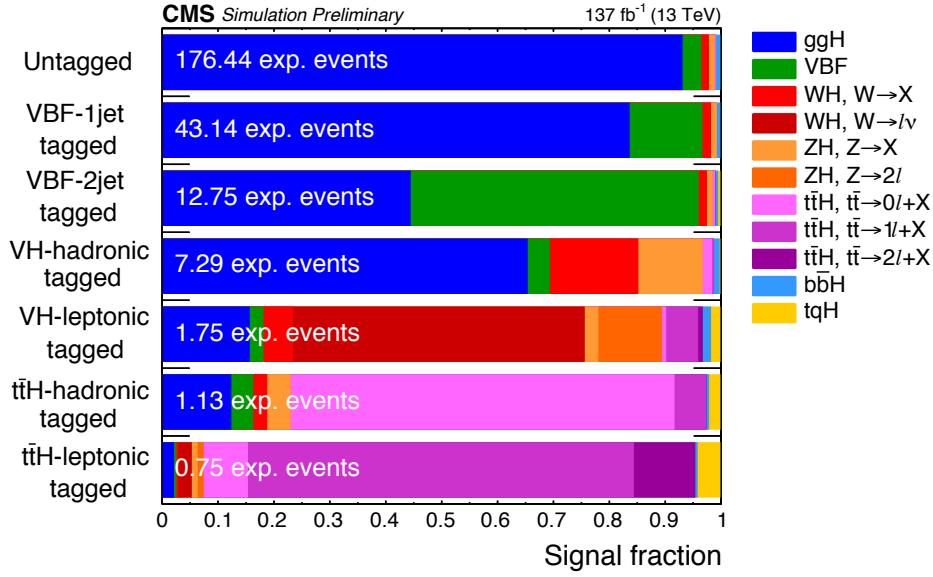


Figure 19: Signal relative purity of the seven event categories in terms of the seven main production mechanisms of the Higgs boson in a $118 < m_{4\ell} < 130$ GeV mass window. The WH, ZH and ttH processes are split according to the decay of associated objects, whereby X denotes anything other than an electron or muon.

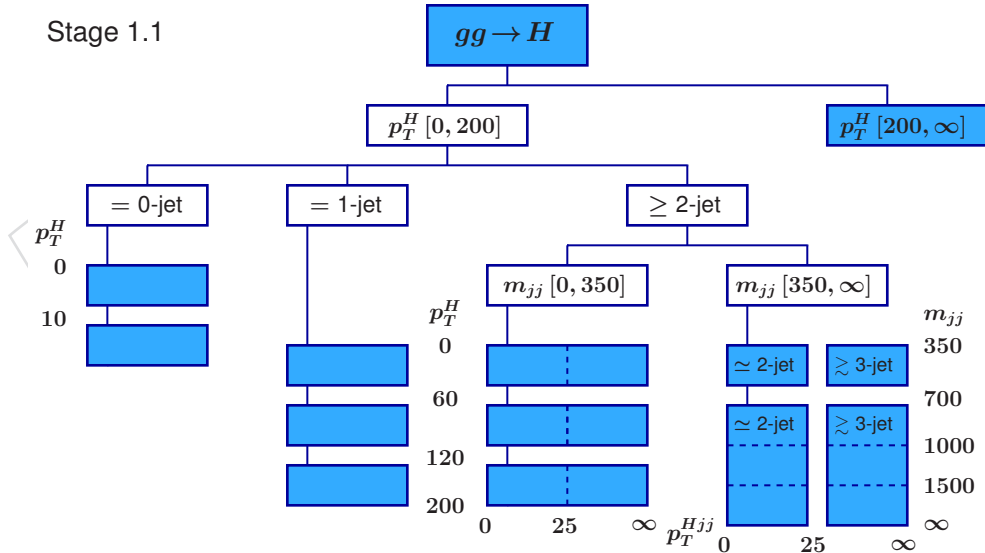


Figure 20: STXS stage 1.1 definition for ggF production mechanism.

556 splitted into 22 categories.

557 • **Untagged category**

558 1. ggH-PTH-200: if p_T of the Higgs is > 200 GeV

559 2. ggH-0J-PTH-0-10: if 0 jet reconstructed and the Higgs p_T is between $[0, 10]$ GeV

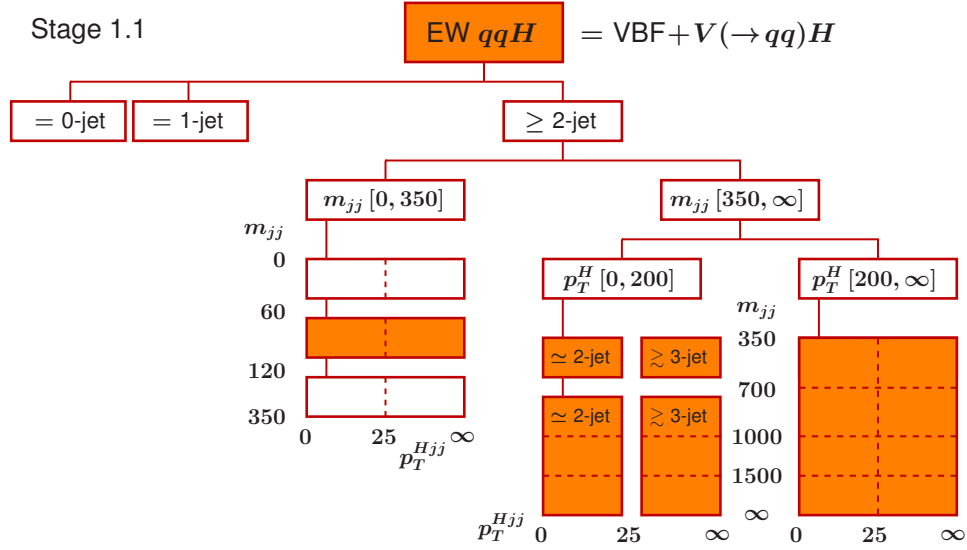


Figure 21: STXS stage 1.1 definition for VBF production mechanism.

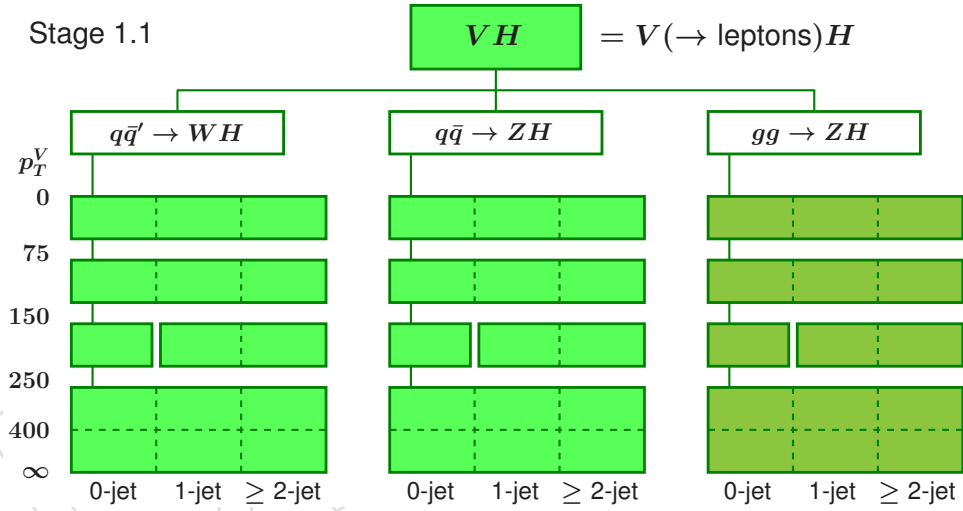


Figure 22: STXS stage 1.1 definition for VH production mechanism.

3. ggH-0J-PTH-10-200: if 0 jet reconstructed and the Higgs p_T is between [10,200] GeV
4. ggH-1J-PTH-0-60: if 1 jet reconstructed and the Higgs p_T is between [0,60] GeV
5. ggH-1J-PTH-60-120: if 1 jet reconstructed and the Higgs p_T is between [60,120] GeV
6. ggH-1J-PTH-120-200: if 1 jet reconstructed and the Higgs p_T is between [120,200] GeV
7. ggH-VBF: if 2 jets reconstructed and $m_{jj} > 350$ GeV
8. ggH-2J-PTH-0-60: if 2 jets reconstructed and the Higgs p_T is between [0,60] GeV and not ggH-VBF
9. ggH-2J-PTH-60-120: if 2 jets reconstructed and the Higgs p_T is between [60,120] GeV

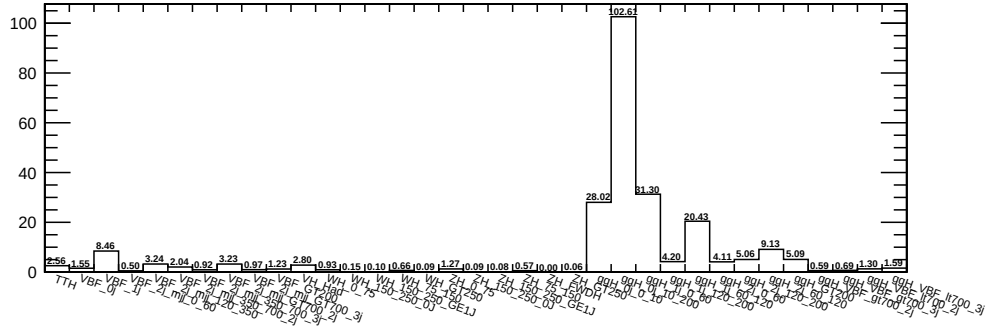


Figure 23: Expected signal event yields at 136.2 fb^{-1} of all stage 1.1 bins reconstructed.

GeV and not ggH-VBF

10. ggH-2J-PTH-120-200: if 2 jets reconstructed and the Higgs p_T is between $[120, 200]$ GeV and not ggH-VBF

- **VBF-1jet-tagged category**

- **VBF-2jet-tagged category**

1. VBF-GT200-2J: if the Higgs p_T is > 200 GeV and $m_{jj} > 350$ GeV
2. VBF-2j-mjj-350-700: if the Higgs p_T is < 200 GeV, $p_T^{Hjj} < 25$ GeV, m_{jj} is between $[350, 700]$ GeV
3. VBF-2j-mjj-GT700j: if the Higgs p_T is < 200 GeV, $p_T^{Hjj} < 25$ GeV, m_{jj} is > 700 GeV
4. VBF-3j: if the Higgs p_T is < 200 GeV, $p_T^{Hjj} > 25$ GeV, $m_{jj} > 350$ GeV
5. VBF-2j: if not above

- **VH-hadronic-tagged category**

1. VH-Had if $60 < m_{jj} < 120$ GeV
2. VBF-rest-VH: not above

- **VH-leptonic-tagged category**

1. VH-lep-0-150: if $p_T^V < 150$ GeV
2. VH-lep-GT150: if $p_T^V > 150$ GeV

- **$t\bar{t}H$ -hadronic-tagged category**

- **$t\bar{t}H$ -leptonic-tagged category**

Figure 24 shows the signal relative purity of all STXS Stage 1.1 Bins in the 22 event sub-categories defined above.

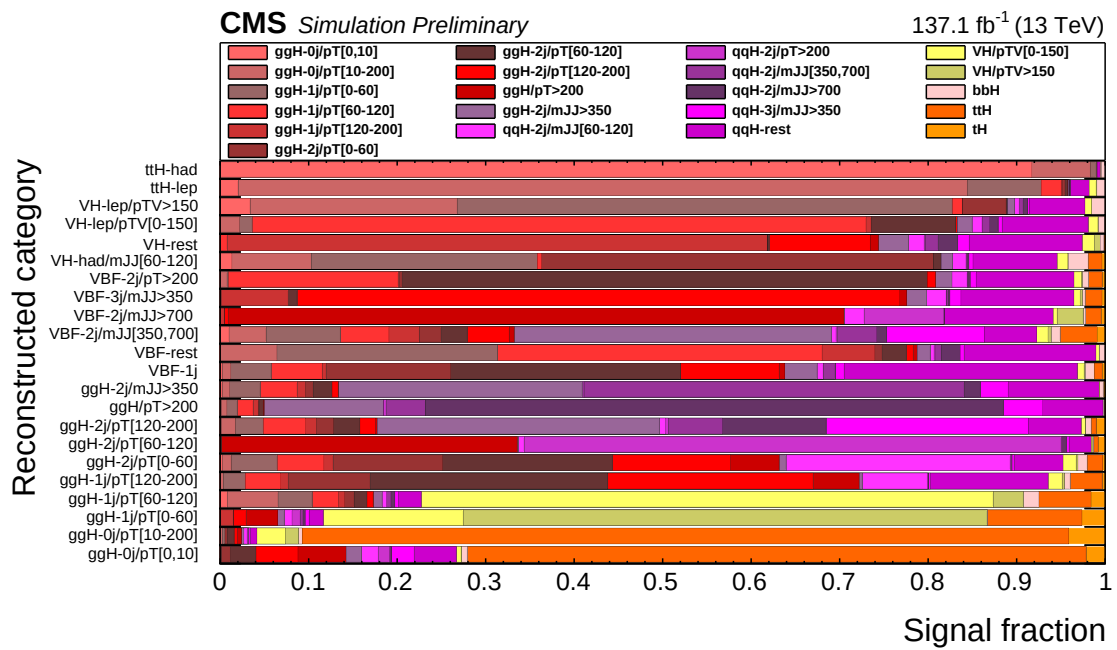


Figure 24: Signal relative purity of the 22 event sub-categories in terms of the STXS Stage 1.1 Bins in a $118 < m_{4\ell} < 130$ GeV mass window.

8 Background Estimation

8.1 Irreducible Backgrounds

8.1.1 $q\bar{q} \rightarrow ZZ$ Modelling

The $q\bar{q} \rightarrow ZZ$ background is generated at NLO, while the fully differential cross section has been computed at NNLO [58], but are not yet available in a partonic level event generator. Therefore NNLO/NLO k -factors for the $q\bar{q} \rightarrow ZZ$ background process are applied to the POWHEG sample. The inclusive cross sections obtained using the same PDF and renormalization and factorization scales as the POWHEG sample at LO, NLO, and NNLO are shown in Table 10. The NNLO/NLO k -factors are applied in the analysis differentially as a function of $m(ZZ)$.

Additional NLO electroweak corrections which depend on the initial state quark flavor and kinematics are also applied to the $q\bar{q} \rightarrow ZZ$ background process in the region $m(ZZ) > 2m(Z)$ where the corrections have been computed. The differential QCD and electroweak k -factors can be seen in Figure 25.

QCD Order	$\sigma_{2\ell 2\ell'}(\text{fb})$	$\sigma_{4\ell}(\text{fb})$
LO	$218.5^{+16\%}_{-15\%}$	$98.4^{+13\%}_{-13\%}$
NLO	$290.7^{+5\%}_{-8\%}$	$129.5^{+4\%}_{-6\%}$
NNLO	$324.0^{+2\%}_{-3\%}$	$141.2^{+2\%}_{-2\%}$

Table 10: Cross sections for $q\bar{q} \rightarrow ZZ$ production at 13 TeV

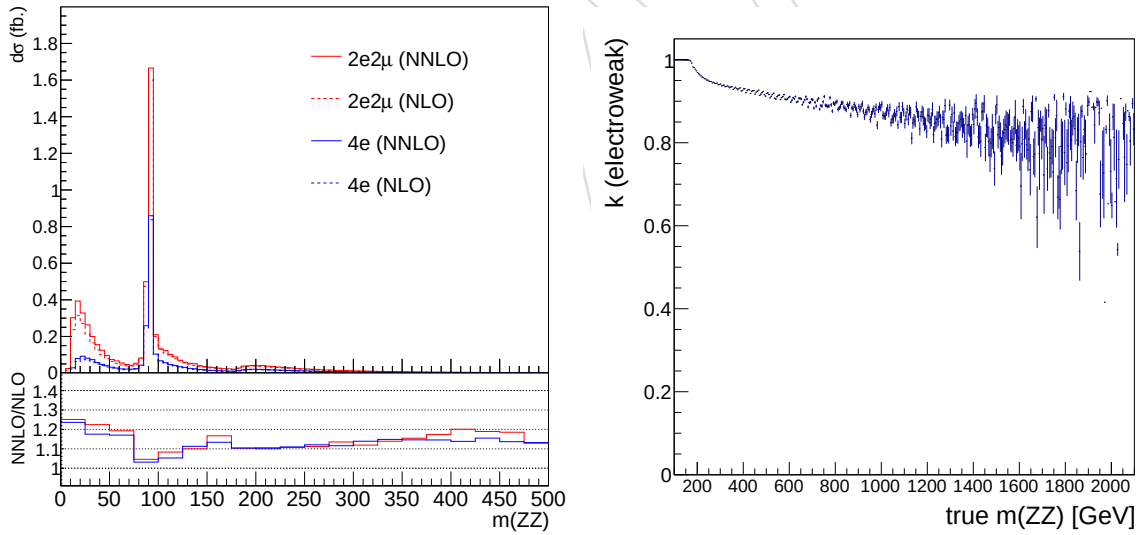


Figure 25: Left: NNLO/NLO QCD k -factor for the $q\bar{q} \rightarrow ZZ$ background as a function of $m(ZZ)$ for the 4ℓ and $2\ell 2\ell'$ final states. Right: NLO/NLO electroweak k -factor for the $q\bar{q} \rightarrow ZZ$ background as a function of $m(ZZ)$.

8.1.2 $gg \rightarrow ZZ$ Modelling

Event simulation for the $gg \rightarrow ZZ$ background is done at LO with the generator MCFM 7.0 [31, 32, 59]. Although no exact calculation exists beyond the LO for the $gg \rightarrow ZZ$ background, it has been recently shown [60] that the soft collinear approximation is able to describe the

background cross section and the interference term at NNLO. Further calculations also show that the K factors are very similar at NLO for the signal and background [61] and at NNLO for the signal and interference terms [62]. Therefore, the same K factor is used for the signal and background [63]. The NNLO K factor for the signal is obtained as a function of $m_{4\ell}$ using the HNNLO v2 Monte Carlo program [64–66] by calculating the NNLO and LO $gg \rightarrow H \rightarrow 2\ell 2\ell'$ cross sections at the small H boson decay width of 4.07 MeV and taking their ratios. The NNLO as well as the NLO K factors and the cross sections from which they are derived are illustrated in Fig. 26, along with the NNLO, NLO and LO cross sections at the SM H boson decay width [67].

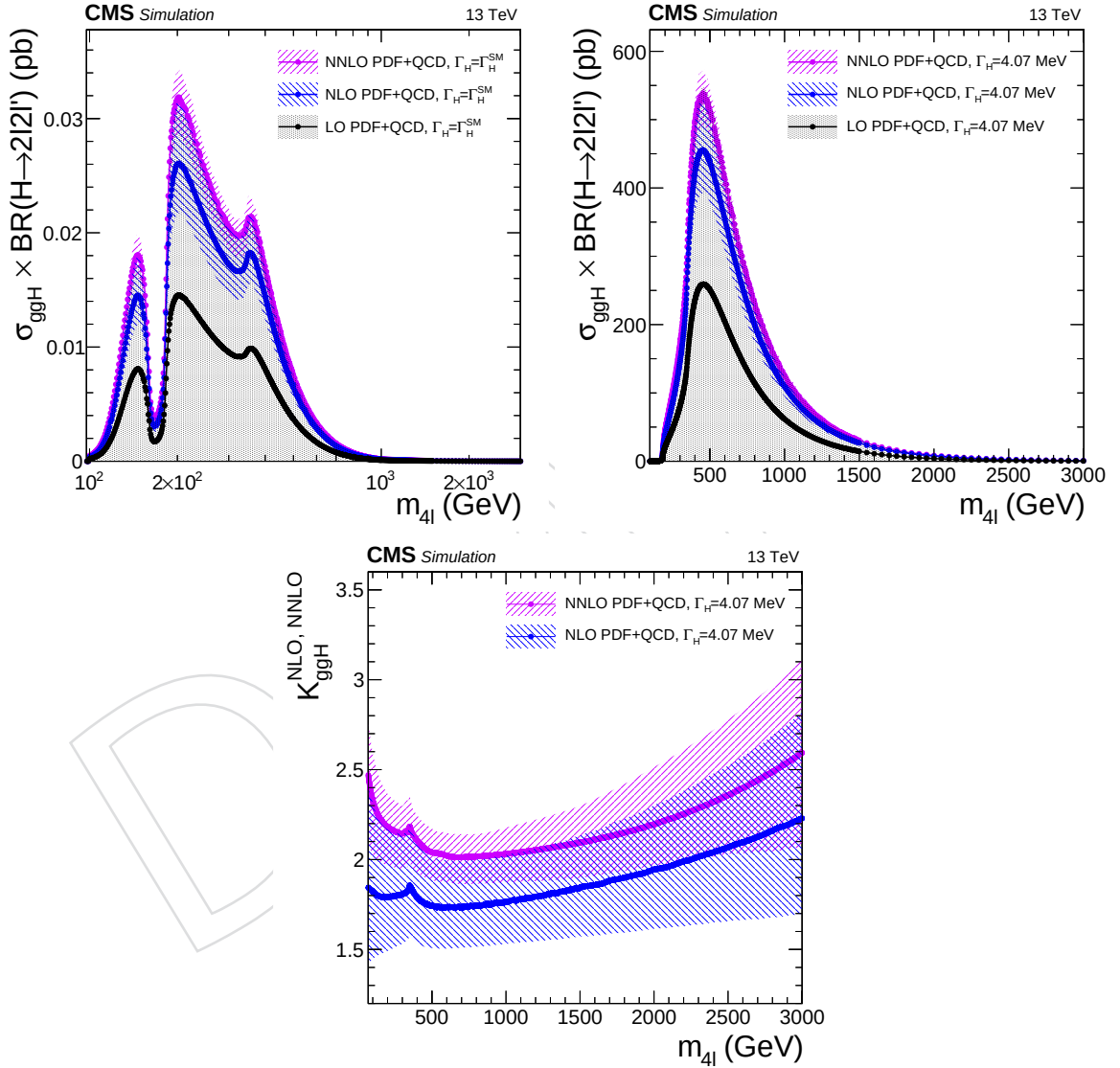


Figure 26: $gg \rightarrow H \rightarrow 2\ell 2\ell'$ cross sections at NNLO, NLO and LO at each H boson pole mass using the SM H boson decay width (top left) or at the fixed and small decay width of 4.07 MeV (top right). The cross sections using the fixed value are used to obtain the K factor for both the signal and the continuum background contributions as a function of $m_{4\ell}$ (bottom).

8.2 Reducible Background

The reducible background for the $H \rightarrow ZZ \rightarrow 4\ell$ analysis, hereafter called $Z + X$, originates from processes which contain one or more non-prompt leptons in the four-lepton final state. The main sources of non-prompt leptons are non-isolated electrons and muons coming from decays of heavy-flavour mesons, mis-reconstructed jets (usually originating from light-flavour quarks) and electrons from γ conversions. In this discussion, we will consider a “fake lepton” any jet mis-reconstructed as a lepton and any lepton originating from a heavy meson decay. Similarly, any electron originating from a photon conversion will be considered a “fake electron”.

In the $H \rightarrow ZZ \rightarrow 4\ell$ analysis, the rate of these background processes is estimated by measuring the f_e and f_μ probabilities for fake electrons and fake muons which do pass the **loose** selection criteria (defined in Section 4.1.1 and 4.2.1) to also pass the final selection criteria (defined in Section 5.3). These probabilities, hereafter referred to as fake ratios or fake rates (FR), are applied in dedicated control samples in order to extract the expected background yield in the signal region.

In the following section, two independent methods are presented to measure both the yields and shapes of the reducible background. The final result combines the outcome of the two approaches. The methods are the same as in the 2016 analysis, although additional cross checks have been performed.

8.2.1 Reducible Background Estimate with Opposite-Sign Leptons

8.2.1.1 Fake Rate Determination (OS Method) In order to measure the lepton fake ratios f_e, f_μ , we select samples of $Z(\ell\ell) + e$ and $Z(\ell\ell) + \mu$ events that are expected to be completely dominated by final states which include a Z boson and a fake lepton. These events are required to have two same flavour, opposite charge leptons with $p_T > 20/10$ GeV passing the tight selection criteria, thus forming the Z candidate. In addition, there is exactly one lepton passing the loose selection criteria as defined above. This lepton is used as the probe lepton for the fake ratio measurement. The invariant mass of this lepton and the opposite sign lepton from the reconstructed Z candidate should satisfy $m_{2l} > 4$ GeV.

The fake ratios are evaluated using the tight requirement $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV, to reduce the contribution from photon (asymmetric) conversions populating low masses. The fake ratios are measured in bins of the transverse momentum of the loose lepton and barrel and endcap region.

The electron and muon fake rates are measured within $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV and $E_T^{miss} < 25$ GeV are shown in Figure 27.

8.2.1.2 Fake Rate Application (OS Method) Two control samples are obtained as subsets of four lepton events which pass the first step of the selection (*First Z* step, see section 5.3), requiring an additional pair of loose leptons of same flavour and opposite charge, that pass the SIP_{3D} cut. The events must satisfy all kinematic cuts applied for the *Higgs phase space* selection (see 5.3). The first control sample is obtained by requiring that the two loose leptons which do not make the Z_1 candidate do not pass the final identification and isolation criteria. The other two leptons pass the final selection criteria by definition of the Z_1 . This sample is denoted as “2 Prompt + 2 Fail” (2P+2F) sample. It is expected to be populated with events that intrinsically have only two prompt leptons (mostly DY , with a small fraction of $t\bar{t}$ and $Z\gamma$ events). The second control sample is obtained by requiring one of the four leptons not to pass the final identification and isolation criteria. The other three leptons should pass the final selection cri-

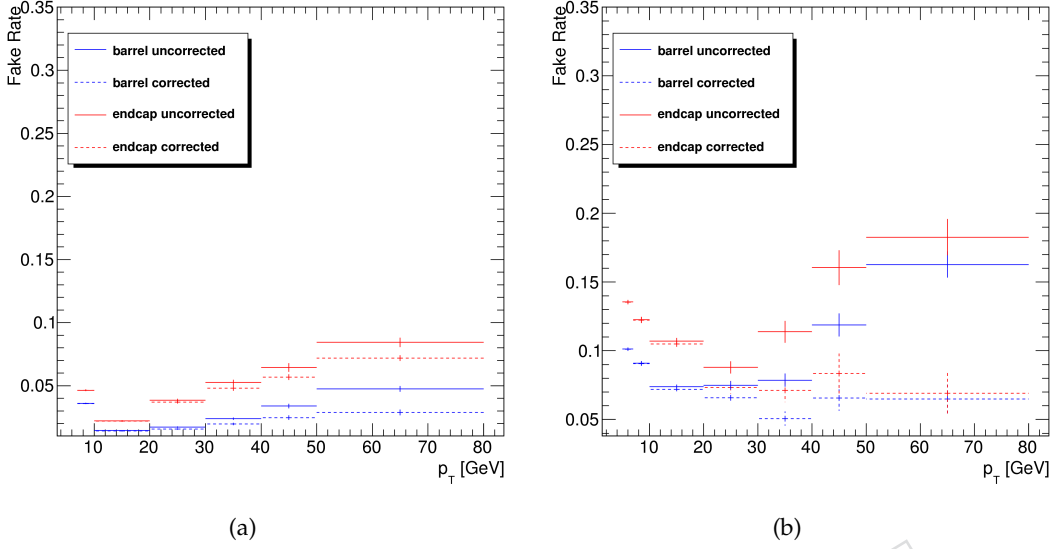


Figure 27: Fake rates as a function of the probe p_T for electrons (a) and muons (b) which satisfy the loose selection criteria, measured in a $Z(\ell\ell) + \ell$ sample in the 13 TeV data. The barrel selection includes electrons (muons) up to $|\eta| = 1.479$ (1.2). The Fake rates are shown before (dotted lines) and after (plain line) removal of WZ contribution from MC.

teria. This control sample is denoted as “3 Prompt + 1 Fail” (3P+1F) sample. It is expected to be populated with the type of events that populate the 2P+2F region, albeit with different relative proportions, as well as with WZ events that intrinsically have three prompt leptons.

The control samples obtained in this way, orthogonal by construction to the signal region, are enriched with fake leptons and are used to estimate the reducible background in the signal region.

The invariant mass distribution of events selected in the 2P+2F control sample is shown in Fig. 28 for the 13 TeV dataset.

The expected number of reducible background events in the 3P+1F region, N_{3P1F}^{bkg} , can be computed from the number of events observed in the 2P+2F control region, N_{2P2F} , by weighting each event in the region with the factor $(\frac{f_i}{1-f_i} + \frac{f_j}{1-f_j})$, where f_i and f_j correspond to the fake ratios of the two loose leptons:

$$N_{3P1F}^{\text{bkg}} = \sum \left(\frac{f_i}{1-f_i} + \frac{f_j}{1-f_j} \right) N_{2P2F} \quad (8)$$

Figure 29 shows the invariant mass distributions of the events selected in the 3P+1F control sample, together with the expected reducible background estimated from Eq. 8, stacked on the distribution of WZ and of irreducible background ($ZZ, Z\gamma^* \rightarrow 4\ell$) taken from the simulation.

Would the fake rates be measured in a sample that has exactly the same background composition as the 2P+2F sample, the difference between the observed number of events in the 3P+1F sample and the expected background predicted from the 2P+2F sample would solely amount to the (small) WZ and $Z\gamma_{\text{conv}}$ contribution. Large differences arise because the fake rates used in Eq. 8 do not properly account for the background composition of the 2P+2F control sample.

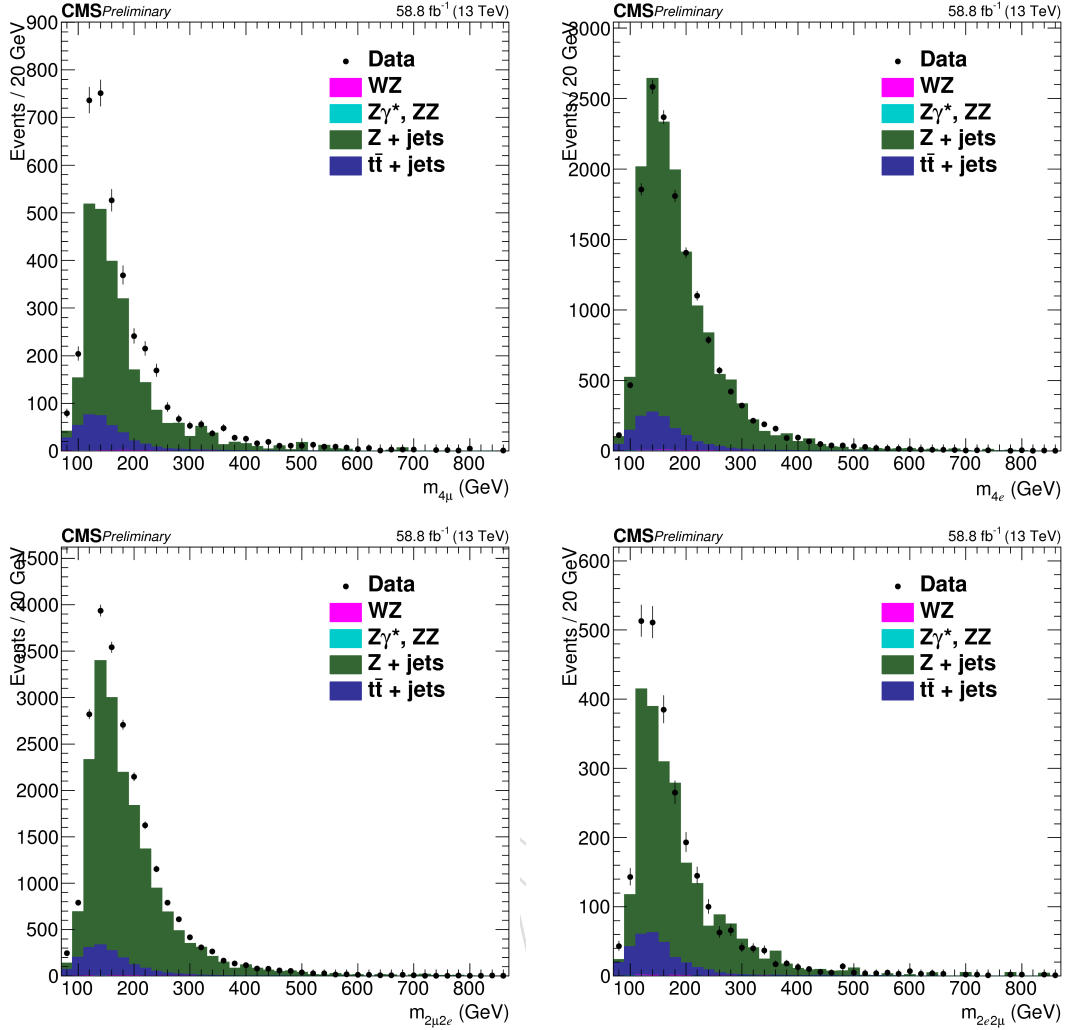


Figure 28: Invariant mass distribution of the events selected in the 2P+2F control sample in the 13 TeV dataset, (top left) 4μ , (top right) $4e$, (bottom left) $2\mu 2e$ and (bottom right) $2e 2\mu$ channels.

In particular, the difference seen in Fig. 29 between the observed 3P+1F distribution and the expectation from 2P+2F, in the channels with loose electrons ($4e$ and $2\mu 2e$), and concentrated at low masses, is due to photon conversions. This is confirmed explicitly by the simulation. The difference between the 3P+1F observation and the prediction from 2P+2F to recover the missing contribution from conversions - and more generally, in principle, to “correct” for the fact that the fake rates do not properly account for the background composition of the 2P+2F sample. More precisely, the expected reducible background in the signal region is given by the sum of two terms :

- a “2P2F component”, obtained from the number of events observed in the 2P+2F control region, N_{2P2F} , by weighting each event in that region with the factor $\frac{f_i}{1-f_i} \frac{f_j}{1-f_j}$, where f_i and f_j correspond to the fake ratios of the two loose leptons.
- a “3P1F component”, obtained from the difference between the number of observed events in the 3P+1F control region, N_{3P1F} , and the expected contribution from the 2P+2F region and ZZ processes in the signal region, $N_{3P1F}^{ZZ} + N_{3P1F}^{bkg}$. The N_{3P1F}^{bkg} is given by equation 8 and N_{3P1F}^{ZZ} is the contribution from ZZ which is taken from the sim-

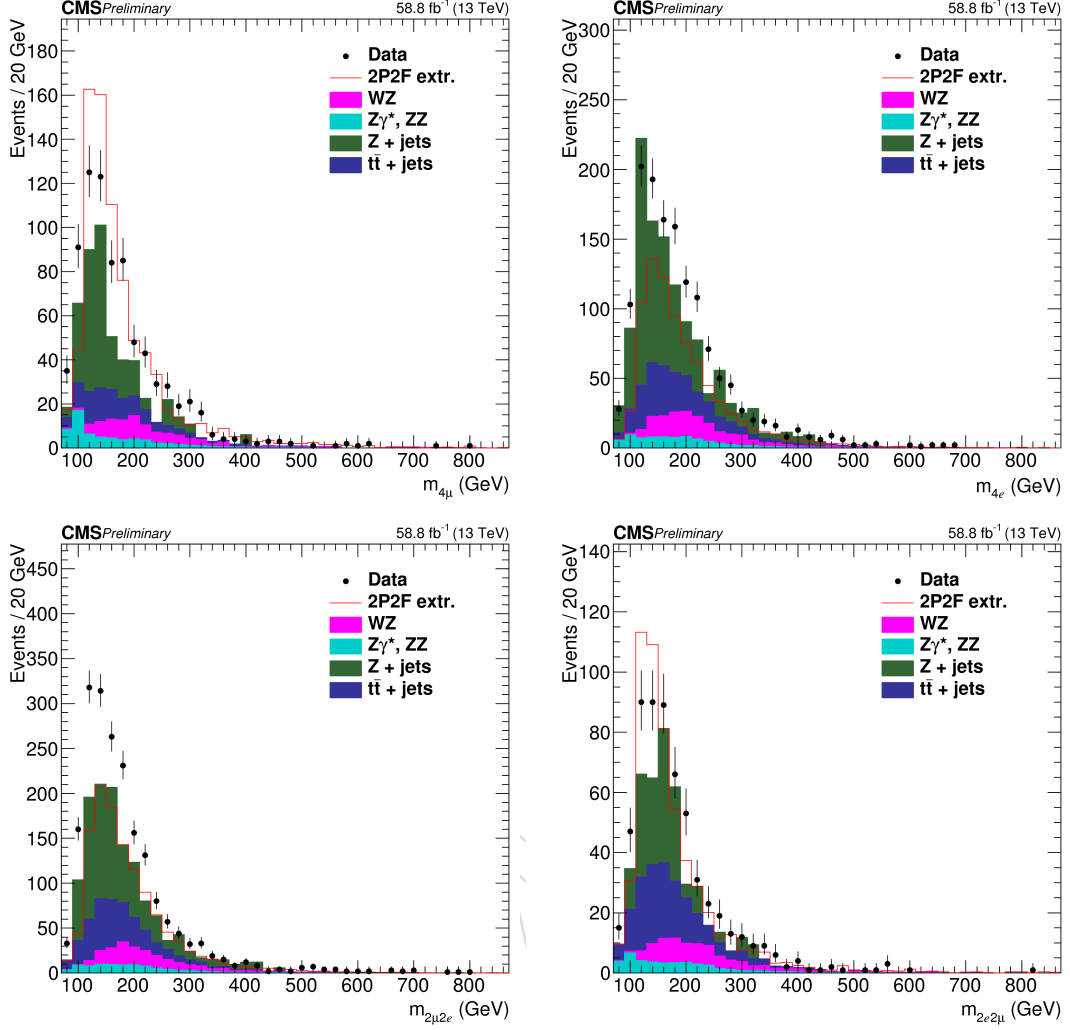


Figure 29: Invariant mass distribution of the events selected in the 3P+1F control sample in the 13 TeV dataset, (top left) 4μ , (top right) $4e$, (bottom left) $2\mu 2e$ and (bottom right) $2e 2\mu$ channels.

ulation. The difference $N_{3P1F} - N_{3P1F}^{bkg} - N_{3P1F}^{ZZ}$, which may be negative, is obtained for each (p_T, η) bin of the “F” lepton, and is weighted by $\frac{f_i}{1-f_i}$, where f_i denotes the fake rate of this lepton. This “3P1F component” accounts for the contribution of reducible background processes with only one fake lepton (like WZ events), and for the contribution of other processes (e.g. photon conversions) that are not properly estimated by the 2P2F component, because of the fake rates used.

Therefore, the full expression for the prediction can be symbolically written as:

$$N_{SR}^{bkg} = \sum \frac{f_i}{(1-f_i)} (N_{3P1F} - N_{3P1F}^{bkg} - N_{3P1F}^{ZZ}) + \sum \frac{f_i}{(1-f_i)} \frac{f_j}{(1-f_j)} N_{2P2F} \quad (9)$$

Previous equation is equivalent to the following:

$$N_{SR}^{bkg} = \left(1 - \frac{N_{3P1F}^{ZZ}}{N_{3P1F}}\right) \sum_j \frac{f_a^j}{1-f_a^j} - \sum_i \frac{f_3^i}{1-f_3^i} \frac{f_4^i}{1-f_4^i} \quad (10)$$

For channels where the Z_2 candidate is made from two electrons, the contribution of the 3P1F component is positive, and amounts to typically 30% of the total predicted background.

For channels with loose muons (4μ and $2e2\mu$), the 3P+1F sample is rather well described by the prediction from 2P+2F, as seen in Fig. 29b, and the 3P1F component is mainly driven by statistical fluctuations in the 3P+1F sample, which are larger than the expectation from WZ production.

Table 11 shows the expected number of events in the signal regions from the reducible background processes for the 13 TeV.

baseline	4e	4 μ	2e2 μ	2 μ 2e
13 TeV	24.9 ± 7.6	49.9 ± 15.5	35.2 ± 11.0	34.0 ± 10.3

Table 11: The contribution of reducible background processes in the signal region predicted from measurements in data using the Opposite-Sign Leptons method. The predictions correspond to 59.7 fb^{-1} of data at 13 TeV.

8.2.2 Reducible Background Estimate with Same-Sign Leptons

This method used to predict the reducible background allows to have an inclusive measurement of the all the main reducible backgrounds at the same time.

The control sample is obtained as a subset of the events that satisfy the first step of the selection (*First Z step*, see section 5.3), requiring an additional pair of loose leptons of same sign (to avoid signal contamination) and same flavour (SS-SF: $e^\pm e^\pm, \mu^\pm \mu^\pm$). The SS-SF leptons are requested to pass the SIP_{3D} cut while no identification or isolation requirements are imposed. The reconstructed invariant mass of the SS-SF leptons has to satisfy $m_{\ell\ell} > 12 \text{ GeV}$, $m_{\ell\ell} < 120 \text{ GeV}$, the reconstructed four-lepton invariant mass is required to satisfy $m_{4\ell} > 100 \text{ GeV}/c^2$, and the QCD suppression cut (see 5.3) is applied. The inclusive number of reducible background events in the signal region is derived from this set of events and from the probability for the two additional leptons to pass the isolation and identification analysis cuts, obtained from the fake rate measurement presented in section 8.2.2.1.

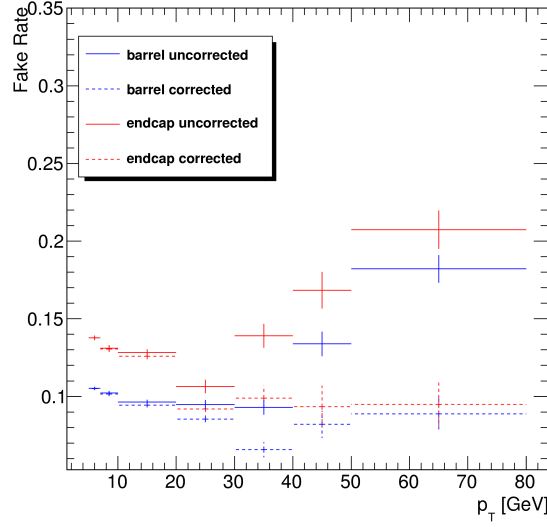
Starting from the control sample previously described, the final reducible background prediction in the signal region is given by the following expression:

$$N_{\text{expect}}^{Z+X} = N^{\text{DATA}} \times \left(\frac{\text{OS}}{\text{SS}}\right)^{\text{MC}} \times f_1 \times f_2 \quad (11)$$

where:

- N^{DATA} is the number of events in the control region,
- $\left(\frac{\text{OS}}{\text{SS}}\right)^{\text{MC}}$ is a correction factor between opposite sign and same sign control samples,
- f_1 and f_2 are the fake rates of each additional loose lepton, parameterised as a function of p_T and η .

8.2.2.1 Fake Rate Determination (SS Method) The lepton fake rates are determined in the very similar way as it was described before for the Opposite Sign method. Samples of $Z(\ell\ell) + e$ and $Z(\ell\ell) + \mu$ events are selected in the same way except that the mass window around the nominal Z mass is set to $40 \text{ GeV} < M_{\text{inv}}(\ell_1, \ell_2) < 120 \text{ GeV}$, as in the signal region ("SS phase space"). Also, despite the cut on the missing transverse energy, a contamination of real leptons from WZ events is still visible at high p_T . This contribution is thus subtracted from



(a)

Figure 30: Fake rates as a function of the probe p_T for muons which satisfy the loose selection criteria, measured in a $Z(\ell\ell) + \ell$ sample in the 13 TeV data. The barrel selection includes electrons (muons) up to $|\eta| = 1.479$ (1.2). The Fake rates are shown before (dotted lines) and after (plain line) the removal of the WZ contribution from the simulation.

the final fake rate, using the estimation given by the Monte Carlo. The fake rates for muon are shown in Fig. 30 for both muons in the barrel ($|\eta| < 1.2$) and in the endcaps ($|\eta| > 1.2$).

For electrons, events where a radiated photon makes an asymmetric conversion, where one low p_T leg is not identified, contribute to the $Z + e$ sample that is used to measure the electron fake rate. While the requirement $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV ("OS phase space") largely suppresses FSR of photons radiated off the lepton legs, these radiations occur at a much larger rate in the "SS phase space". As a result of this enhanced contribution from conversions, the electron fake rates measured within the SR phase space are larger than the "OS fake rates".

However, the relative fraction of FSR conversions is not the same in the "SS phase space" and in the control sample (defined below) where the fake rates will be applied. A correction accounting for this difference must be applied to the fake rates measured within the SS phase space, in order to obtain average fake rates that are appropriate for the control sample. To determine this correction, several fake rate samples of $Z(\ell\ell) + e$ events are defined by applying different cuts on $M_{inv}(\ell_1, \ell_2)$ or on $M_{inv}(\ell_1, \ell_2, e)$. In addition to the "OS fake rate sample", where $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV ensures a minimal amount of conversions from FSR photons, one defines a sample that is maximally enriched in FSR conversions by $|M_{inv}(\ell_1, \ell_2, e) - M_Z| < 5$ GeV, as well as samples with an intermediate contamination from FSR conversions ($60 \text{ GeV} < M_{inv}(\ell_1, \ell_2) < 120 \text{ GeV}$). In each sample, one determines, in several (p_T, η) bins for the loose electron:

- the fake rate ratio;
- the average value of the expected inner missing hits ($N_{miss.hits}$ in the following), variable useful to tag conversions.

In a given (p_T, η) bin, both the measured fake rate and the average $\langle N_{miss.hits} \rangle$ are expected to grow linearly with the fraction of conversions. Hence, one expects a linear dependence of

the fake rate with respect to $\langle N_{miss.hits} \rangle$. This linear behaviour is demonstrated in Fig. 31, and linear fits are made in each (p_T, η) bin, which relate the fake rate to $\langle N_{miss.hits} \rangle$.

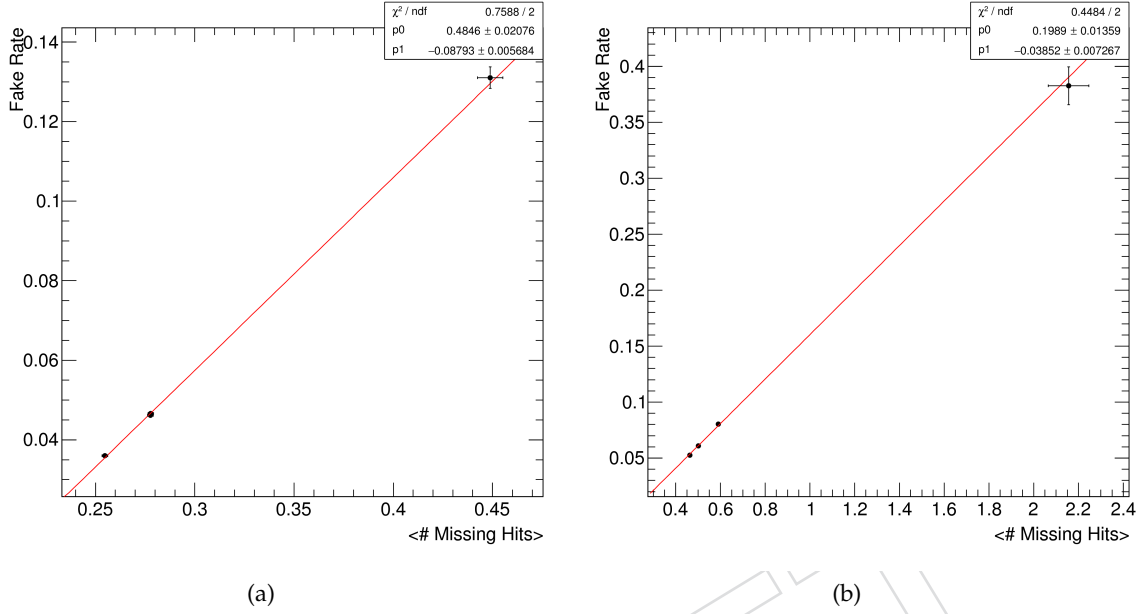


Figure 31: Examples of the correlation between the fake rate and the fraction of loose electrons for which the track has one missing hit in the pixel detector for $7 < p_T < 10$ GeV electrons in the barrel (left) and for $20 < p_T < 30$ GeV electrons in the endcaps (right). Each dot shows the measurements made in a given $Z + \text{loose } e$ sample. Results are produced at 13 TeV with 36.8 fb^{-1} data.

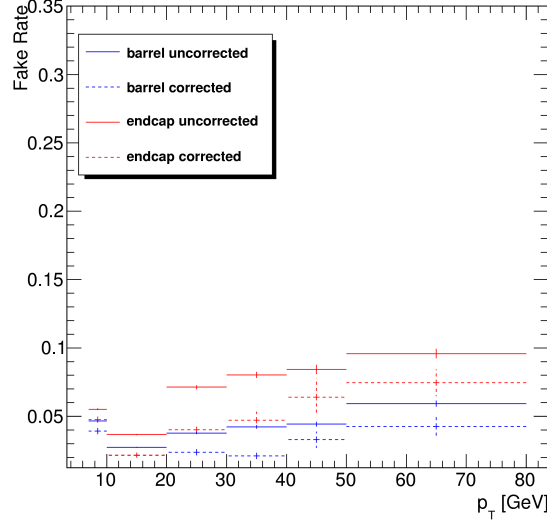
Finally, one looks at the loose electrons in the control sample where the fake rate will be applied, and one measures, in each (p_T, η) bin, $\langle N_{miss.hits} \rangle$. The proper average fake rate corresponding to the control sample is then determined from the linear relation derived previously. Figure 32 shows examples of the resulting corrected fake rates, together with the fake rates measured in the SS phase space. It can be seen that the correction for the actual fraction of conversions that is present in the control sample lowers the fake rates considerably with respect to what is measured in the SS phase space. The determination of these corrected fake rates mostly suffers from the limited statistics of the control sample, which translates into a large uncertainty on $\langle N_{miss.hits} \rangle$, the error on each dot being the fake rate obtained from the linear relations using the error on the $\langle N_{miss.hits} \rangle$.

8.2.2.2 Fake Rate Application (SS Method) Figure 33 shows the invariant mass distribution of events in control samples with SS-SF loose leptons, for channels with loose electrons ($4e$ and $2\mu 2e$) and channels with loose muons (4μ and $2e 2\mu$), for the 13 TeV dataset. The prediction from the Monte-Carlo simulation is also shown.

A good agreement is achieved between data and MC for the channels with loose electrons. The agreement is not as good for loose muons but this has no impact on the final result as only data are used in the end.

The differences in rates between OS and SS samples are used to compute the correction factor in equation 11 for the final data-driven estimation. They are given in table 12.

The event yields expected from $Z+X$ in the signal region, in the mass range $m_{4\ell} > 70 \text{ GeV}/c^2$,



(a)

Figure 32: Average fake rates to be applied to the control sample of SS Method (plain line), compared to the fake rates measured in the SS phase space (dotted line), for electrons in the barrel (blue) and in the endcaps (red). Results are produced at 13 TeV with 2018 data.

channel	4e	2μ 2e	4 μ	2e 2 μ
OS/SS	1.00 ± 0.01	1.00 ± 0.01	1.03 ± 0.02	1.03 ± 0.03

Table 12: The OS / SS ratios used in the Same-Sign Method.

are calculated for each final state and for each of the 22 STXS Stage 1.1. categories defined in Section 7.2. An example for the 4e final state is shown in Table 13. The statistical error is due to the event statistics in the control region. The systematic is due to error introduced with the missing hits based correction for electrons and the uncertainty in the fake rates in the muon case. The background is due to the systematic introduced when estimating the background composition. The total error is obtained with a quadrature sum for the statistical, background composition and correction systematics.

8.2.3 Uncertainties on Reducible Background estimation Methods

Source of systematic uncertainty of the methods presented above potentially arises from the different composition of background processes (DY , $t\bar{t}$, WZ , $Z\gamma$) in the region where we measure and where we apply the fake ratios. OS method corrects for the resulting bias via the “3P1F component” of its prediction. SS method corrects explicitly the electron fake rates for the correct fraction of photon conversions, but no attempt is made to correct the muon fake rates. The closure tests presented here are used to assess a possible residual bias in the methods.

8.2.3.1 Statistics in 4l Control Sample

The limited size of the samples in the control regions where we measure and where we apply the fake ratios is the source of the statistical uncertainties of the method. The dominating statistical uncertainty is driven by the number of events in the control region and is typically in the range of 1-10%.

8.2.3.2 Sensitivity of Fake Ratios to Background Composition

Compositions of reducible background processes (DY , $t\bar{t}$, WZ , $Z\gamma^{(*)}$) in the region where we measure and where

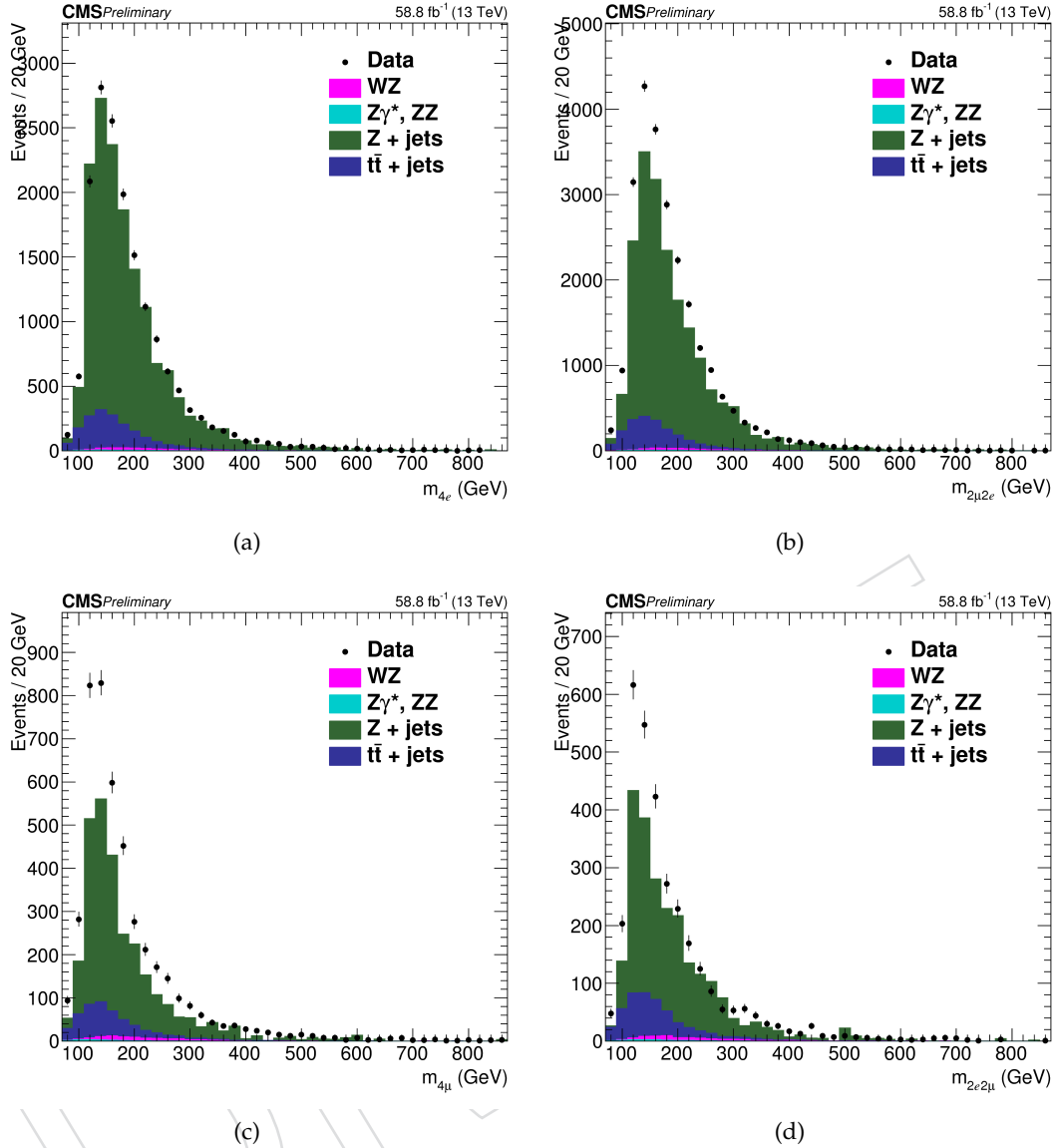


Figure 33: DATA-MC comparison of the SS-SF samples in the Z+X background control samples. (a) and (b) $4e$ and $2\mu 2e$, (c) and (d) 4μ and $2e 2\mu$ in 2018 data.

we apply the fake ratio method are typically not the same. This is the main source of the systematic uncertainty of the fake ratio method.

This uncertainty can be estimated by measuring the fake ratios for individual background processes in the $Z + 1L$ region in simulation. The weighted average of these individual fake ratios is the fake ratio that we measure in this sample (in simulation). The exact composition of the background processes in the $2P+2F$ region where we plan to apply the fake ratios can be determined from simulation, and one can reweigh the individual fake ratios according to the $2P+2F$ composition. The difference between the reweighed fake ratio and the average one can be used as a measure of the uncertainty on the measurement of the fake ratios. The fake ratios for individual processes, the average fake ratio and the reweighed fake ratio determined by simulation are shown in table below. The effect of this systematic uncertainty is propagated to the final estimates, and it amounts to about 32% for $4e$, 33% for $2e 2\mu$ and 35% for 4μ final state.

Category	exp. N^{Z+X} in SR	(stat.)	(syst.)	Total unc.
ggH.0J_PTH.0_10	0.81	± 0.22	± 0.02	± 0.33
ggH.0J_PTH.10_200	10.8	± 0.79	± 0.20	± 3.33
ggH.1J_PTH.0_60	2.93	± 0.39	± 0.059	± 0.97
ggH.1J_PTH.60_120	2.81	± 0.38	± 0.08	± 0.93
ggH.2J_PTH.0_60	0.89	± 0.20	± 0.02	± 0.33
ggH.2J_PTH.60_120	1.25	± 0.24	± 0.02	± 0.45
ggH.2J_PTH.120_200	0.70	± 0.18	± 0.02	± 0.28
ggH.PTH.200	0.72	± 0.16	± 0.03	± 0.27
ggH.VBF	0.65	± 0.16	± 0.02	± 0.25
VBF.1j	1.15	± 0.23	± 0.03	± 0.42
VBF.2j	0.34	± 0.11	± 0.01	± 0.15
VBF.2j_mjj_350_700_2j	0.04	± 0.04	± 0.00	± 0.04
VBF.2j_mjj_GT700_2j	0.06	± 0.04	± 0.00	± 0.04
VBF.2j_mjj_GT350_3j	0.63	± 0.17	± 0.02	± 0.26
VBF_GT200_2j	0.06	± 0.03	± 0.00	± 0.04
VH.Had	1.10	± 0.21	± 0.03	± 0.39
VBF_rest_VH	0.15	± 0.06	± 0.01	± 0.08
VH.lep_0_150	0.69	± 0.18	± 0.02	± 0.27
VH.Lep_GT150	0.11	± 0.06	± 0.00	± 0.07
ttH.Lep	0.32	± 0.11	± 0.01	± 0.14
ttH.Had	0.51	± 0.13	± 0.01	± 0.20
Inclusive	24.92	± 1.18	± 0.42	± 7.58

Table 13: Number of 4e events in the SS-SF control region for 59.7 fb^{-1} of 2018 data, and expected number of $Z+X \rightarrow 4e$ events in the signal region as predicted from the SS method. Systematical error shown listed in table is only due to fake rate variation. Background composition component of 30% is taken into account in the total uncertainty.

8.2.3.3 Shape Uncertainty

In order to estimate the uncertainty on the m_{4l} shape we have looked at the differences between the shapes of predicted background distributions for all three channels, and between both of the two methods. The envelope of differences between these shapes of distributions is used as an estimate of the shape uncertainty. The uncertainty is estimated to be roughly in the range of 5% - 15%. Since the difference of the shapes slowly varies with m_{4l} , it is taken as a constant versus m_{4l} and is absorbed in the much larger uncertainty on the predicted yield of backgrounds.

8.2.4 Combination

Two methods for the reducible background estimation are presented with associated uncertainties in the previous sections. Table 14 shows the summary of the results of the methods. The shape systematics is absorbed in the statistical and systematic uncertainties on the estimated yields, and the total uncertainties are found to be of the order of 35–45%. Predictions of the two methods are combined assuming no correlation between the uncertainties (independent control regions, partially different sources systematics), and combined mean values are obtained by weighting the individual means according to the corresponding variances.

Table 14: The summary of the results of two methods for the prediction of the contribution of reducible background processes in the signal region. The weighted mean value of the results of the two methods is taken as the final estimate of this contribution, while the uncertainty of the result is the one that covers the uncertainties of both methods. The table shows symmetric individual uncertainties for two methods, while in the combination and statistical analysis these are treated as asymmetric. Results are given for an integrated luminosity of 59.7 fb^{-1} .

13 TeV	4e	4 μ	2e2 μ
Method OS	$24.6 \pm 7.6_{\text{tot.}}$	$49.9 \pm 15.5_{\text{tot.}}$	$69.2 \pm 15.1_{\text{tot.}}$
Method SS	$16.0 \pm 5.9_{\text{tot.}}$	$51.8 \pm 15.8_{\text{tot.}}$	$60.5 \pm 14.1_{\text{tot.}}$
Combined	19.3	50.8	64.6

9 Signal Modelling

9.1 Signal modeling of the SM Higgs

9.1.1 Signal Corrections

For the dominant gluon fusion production mode for the signal, the p_T spectrum has been tuned using the “hfact” parameter of POWHEG to match closely the the Higgs boson p_T spectrum in the full phase space as predicted by the HRES generator at $\sqrt{s} = 13$ TeV. The agreement can be seen in Fig. 34. For other mass points, the “hfact” parameter is parametrized as a function of $m(H)$ as $\text{hfact} = 0.1 \cdot m(H) + 37.5$.

We use weights defined in bins of generated Higgs p_T and number of generated jets to reweight our POWHEG signal sample to the NNLOPS one.

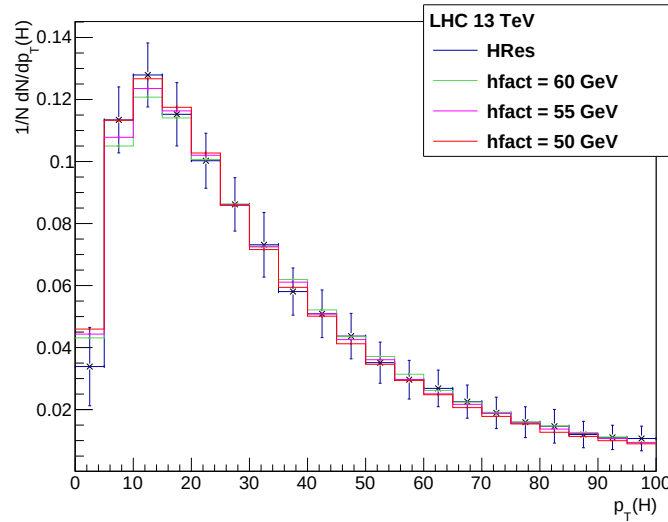


Figure 34: Comparison of POWHEG and HRES p_T spectrum at 13 TeV for different hfact.

9.1.2 Signal Normalization

The normalization of the Higgs boson signal in the peak region is directly taken from simulation. Using simulated samples for five m_H points (120, 124, 125, 126 and 130 GeV), polynomial fits of the expected signal yields in a $[105, 140]$ GeV $m_{4\ell}$ window around the Higgs boson peak are performed separately for the stage1 bins, in the three final states and the twenty two categories. Some examples of the fits are shown in Fig. 35 for the different stage1 bins.

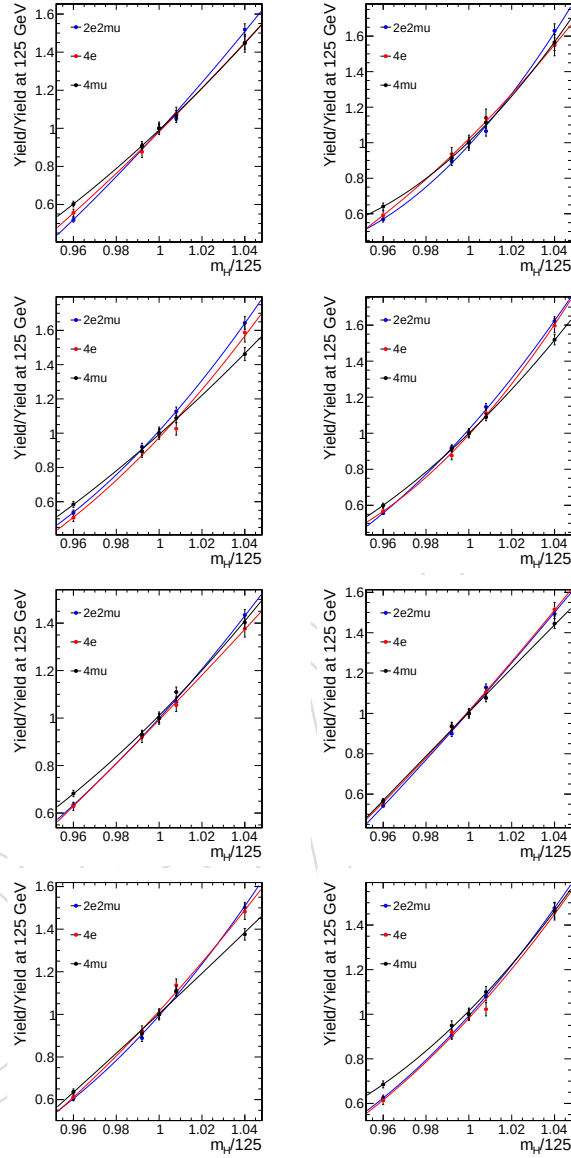


Figure 35: Fits of the expected signal yields after full event selection in a $[105, 140]$ GeV $m_{4\ell}$ window, for the ggH_1j_60_200 stage1 bin in ggH_1j_60_200 and VBF_1j categories (rows 1), VBF_Rest stage1 bins in ggH_1j_0_60 and VBF 1jet categories (row 2), VH-hadronic and VH-leptonic (row 3) in their dedicated categories and ttH production mode in ttH hadronic and leptonic categories (row 4) in the $[118, 130]$ GeV range of generated m_H .

9.1.3 Signal Lineshape

In this section we describe the PDF used to describe the signal shapes for different masses in different Event Categories as explained before. The resolution function model depends only on the leptons quality, while the exact shape depends on the lepton kinematics, and so depend also on the Higgs mass. A Double Crystal-Ball function $f_{dCB}(m_{4\ell} | m_H)$:

$$dCB(\xi) = N \cdot \begin{cases} A \cdot (B + |\xi|)^{-n_L}, & \text{for } \xi < \alpha_L \\ A \cdot (B + |\xi|)^{-n_R}, & \text{for } \xi > \alpha_R \\ \exp(-\xi^2/2), & \text{for } \alpha_L \leq \xi \leq \alpha_R \end{cases} \quad (12)$$

where $\xi = (m_{4\ell} - m_H - \Delta m_H)/\sigma_m$. This function has six independent parameters, and is intended to capture the Gaussian core (σ_m) of the four-lepton mass resolution function, systematic mass shift Δm_H of the peak, and the left- and right-hand tail originating from leptons emitting brem in the tracker material, present for both electrons and muons, and from the non-Gaussian mis-measurements specific to interactions of electrons with the detector material (two parameters, n and α , for each side of the mean): The prominence of the left-,right-hand tail is defined the power n_L, n_R , respectively. The parameters α_L, α_R define where the splicing of the tails and the core are made, in units of σ_m . Parameters A and B are not independent; they are defined by requiring the continuity of the function itself and its first derivatives. N is the normalizing constant.

We model each stage1 bin shapes in three final states at various mass points (120-130). The fitting strategy used to deal with this situation is to use the linear approximation of all the Double Crystal Ball parameters varying with m_H .

$$params = params_{CB0} + params_{CB1} \times (m_H - 125\text{GeV}) + params_{CB2} \times (m_H - 125\text{GeV})^2 \quad (13)$$

Simultaneous fits are performed using 5 mass samples to extract the $params_{CB0}$, $params_{CB1}$ and $params_{CB2}$. The initial value for the $params_{CB0}$ is obtained by fitting 125 GeV mass sample alone, and refitted in the simultaneous fits. Some examples of the fits are shown in Fig. 36 and 37 for the ggH and VH Stage 1.1 bins. The simultaneous fit at different mass points are shown in Fig. 38.

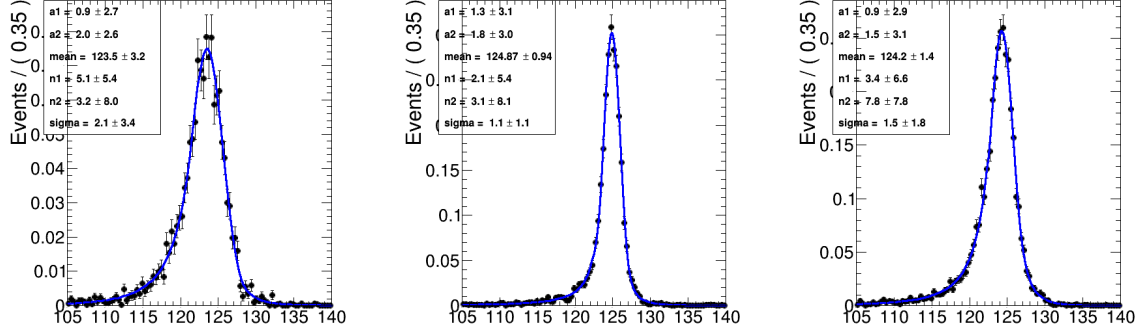


Figure 36: Probability density functions $f(m_{4l}|m_H)$ for signal with $m_H=125$ GeV at the reconstruction level after the full lepton and event selections are applied. The distributions obtained from 13 TeV ggH MC samples are fitted with the model described in the text for 4μ (left), $4e$ (center) and $2e2\mu$ (right) events. The distribution show the events in the stage 1.1 bin ggH.0j_0-10.

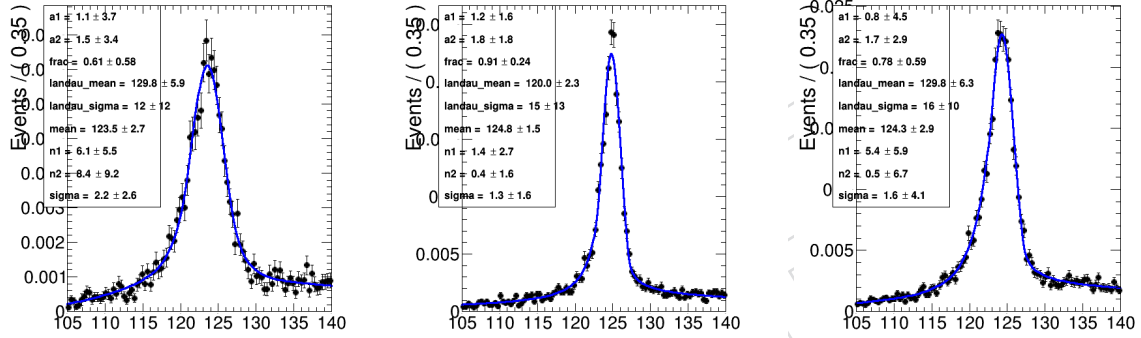


Figure 37: The sum of the Probability density functions $f(m_{4l}|m_H)$ for signal with $m_H=125$ GeV at the reconstruction level after the full lepton and event selections are applied. The distributions obtained from 13 TeV VH MC samples are fitted with the model described in the text for 4μ (left), $4e$ (center) and $2e2\mu$ (right) in the Untagged Category. The distribution show the events in the stage 1.1 bin VH.Lep_0-150.

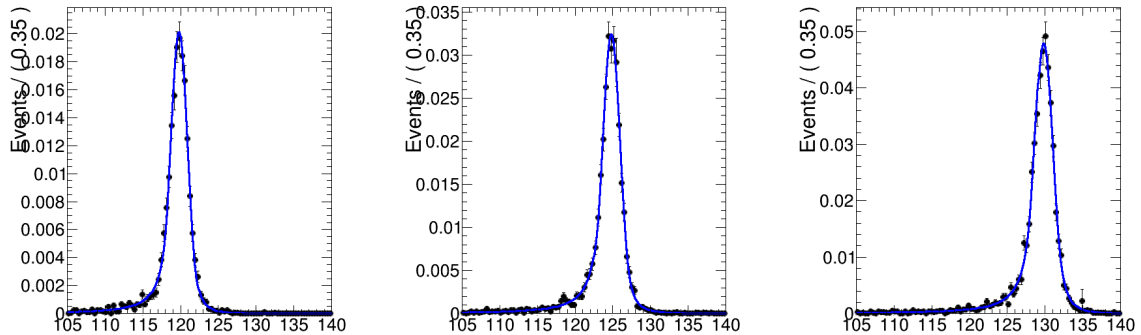


Figure 38: Simultaneous fits for $f(m_{4l}|m_H)$ for VBF signal with m_{jj} above 700 GeV with $m_H=120, 125, 130$ GeV at the reconstruction level after the full lepton and event selections are applied. The distributions obtained from 13 TeV VBF MC samples are fitted with the model described in the text for 4μ events.

10 Systematic uncertainties

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The main experimental uncertainties which affect both signal and background are the uncertainty on the integrated luminosity (2.6%) and the uncertainty on the lepton identification and reconstruction efficiency (ranging from 2.5–9% on the overall event yield for the 4μ and $4e$ channels, respectively). Experimental uncertainties for the reducible background estimation, described in Section 8.1.1, vary between 6% to 32% from the variation of the fake rates. The mismatch in the composition of backgrounds between the samples where the fake rate is derived and where it is applied accounts for 32% to 35%.

The uncertainty of the lepton energy scale is assessed by propagating down to the four-leptons invariant mass the uncertainty associated the lepton momentum corrections on each individual lepton. The uncertainty is treated as uncorrelated (all up or all down). The new four-leptons invariant mass obtained this way is then fitted with only the mean value floating. The difference between the new mean value and the default one gives an estimation of the impact of the lepton energy scale. The values are reported in the Table 15, together with the estimation of the impact of lepton momentum resolution, following a similar procedure.

Table 15: Difference (in %) between the nominal mean and width of the four-leptons Higgs mass distribution and those where the lepton momentum scale and resolution uncertainties have been propagated.

	4e	4μ	$2e2\mu$
Scale Difference (%)	0.087	0.049	0.069
Resolution Difference (%)	17	12	17

The results obtained this way suggest that the uncertainties from 2016 (20% on the resolution, 0.3% for the $4e$ channel for scale) are still valid.

Theoretical uncertainties which affect both the signal and background estimation include uncertainties from the renormalization and factorization scale and choice of PDF set. The uncertainty from the renormalization and factorization scale is determined by varying these scales between 0.5 and 2 times their nominal value while keeping their ratio between 0.5 and 2. In the case of ggF production mode, the QCD scale uncertainty is broke down to 9 different sources. The renormalization and factorization scale affect the overall cross section, the migration of the stage1.1 bins in differnt Higgs pT and number of jets, top loop effects are treated as independent uncertainties, and correclated among different stage 1.1 bins. The numbers are shown in Fig 39. The uncertainty from the PDF set is determined by taking the root mean square of the variation when using different replicas of the default NNPDF set. PDF uncertainty arises from uncertainty is also taken into account On the background, an additional uncertainty of 10% on the K factor used for the $gg \rightarrow ZZ$ prediction is applied as described in Section 8.1. A systematic uncertainty of 2% on the branching ratio of $H \rightarrow ZZ \rightarrow 4\ell$ only affects the signal yield. In the case of event categorization, experimental and theoretical uncertainties which account for possible migration of signal and background events between categories are included. The main sources of uncertainty on the event categorization include the QCD scale, PDF set, and the modeling of hadronization and the underlying event. These uncertainties amount to between 4–20% for the signal and 3–20% for the background depending on the category. The lower range corresponds to the VBF and VH processes and the upper range corresponds to the $gg \rightarrow H$ process yield in the VBF-2jet-tagged category. Additional uncertainties come from the imprecise knowledge of the jet energy scale (from 2% for the $gg \rightarrow H$ yield in the untagged category to 22% for $gg \rightarrow H$ yield in the VBF-2jet-tagged category) and b-tagging efficiency and

mistag rate (up to 10% in the $t\bar{t}H$ -tagged category). In the cross section measurement, the signal cross section uncertainties arise from theoretical sources are removed, while they are kept in the signal strength measurement. The exact numbers are shown in Table ?? and Table ??.

Uncertainties affecting the shape of the kinematic discriminants are taken into account. While none is affecting $\mathcal{D}_{\text{bkg}}^{\text{kin}}$, there are several sources that could affect the shape of $\mathcal{D}_{\text{bkg}}^{\text{VBF+dec}}$ or $\mathcal{D}_{\text{bkg}}^{\text{VH+dec}}$: jet energy scale or resolution uncertainty, Pythia tune, variations on QCD renormalization scale or PDF weights, etc... Only the leading one in each category and process is taken. These uncertainties are modeled as alternative templates of the kinematic discriminants in the final fit. The Fig ?? shows examples of nominal and alternative templates for the VBF-2 jets tagged category.

Table 16: Summary of the experimental systematic uncertainties in the $H \rightarrow 4\ell$ measurements of 2016 data.

Summary of relative systematic uncertainties	
Common experimental uncertainties	
Luminosity	2.6 %
Lepton identification/reconstruction efficiencies	2.5 – 9 %
Background related uncertainties	
Reducible background (Z+X)	36 – 43 %
Signal related uncertainties	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

Table 17: Summary of the experimental systematic uncertainties in the $H \rightarrow 4\ell$ measurements of 2017 data.

Summary of relative systematic uncertainties	
Common experimental uncertainties	
Luminosity	2.3 %
Lepton identification/reconstruction efficiencies	3. – 12.5 %
Background related uncertainties	
Reducible background fake rate variation (Z+X)	6 – 32 %
Signal related uncertainties	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

Theoretical systematics on scale and category migrations due to QCD scale in gluon fusion are calculated using the so called WG1 uncertainty scheme. There are 9 uncertainties in total, 2 of which correspond to scale change (THU_ggH_Mu and THU_ggH_Res) and 7 others to category migrations. They give largest contribution to the overall systematics from the theoretical part while all other theoretical uncertainties mentioned here do not have a significant impact as it can be seen on the Fig. 40.

10.1 Systematic uncertainties treatment when combining the data sets

In this Section we describe how different systematic uncertainties are treated when combining the data sets. The theoretical uncertainties are correlated. All the experimental systematics are treated independently.

Table 18: **FIXME:** TO BE UPDATED. COPY PASTE NUMBERS OF 2017 FOR NOW. Summary of the experimental systematic uncertainties in the $H \rightarrow 4\ell$ measurements of 2018 data.

Summary of relative systematic uncertainties	
Common experimental uncertainties	
Luminosity	2.3 %
Lepton identification/reconstruction efficiencies	3. – 12.5 %
Background related uncertainties	
Reducible background fake rate variation (Z+X)	6 – 32 %
Signal related uncertainties	
Lepton energy scale	0.04 – 0.3 %
Lepton energy resolution	20 %

Table 19: Summary of the theory systematic uncertainties in the $H \rightarrow 4\ell$ measurements for the inclusive analysis

Summary of inclusive theory uncertainties	
BR($H \rightarrow ZZ \rightarrow 4\ell$)	2 %
QCD scale ($q\bar{q} \rightarrow ZZ$)	+3.2/-4.2 % %
PDF set ($q\bar{q} \rightarrow ZZ$)	+3.1/-3.4 %
Electroweak corrections ($q\bar{q} \rightarrow ZZ$)	± 0.1 %

Cross sections and fractional uncertainties												
	sig	stat	mu	res	mig01	mig12	VBF2j	VBF3j	pT60	pT120	qm_top	Tot
Incl	48.52 +/- 0.00		+4.6%	+2.1%	-0.0%	-0.0%	+0.3%	-0.0%	+0.0%	+0.2%	+0.2%	+5.1%
FWDH	4.27 +/- 0.01		+4.5%	+1.9%	-0.5%	-0.2%	+0.0%	+0.0%	-0.3%	-0.1%	+0.0%	+4.9%
VBF_J3V	0.27 +/- 0.00		+0.0%	+0.0%	+0.0%	+0.0%	+20.0%	-32.0%	-1.6%	+1.1%	+0.1%	+37.8%
VBF_J3	0.36 +/- 0.00		+0.0%	+0.0%	+0.0%	+0.0%	+20.0%	+23.5%	-0.2%	+2.5%	+0.2%	+31.0%
=0J	27.25 +/- 0.03		+3.8%	+0.1%	-4.1%	+0.0%	+0.0%	+0.0%	+0.0%	+0.0%	+0.0%	+5.6%
=1J_0-60	6.49 +/- 0.01		+5.2%	+4.5%	+7.9%	-6.8%	+0.0%	+0.0%	-4.8%	-1.6%	+0.0%	+13.5%
=1J_60-120	4.50 +/- 0.01		+5.2%	+4.5%	+7.9%	-6.8%	+0.0%	+0.0%	+4.8%	-0.9%	+0.0%	+13.4%
=1J_120-200	0.74 +/- 0.00		+5.2%	+4.5%	+7.9%	-6.8%	+0.0%	+0.0%	+10.0%	+10.1%	+0.5%	+18.9%
=1J_200->	0.15 +/- 0.00		+5.2%	+4.5%	+7.9%	-6.8%	+0.0%	+0.0%	+10.0%	+14.0%	+10.5%	+23.7%
>=2J_0-60	1.22 +/- 0.01		+8.9%	+8.9%	+4.4%	+18.2%	+0.0%	+0.0%	-5.9%	-1.6%	+0.0%	+23.3%
>=2J_60-120	1.86 +/- 0.01		+8.9%	+8.9%	+4.4%	+18.2%	+0.0%	+0.0%	-0.2%	-0.2%	+0.0%	+22.5%
>=2J_120-200	0.99 +/- 0.00		+8.9%	+8.9%	+4.4%	+18.2%	+0.0%	+0.0%	+6.6%	+10.6%	+0.6%	+25.8%
>=2J_200->	0.42 +/- 0.00		+8.9%	+8.9%	+4.4%	+18.2%	+0.0%	+0.0%	+10.0%	+14.0%	+11.8%	+30.7%
=0J	30.12 +/- 0.03		+3.8%	+0.1%	-4.1%	+0.0%	+0.0%	+0.0%	+0.0%	+0.0%	+0.0%	+5.6%
=1J	12.92 +/- 0.02		+5.2%	+4.5%	+7.9%	-6.8%	+0.0%	+0.0%	-0.1%	-0.4%	+0.2%	+12.5%
>=2J	5.47 +/- 0.01		+7.8%	+7.8%	+3.9%	+16.1%	+2.3%	-0.0%	+0.4%	+2.9%	+1.1%	+20.3%
>=1J_60-200	9.09 +/- 0.01		+6.2%	+5.8%	+6.4%	+1.9%	+0.9%	+0.1%	+4.2%	+1.7%	+0.1%	+11.8%
>=1J_120-200	1.96 +/- 0.01		+6.8%	+6.5%	+5.5%	+6.9%	+1.5%	+0.4%	+8.0%	+10.4%	+0.6%	+18.5%
>=1J_>200	0.58 +/- 0.00		+7.9%	+7.7%	+5.4%	+11.6%	+0.0%	+0.0%	+10.0%	+14.0%	+11.4%	+26.7%
>=1J_>60	9.68 +/- 0.01		+6.3%	+5.9%	+6.3%	+2.5%	+0.8%	+0.1%	+4.6%	+2.5%	+0.8%	+12.2%
>=1J_>120	2.54 +/- 0.01		+7.0%	+6.8%	+5.5%	+8.0%	+1.2%	+0.3%	+8.4%	+11.2%	+3.0%	+19.9%
>=1	18.40 +/- 0.02		+6.0%	+5.5%	+6.7%	-0.0%	+0.7%	-0.0%	+0.0%	+0.5%	+0.4%	+10.6%

Figure 39: Summary of the theory systematic uncertainties of the ggH production modes in $H \rightarrow 4\ell$ measurements for the stage 1.1 analysis

Table 20: Summary of the theory systematic uncertainties of the non-ggH production modes in $H \rightarrow 4\ell$ measurements for the stage 1.1 analysis

	QCD scale μ_R	QCD scale μ_F	PDF set	PDF AlphaS
VBF_2j_mjj_350_700_2j	0.9	1.0	3.7	0.6
VBF_2j_mjj_GT700_2j	0.4	1.0	5.6	0.5
VBF_2j_mjj_GT350_3j	0.1	0.1	5.6	0.6
VBF_Rest	0.5	0.7	4.4	0.7
VBF_GT200	0.7	1.0	6.7	0.4
VH_Had	4.8	5.2	3.9	0.7
VH_Lep_0.150	3.6	1.6	3.7	0.8
VH_Lep_GT150	3.7	4.9	3.9	0.9
ttH	4.3	2.8	3.2	1.5

Table 21: Summary of the systematic uncertainties in the $H \rightarrow 4\ell$ measurements on the categorization migration

Summary of the systematic uncertainties on the categorization migration	
QCD scale μ_R (gg)	0.1–6 %
QCD scale μ_F (gg)	0.1–6 %
PDF set (gg)	0.1–2 %
PDF AlphaS(gg)	0.1–2 %
QCD scale μ_R (VBF, WH, ZH)	0.1–12 %
QCD scale μ_F (VBF, WH, ZH)	0.1–6 %
PDF set (VBF, WH, ZH)	0.1–5 %
PDF AlphaS(VBF, WH, ZH)	0.1–2 %
QCD scale μ_R (ttH)	0.1–8 %
QCD scale μ_F (ttH)	0.1–3 %
PDF set (ttH)	0.2–1 %
PDF AlphaS(ttH)	0.1–0.7 %
JES	1–200 %
JER	1–55 %
Btag	1–22 %

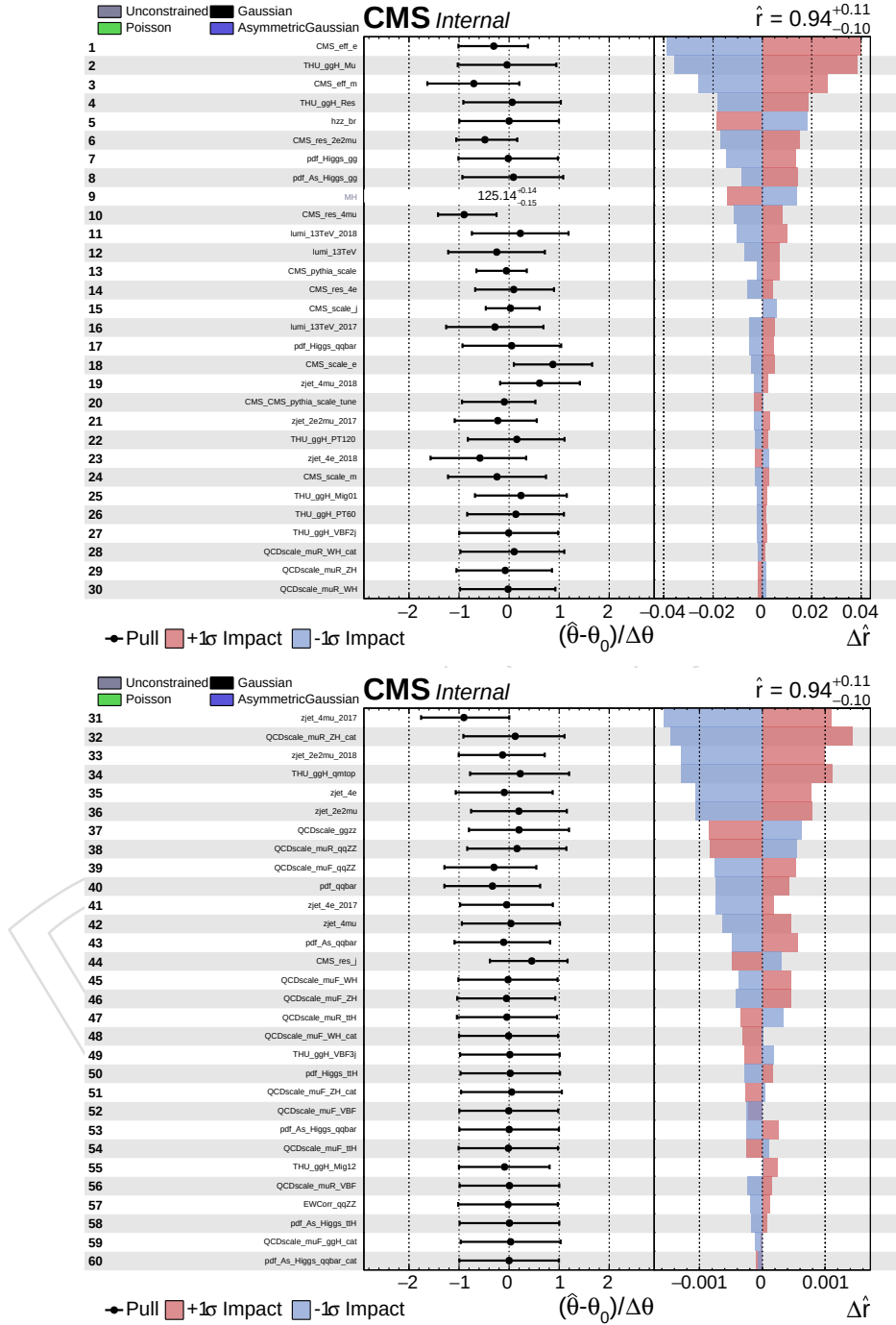


Figure 40: Pulls

11 Yields and distributions

In this section we summarize the status of the analysis after selection, showing the inputs to the final results, namely the event yields and errors in the full signal region and in a restricted $m_{4\ell}$ range, and the distributions of the main kinematic variables in data and MC.

11.1 Signal Region Yields

The number of candidates observed in data and the expected yields for the backgrounds and Higgs boson signal after the full event selection are reported in Table 22 for the full range of $m_{4\ell}$ and in Table 23 for a $118 < m_{4\ell} < 130$ GeV mass window around the Higgs boson peak. Table 24 shows the expected and observed yields for each of the 22 event categories.

Table 22: The number of expected background and signal events and number observed candidates after full analysis selection, for each final state, for the full mass range $m_{4\ell} > 70$ GeV, for an integrated luminosity of 59.7 fb^{-1} . Signal and ZZ backgrounds are estimated from Monte Carlo simulation, Z+X is estimated from data.

Channel	4e	4 μ	2e2 μ	4 ℓ
$q\bar{q} \rightarrow ZZ$	$333.01^{+y.y}_{-z.z}$	$622.20^{+y.y}_{-z.z}$	$815.20^{+y.y}_{-z.z}$	$1770.41^{+y.y}_{-z.z}$
$gg \rightarrow ZZ$	$75.11^{+y.y}_{-z.z}$	$116.55^{+y.y}_{-z.z}$	$176.86^{+y.y}_{-z.z}$	$368.52^{+y.y}_{-z.z}$
Z + X	$19.37^{+y.y}_{-z.z}$	$50.75^{+y.y}_{-z.z}$	$64.84^{+y.y}_{-z.z}$	$134.96^{+y.y}_{-z.z}$
Sum of backgrounds	$427.50^{+y.y}_{-z.z}$	$789.50^{+y.y}_{-z.z}$	$1056.89^{+y.y}_{-z.z}$	$2273.89^{+y.y}_{-z.z}$
Signal ($m_H = 125$ GeV)	$19.58^{+y.y}_{-z.z}$	$40.83^{+y.y}_{-z.z}$	$50.68^{+y.y}_{-z.z}$	$111.09^{+y.y}_{-z.z}$
Total expected	$447.08^{+y.y}_{-z.z}$	$830.33^{+y.y}_{-z.z}$	$1107.57^{+y.y}_{-z.z}$	$2384.98^{+y.y}_{-z.z}$
Observed	462	850	1130	2442

Table 23: The number of expected background and signal events and number of observed candidates after full analysis selection, for each event category, for the mass range $118 < m_{4\ell} < 130$ GeV, for an integrated luminosity of 59.7 fb^{-1} . The yields are given for the different production modes. Signal and ZZ backgrounds are estimated from Monte Carlo simulation, Z+X is estimated from data.

Channel	4 μ	4e	2e2 μ
Signal ($m_H = 125$ GeV)	37.09	16.35	44.16
qqZZ	14.10	5.33	16.43
ggZZ	1.30	0.63	1.15
ZX	6.53	1.72	7.22
Total expected	59.02	24.03	68.96
Data	50	23	63

11.2 Signal Region Distributions

The reconstructed four-lepton invariant mass distribution is shown in Figure 41 for the full dataset, and compared to expectations from the SM backgrounds, first for the full mass range, and then zooming on the low-mass range and high-mass range. In Figure 42, the same distributions are shown split by final state (4e, 4 μ , and 2e2 μ), for the two same mass ranges. In Figure 43, they are split by event category, for the low-mass range. The SM background distributions are obtained combining the rate normalization from data-driven methods and knowledge on shapes taken from the MC samples.

Table 24: The number of expected background and signal events and number of observed candidates after full analysis selection, for each event category, for the mass range $118 < m_{4\ell} < 130$ GeV, for an integrated luminosity of 137.1 fb^{-1} . The yields are given for the different production modes. Signal and ZZ backgrounds are estimated from Monte Carlo simulation, Z+X is estimated from data.

Event category	Signal							Background			Expected		Observed
	gg $\rightarrow$ H	VBF	WH	ZH	ttH	bbH	tqH	q $\bar{q} \rightarrow$ ZZ	gg $\rightarrow$ ZZ	Z + X	signal	total	
ggH-0j/pT[0,10]	25.31	0.08	0.02	0.02	0.00	0.14	0.00	26.46	0.97	1.19	25.57	54.18	61.00
ggH-0j/pT[10-200]	86.80	1.69	0.54	0.86	0.00	0.90	0.00	35.42	3.79	15.48	90.80	145.49	153.00
ggH-1j/pT[0-60]	26.24	1.43	0.50	0.45	0.01	0.43	0.01	10.26	1.19	5.54	29.06	46.05	40.00
ggH-1j/pT[60-120]	12.35	1.24	0.45	0.47	0.01	0.10	0.01	2.76	0.16	3.21	14.63	20.76	17.00
ggH-1j/pT[120-200]	3.31	0.62	0.17	0.26	0.00	0.02	0.00	0.38	0.00	0.52	4.38	5.28	6.00
ggH-2j/pT[0-60]	3.68	0.29	0.14	0.14	0.06	0.09	0.02	0.97	0.15	2.07	4.42	7.60	9.00
ggH-2j/pT[60-120]	5.17	0.54	0.22	0.22	0.09	0.04	0.02	0.84	0.07	1.86	6.30	9.06	12.00
ggH-2j/pT[120-200]	2.90	0.40	0.15	0.17	0.07	0.01	0.02	0.26	0.00	0.40	3.71	4.37	5.00
ggH/pT>200	2.72	0.65	0.21	0.24	0.06	0.01	0.02	0.16	0.00	0.21	3.91	4.28	2.00
ggH-2j/mJJ>350	0.82	0.17	0.06	0.05	0.04	0.01	0.01	0.16	0.02	0.65	1.16	1.98	3.00
VBF-1j	14.17	2.94	0.20	0.18	0.00	0.12	0.01	2.37	0.43	1.05	17.61	21.46	20.00
VBF-2j/mJJ[350,700]	0.80	1.11	0.01	0.01	0.00	0.01	0.00	0.08	0.02	0.04	1.95	2.09	2.00
VBF-2j/mJJ>700	0.43	1.80	0.00	0.00	0.00	0.00	0.00	0.02	0.01	0.03	2.25	2.31	2.00
VBF-3j/mJJ>350	2.43	2.15	0.06	0.07	0.02	0.03	0.05	0.24	0.06	0.96	4.81	6.07	6.00
VBF-2j/pT>200	0.42	0.76	0.01	0.01	0.01	0.00	0.01	0.01	0.00	0.03	1.22	1.26	0.00
VBF-rest	2.40	0.87	0.11	0.10	0.03	0.04	0.01	0.34	0.06	0.74	3.56	4.70	2.00
VH-lep/pTV[0-150]	0.24	0.04	0.71	0.25	0.08	0.02	0.02	0.82	0.14	0.40	1.37	2.72	5.00
VH-lep/pTV>150	0.02	0.01	0.21	0.08	0.04	0.00	0.01	0.01	0.00	0.02	0.36	0.40	0.00
VH-had/mJJ[60-120]	4.11	0.25	1.01	1.20	0.11	0.07	0.02	0.70	0.05	1.36	6.77	8.89	8.00
VH-rest	0.56	0.04	0.08	0.07	0.03	0.00	0.00	0.08	0.00	0.15	0.77	1.01	1.00
ttH-had	0.19	0.05	0.03	0.06	0.82	0.01	0.03	0.01	0.00	0.45	1.19	1.66	2.00
ttH-lep	0.02	0.00	0.02	0.02	0.60	0.00	0.03	0.03	0.00	0.12	0.70	0.85	0.00

The reconstructed dilepton invariant masses selected as Z_1 and Z_2 are shown in Figures 44 together with their correlation, both full range of $m_{4\ell}$ and focusing on a $118 < m_{4\ell} < 130$ GeV mass window around the Higgs boson peak. Similarly, the decay discriminant $\mathcal{D}_{\text{bkg}}^{\text{kin}}$, $\mathcal{D}_{\text{bkg}}^{\text{VBF+dec}}$ and $\mathcal{D}_{\text{bkg}}^{\text{VH+dec}}$ are shown in Fig. ??, 45 and 46 in this window. The four production mechanism discriminant $\mathcal{D}_{2\text{jet}}$, $\mathcal{D}_{1\text{jet}}$, $\mathcal{D}_{\text{WH}}$, and $\mathcal{D}_{\text{ZH}}$ are shown in Fig. 47 in the $118 < m_{4\ell} < 130$ GeV mass window, for events with at least two selected jets (except $\mathcal{D}_{1\text{jet}}$ which is for events with exactly one selected jet). Their correlations with $m_{4\ell}$ are shown in Fig. 48.

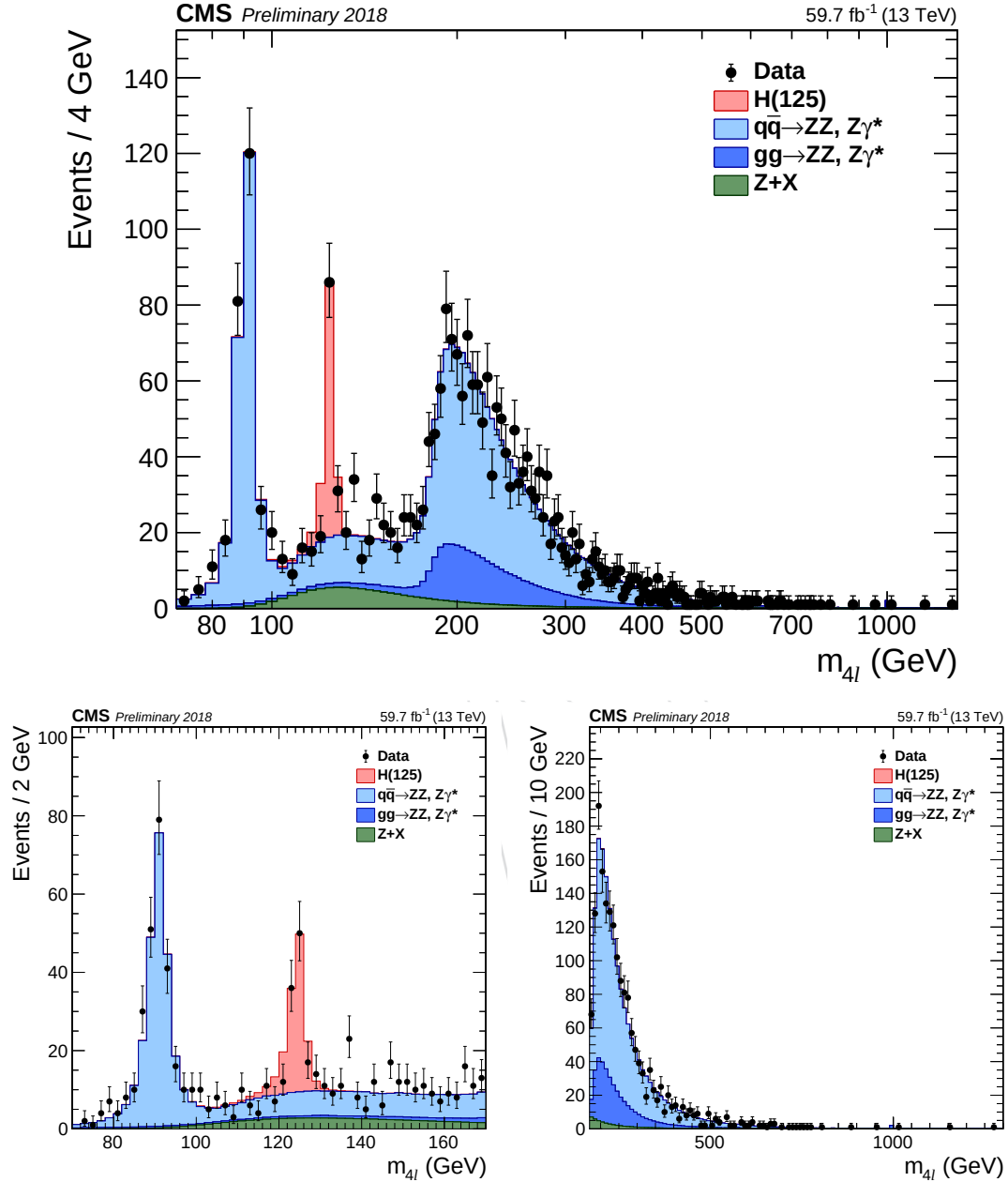


Figure 41: Distribution of the four-lepton reconstructed invariant mass $m_{4\ell}$ in the full mass range (top) and the low-mass range (bottom left) and high-mass range (bottom right). Points with error bars represent the data and stacked histograms represent expected distributions. The 125 GeV Higgs boson signal and the ZZ backgrounds are normalized to the SM expectation, the Z+X background to the estimation from data.

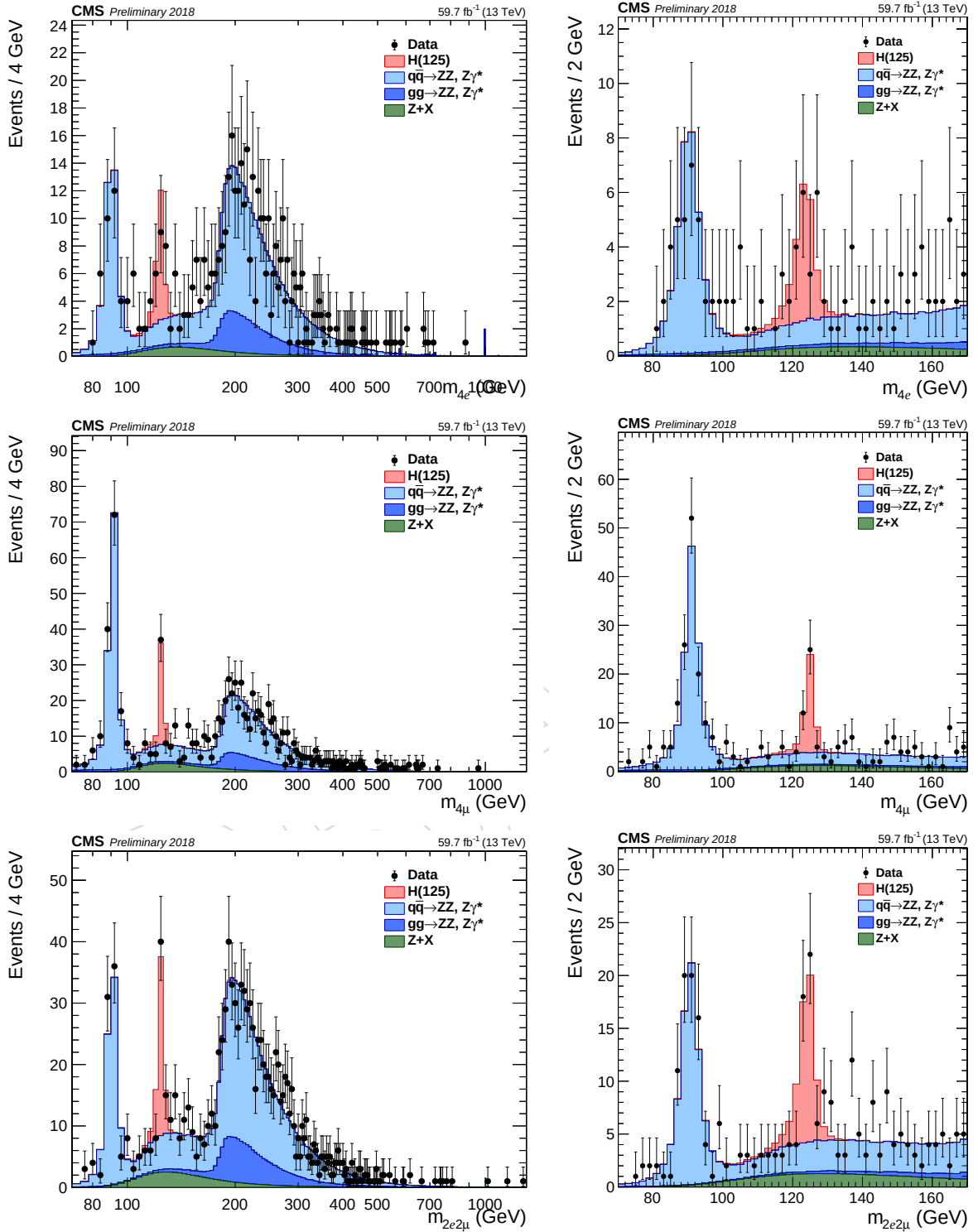


Figure 42: Distribution of the four-lepton reconstructed mass in several sub-channels: $4e$ (top), 4μ (middle), $2e2\mu$ for the low-mass range (bottom) for the full mass range (left) and the low-mass range (right).

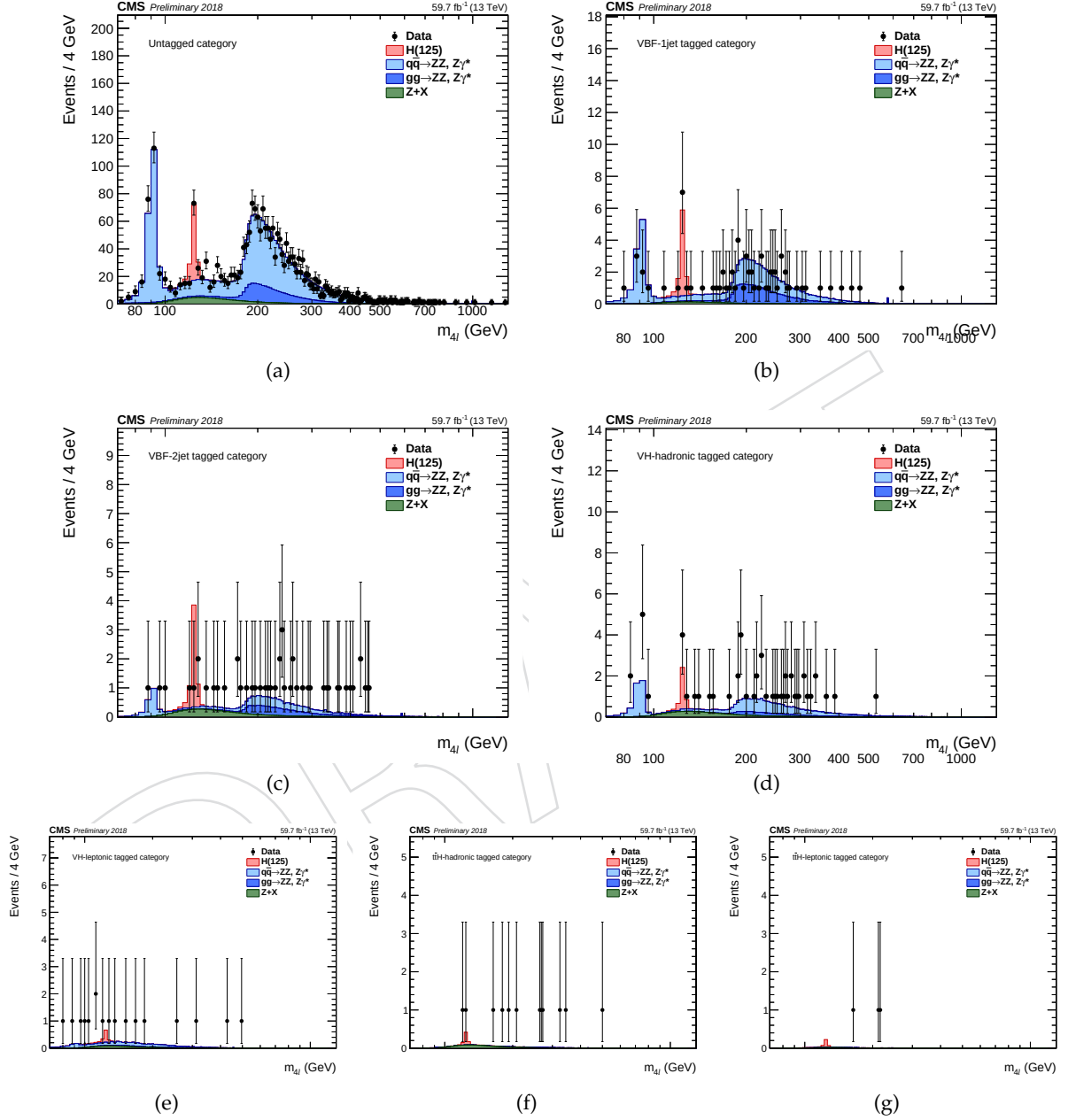


Figure 43: Distribution of the four-lepton reconstructed mass in the seven event categories for the low-mass range. (a) untagged category (b) VBF-1jet-tagged category (c) VBF-2jet-tagged category (d) VH-hadronic-tagged category (e) VH-leptonic-tagged category (f) $t\bar{t}H$ -hadronic-tagged category (g) $t\bar{t}H$ -leptonic-tagged category.

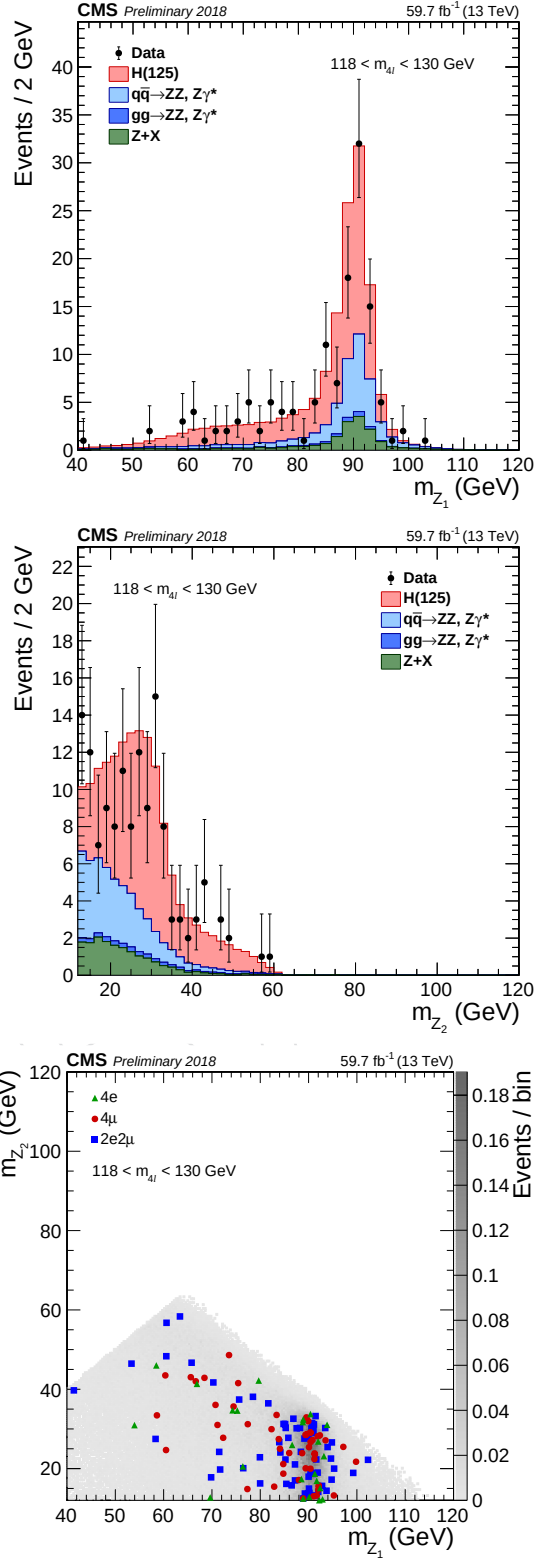
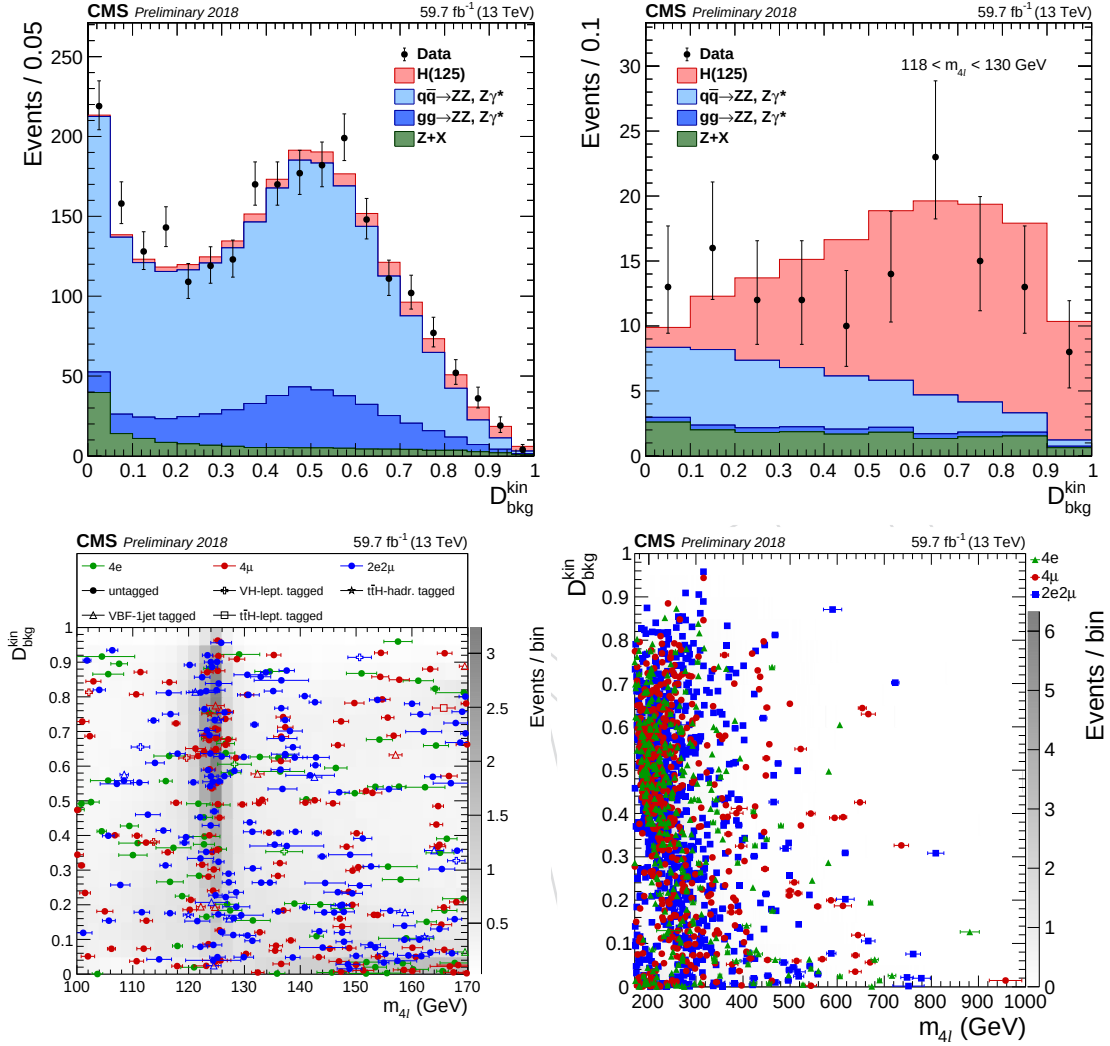


Figure 44: Distribution of the Z_1 (top), Z_2 (middle) and Z_1 vs Z_2 (bottom) reconstructed invariant masses for the full mass range (left) and the low mass ($118 < m_{4\ell} < 130$ GeV) range (right). The stacked histograms and the gray scale represent expected distributions, and points represent the data. The 125 GeV Higgs boson signal and the ZZ backgrounds are normalized to the SM expectation, the Z+X background to the estimation from data.



Top row: One-dimensional distribution of the kinematic discriminant D_{bkg}^{kin} in the full mass range (left) and in the mass region $118 < m_{4l} < 130$ GeV (right). Points with error bars represent the data and stacked histograms represent expected distributions. The 125 GeV Higgs boson signal and the ZZ backgrounds are normalized to the SM expectation, the $Z+X$ background to the estimation from data. Bottom row: Distribution of the kinematic discriminant D_{bkg}^{kin} versus the four-lepton reconstructed mass m_{4l} in the low-mass region (left) and in the high-mass region (right). The gray scale represents the expected relative density of ZZ background plus Higgs boson signal for $m_H = 125$ GeV. The points show the data and the horizontal bars represent the measured mass uncertainties.

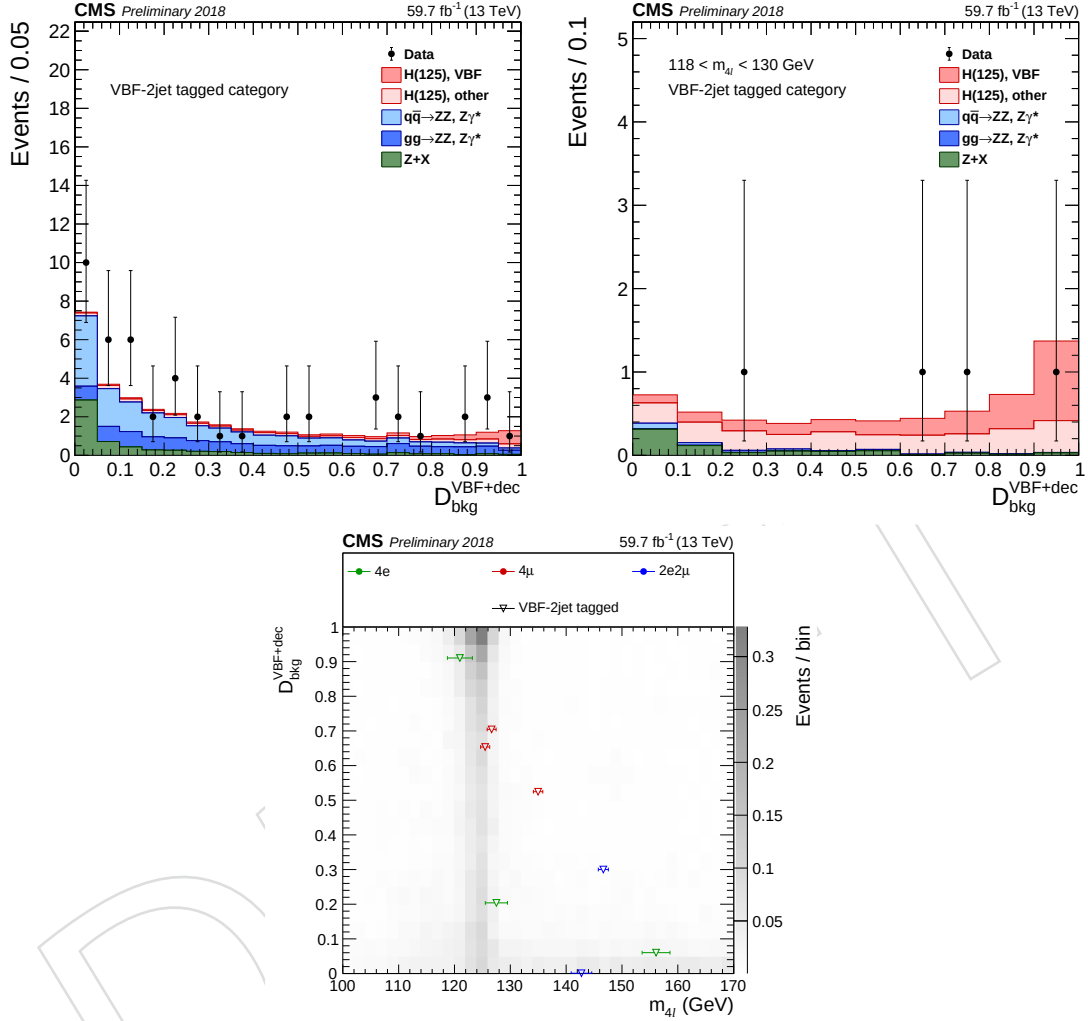


Figure 45: Top row: One-dimensional distribution of the kinematic discriminant $D_{\text{bkg}}^{\text{VBF+dec}}$ in the full mass range (left) and in the mass region $118 < m_{4\ell} < 130$ GeV (right). Points with error bars represent the data and stacked histograms represent expected distributions. The 125 GeV Higgs boson signal and the ZZ backgrounds are normalized to the SM expectation, the Z+X background to the estimation from data. Bottom row: Distribution of the kinematic discriminant $D_{\text{bkg}}^{\text{VBF+dec}}$ versus the four-lepton reconstructed mass $m_{4\ell}$ in the low-mass region. The gray scale represents the expected relative density of ZZ background plus Higgs boson signal for $m_H = 125$ GeV. The points show the data and the horizontal bars represent the measured mass uncertainties.

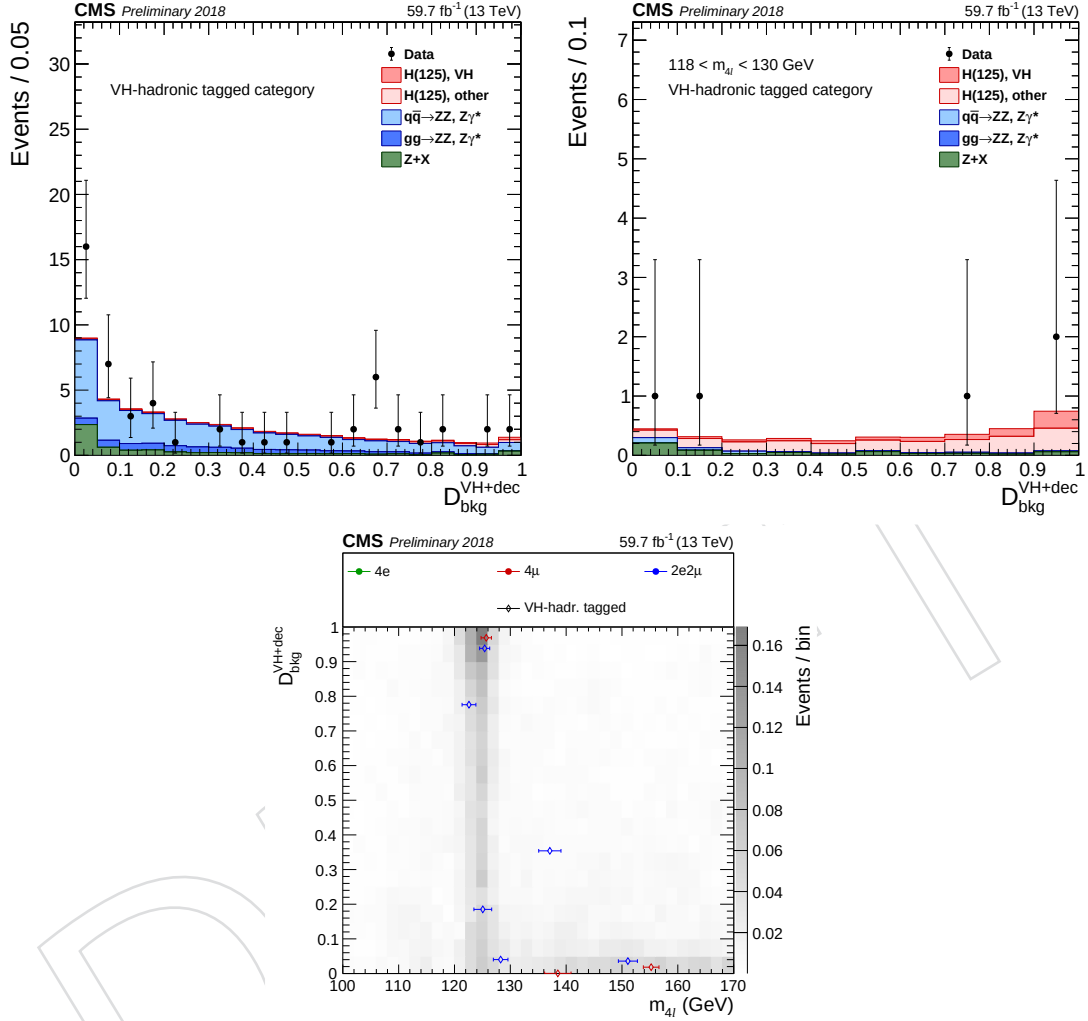


Figure 46: Top row: One-dimensional distribution of the kinematic discriminant $D_{\text{bkg}}^{\text{VH+dec}}$ in the full mass range (left) and in the mass region $118 < m_{4\ell} < 130$ GeV (right). Points with error bars represent the data and stacked histograms represent expected distributions. The 125 GeV Higgs boson signal and the ZZ backgrounds are normalized to the SM expectation, the Z+X background to the estimation from data. Bottom row: Distribution of the kinematic discriminant $D_{\text{bkg}}^{\text{VH+dec}}$ versus the four-lepton reconstructed mass $m_{4\ell}$ in the low-mass region. The gray scale represents the expected relative density of ZZ background plus Higgs boson signal for $m_H = 125$ GeV. The points show the data and the horizontal bars represent the measured mass uncertainties.

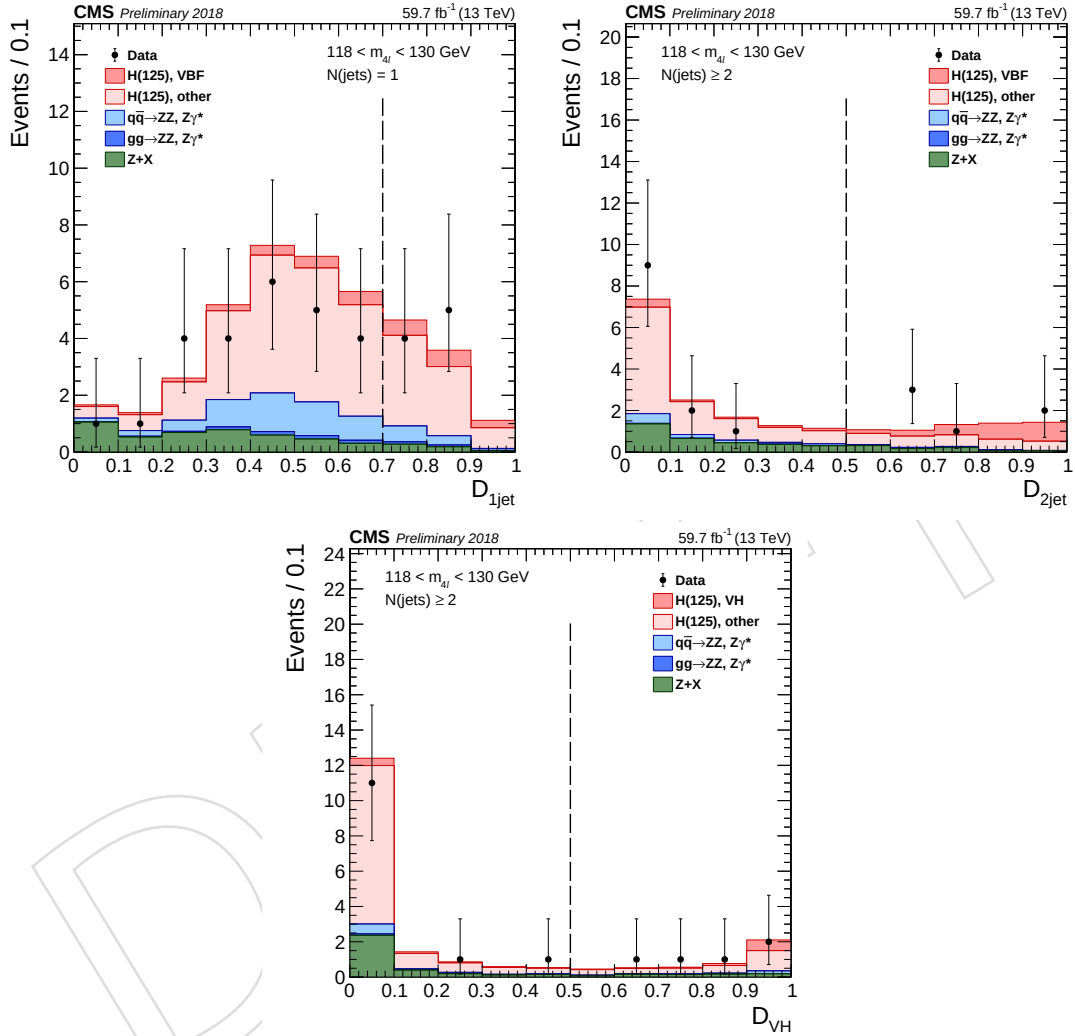


Figure 47: Distribution of the three production discriminants used for event categorization, in the mass region $118 < m_{4\ell} < 130$ GeV. Points with error bars represent the data and stacked histograms represent expected distributions. The 125 GeV Higgs boson signal and the ZZ backgrounds are normalized to the SM expectation, the Z+X background to the estimation from data.

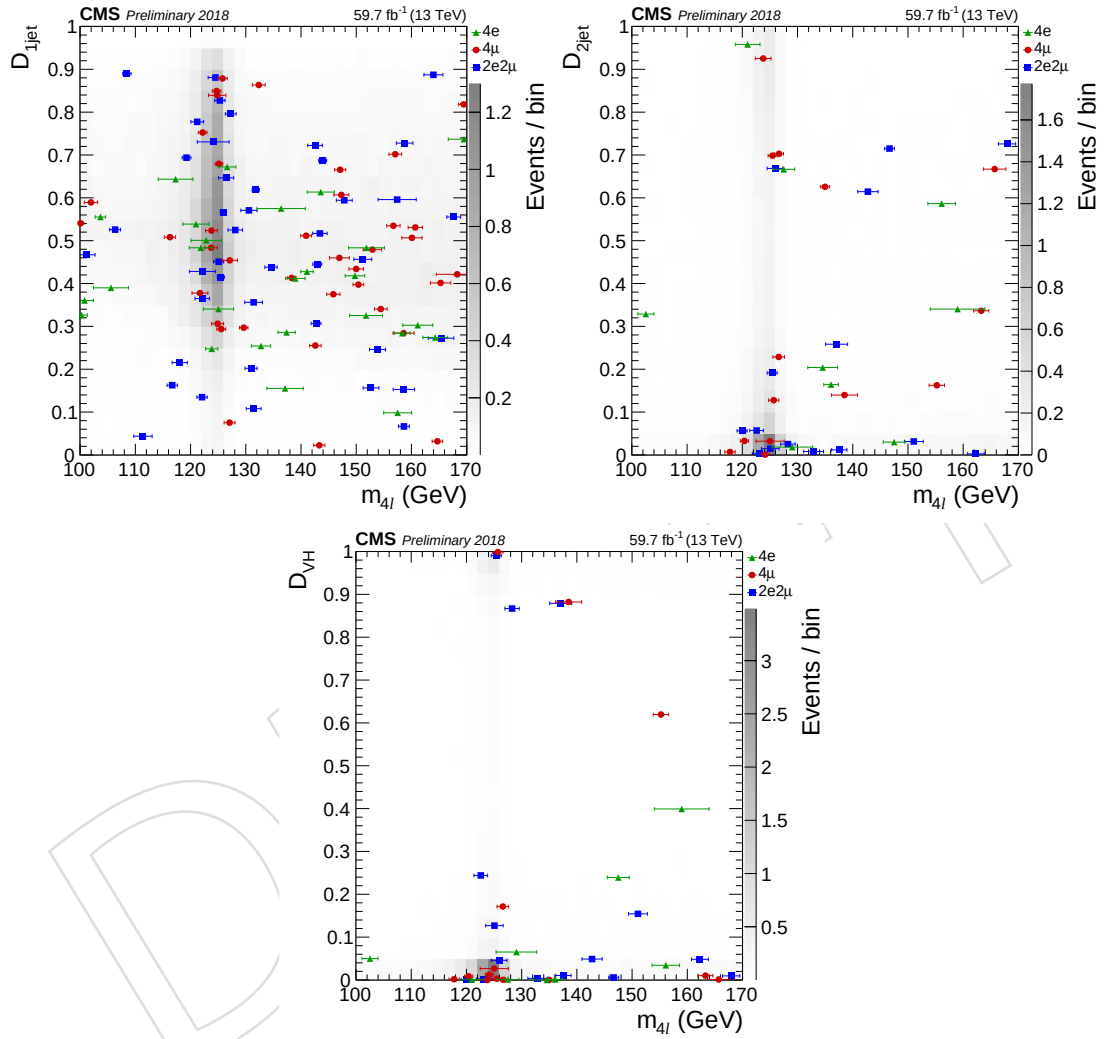


Figure 48: Distribution of the three production discriminants used for event categorization versus the four-lepton reconstructed mass $m_{4\ell}$ in the low-mass region. The gray scale represents the expected relative density of ZZ background plus Higgs boson signal for $m_H = 125$ GeV. The points show the data and the horizontal bars represent the measured mass uncertainties.

952 The reconstructed four-lepton invariant mass distribution is shown for the full Run 2 in Fig. 49
 953 for the sum of the $4e$, 4μ and $2e2\mu$ subchannels, and compared with the expectations from
 954 signal and background processes. The error bars on the data points correspond to the so-called
 955 Garwood confidence intervals at 68% confidence level (CL) [?]. The observed distribution
 956 agrees with the expectation within the statistical uncertainties over the whole spectrum. The
 957 Fig.50 shows number of expected and observed events in all Stage 1.1 sub-categories for full
 958 Run 2.

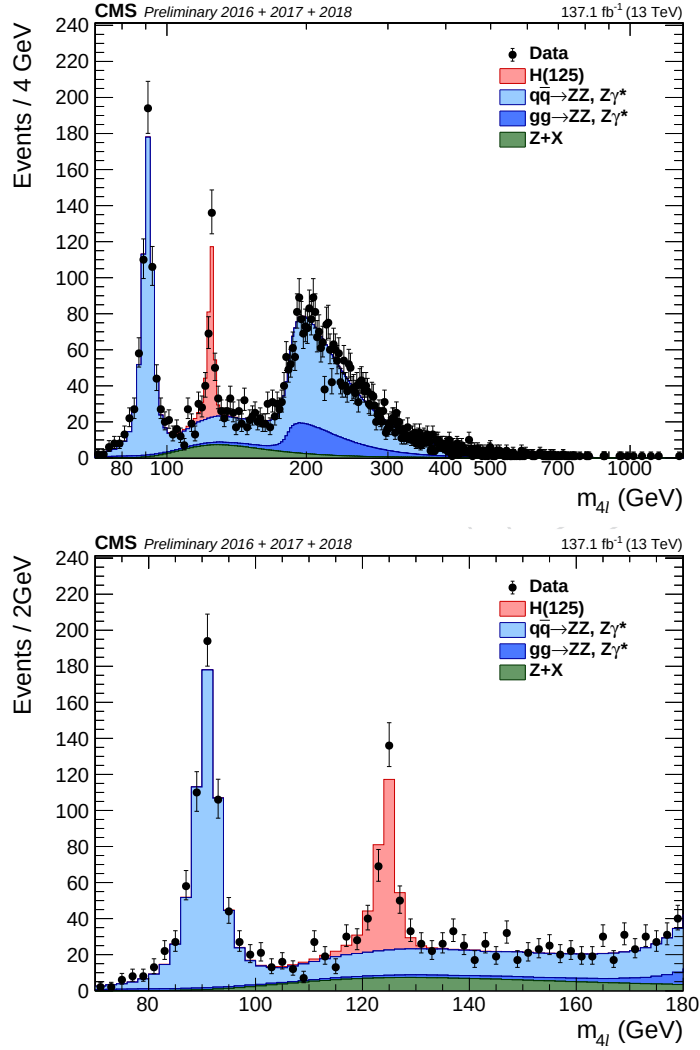


Figure 49: Distribution of the reconstructed four-lepton invariant mass $m_{4\ell}$ up to 500 GeV (left) and the low-mass range (right), with Run 2 data. Points with error bars represent the data and stacked histograms represent expected distributions of the signal and background processes. The SM Higgs boson signal with $m_H = 125$ GeV, denoted as H(125), and the ZZ backgrounds are normalized to the SM expectation, the Z+X background to the estimation from data.

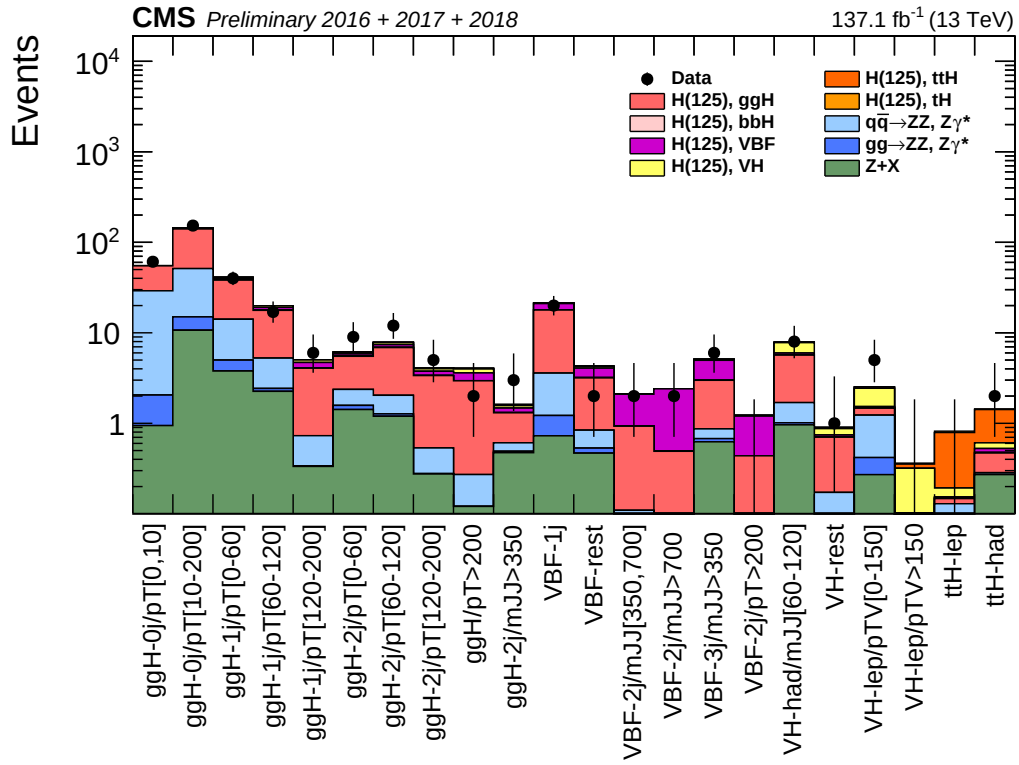


Figure 50: Distributions of the expected and observed number of events for all Stage 1.1 sub-categories described in Section ?? in the mass region $118 < m_{4\ell} < 130$ GeV with Run 2 data. Points with error bars represent the data and stacked histograms represent the expected numbers of the signal and background events. The different SM Higgs boson signal production modes with $m_H = 125$ GeV, denoted as H(125), and the ZZ backgrounds are normalized to the SM expectation, the Z+X background to the estimation from data.

12 Results

12.1 Signal Strength

In this section, we describe the methods used for the estimation of the significance of the excess of events above the SM backgrounds observed in the low-mass region, and of the signal strength, i.e. its cross section normalized to the one expected for a SM Higgs.

To exploit all the properties of the resonance under study or search, a multi-dimensional fit is implemented. For each of the categories defined in Section 7, two variables are used in the maximum likelihood fit, namely:

1. The four-lepton mass without kinematic refitting $m_{4\ell}$,
2. A kinematical discriminant K_D , where $K_D = \mathcal{D}_{\text{bkg}}^{\text{VBF+dec}}$ for VBF-2 jets tagged category, $\mathcal{D}_{\text{bkg}}^{\text{VH+dec}}$ for VH-hadronic tagged category and $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ for all other categories.

To account for the strong correlation of the kinematic discriminant with the mass, a 2D histogram template of K_D vs. $m_{4\ell}$ is implemented. Due to the small number of expected events in the mass peak, the mass dimension is unbinned and the resolution model is used as described in section 9.1.3. The total PDF is defined as:

$$\mathcal{L}_{2D}(m_{4\ell}, K_D) = \mathcal{L}(m_{4\ell})\mathcal{L}(K_D|m_{4\ell}), \quad (14)$$

where the first factor corresponds to the 1D mass PDF and the second factor to the 2D template of mass vs. kinematic discriminant. The conditional term in the second factor is implemented in the template by normalizing all columns corresponding to the same mass to the same value each. Therefore, the 2D template doesn't include any information on the mass, but given the mass, it provides information on the kinematic discriminant. 2D Templates for $\mathcal{D}_{\text{bkg}}^{\text{VBF+dec}}$ and $\mathcal{D}_{\text{bkg}}^{\text{VH+dec}}$ vs $m_{4\ell}$ are computed in 7x20 bins, for all processes. 2D Templates for $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ vs $m_{4\ell}$ are computed in 35x30 bins in the Untagged categories. The same templates are also used for VH-leptonic and ttH tagged categories. Based on the seven event categories and the three final states (4μ , $4e$, $2e2\mu$), the $(m_{4\ell}, K_D)$ unbinned distributions of selected events are split into $22 \times 3 = 66$ categories.

With a 2D fit, the signal strength is measured to be $\mu = \sigma/\sigma_{\text{SM}} = 0.94_{-0.10}^{+0.11}$ with m_H profiled in the fit. The results are compared to the expected signal-strength modifiers in Table 25.

A fit is performed for four main signal-strength modifiers ($\mu_{\text{ggH}, \text{b}\bar{\text{b}}\text{H}}$, μ_{VBF} , μ_{VH} , and μ_{ttH}) controlling the contribution of the main SM Higgs boson production modes. The WH and ZH processes are merged into VH. Contributions of the $\text{b}\bar{\text{b}}\text{H}$ and production modes are also taken into account in the fit. The $\text{b}\bar{\text{b}}\text{H}$ contribution is floated together with gluon fusion and production mode is floated with ttH.

The results are reported in Fig. ?? (left) and compared to the expected signal-strength modifiers in Table 25.

Table 25: Expected and observed signal-strength modifiers with Run 2 data. The observed uncertainty numbers are broken into statistical (first) and systematic (second) sources.

	Inclusive	μ_{ggH}	μ_{VBF}	μ_{VH}	μ_{ttH}
Expected	$1.00_{-0.08}^{+0.08}(\text{stat.})_{-0.07}^{+0.09}(\text{syst.})$	$1.00_{-0.10}^{+0.10}(\text{stat.})_{-0.07}^{+0.09}(\text{syst.})$	$1.00_{-0.45}^{+0.54}(\text{stat.})_{-0.14}^{+0.27}(\text{syst.})$	$1.00_{-0.72}^{+0.91}(\text{stat.})_{-0.16}^{+0.29}(\text{syst.})$	$1.00_{-0.73}^{+1.16}(\text{stat.})_{-0.04}^{+0.19}(\text{syst.})$
Observed	$0.94_{-0.07}^{+0.07}(\text{stat.})_{-0.07}^{+0.08}(\text{syst.})$	$0.97_{-0.09}^{+0.09}(\text{stat.})_{-0.07}^{+0.09}(\text{syst.})$	$0.64_{-0.36}^{+0.45}(\text{stat.})_{-0.09}^{+0.16}(\text{syst.})$	$1.15_{-0.72}^{+0.89}(\text{stat.})_{-0.16}^{+0.26}(\text{syst.})$	$0.13_{-0.13}^{+0.92}(\text{stat.})_{-0.00}^{+0.11}(\text{syst.})$

988

989 Two signal-strength modifiers $\mu_{ggH, t\bar{t}H}$ and $\mu_{VBF, VH}$ are introduced as scale factors for the
 990 fermion and vector-boson induced contribution to the expected SM cross section. A two-
 991 dimensional fit is performed, profiling m_H , leading to the measurements of $\mu_{ggH, t\bar{t}H} = 0.96^{+0.11}_{-0.12}$
 992 and $\mu_{VBF, VH} = 0.83^{+0.29}_{-0.35}$. The 68% and 95% CL contours in the $(\mu_{ggH, t\bar{t}H}, \mu_{VBF, VH})$ plane are
 993 shown in Fig. 51.

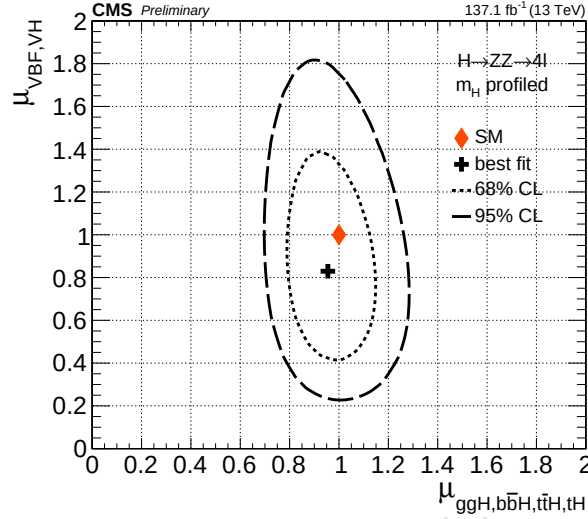


Figure 51: Result of the 2D likelihood scan for the $\mu_{ggH, t\bar{t}H}$ and $\mu_{VBF, VH}$ signal-strength modifiers. The solid and dashed contours show the 68% and 95% CL regions, respectively. The cross indicates the best-fit values, and the diamond represents the expected values for the SM Higgs boson.

994 Signal-strength modifiers for the five main Higgs production modes, μ_{ggH} , μ_{VBF} , μ_{WH} , μ_{ZH}
 995 and $\mu_{t\bar{t}H}$ are also introduced as scale factors to the expected SM cross section. Similarly, the fits
 996 are performed fixing the mass to $m_H = 125$ GeV and profiling the likelihood for all nuisance
 997 parameters, leading to the results illustrated in Fig. 52, where μ_{WH} and μ_{ZH} are merged to μ_{VH} .

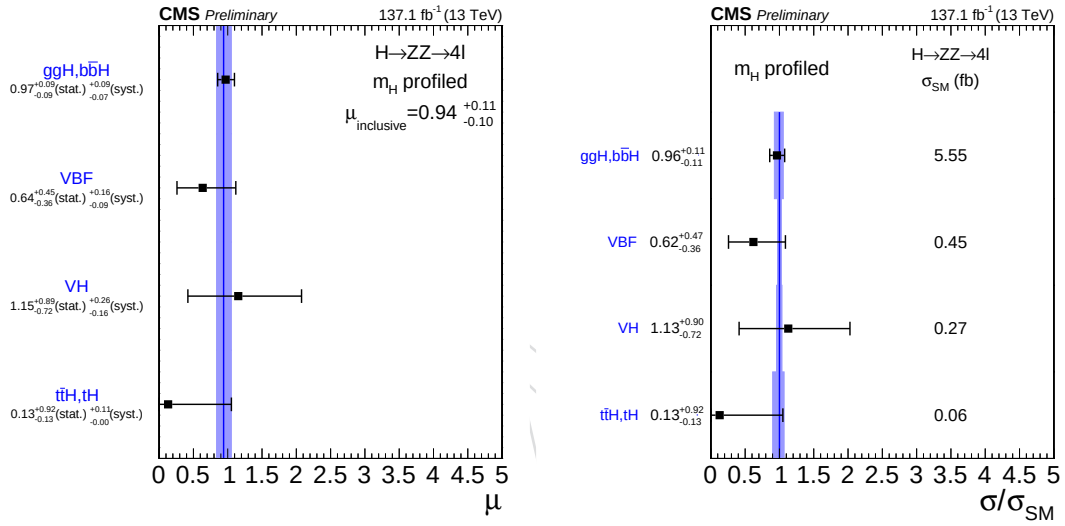


Figure 52: Results of likelihood scans for the signal strength modifiers (left) and cross section (right) corresponding to the four main Higgs boson production modes. On the left figure, the vertical line shows the combined μ , together with its associated $\pm 1\sigma$ uncertainties shown as filled band. On the right figure, the band of the vertical line shows the theoretical uncertainties on the SM cross sections. The horizontal bars indicate the $\pm 1\sigma$ uncertainties on μ for the different production modes. The uncertainties include both statistical and systematic sources.

12.2 Simplified Template Cross Sections

In this section we present the results for simplified template cross sections Stage 1.1, a measurement strategy detailed in the CERN Yellow Report 4 of the LHC-HXSWG [68]. The reduced stage 1.1 bins are measured in forms of signal strength modifier. Figure 53 shows the results. The covariance matrix of the fitted results are shown in Figure 54

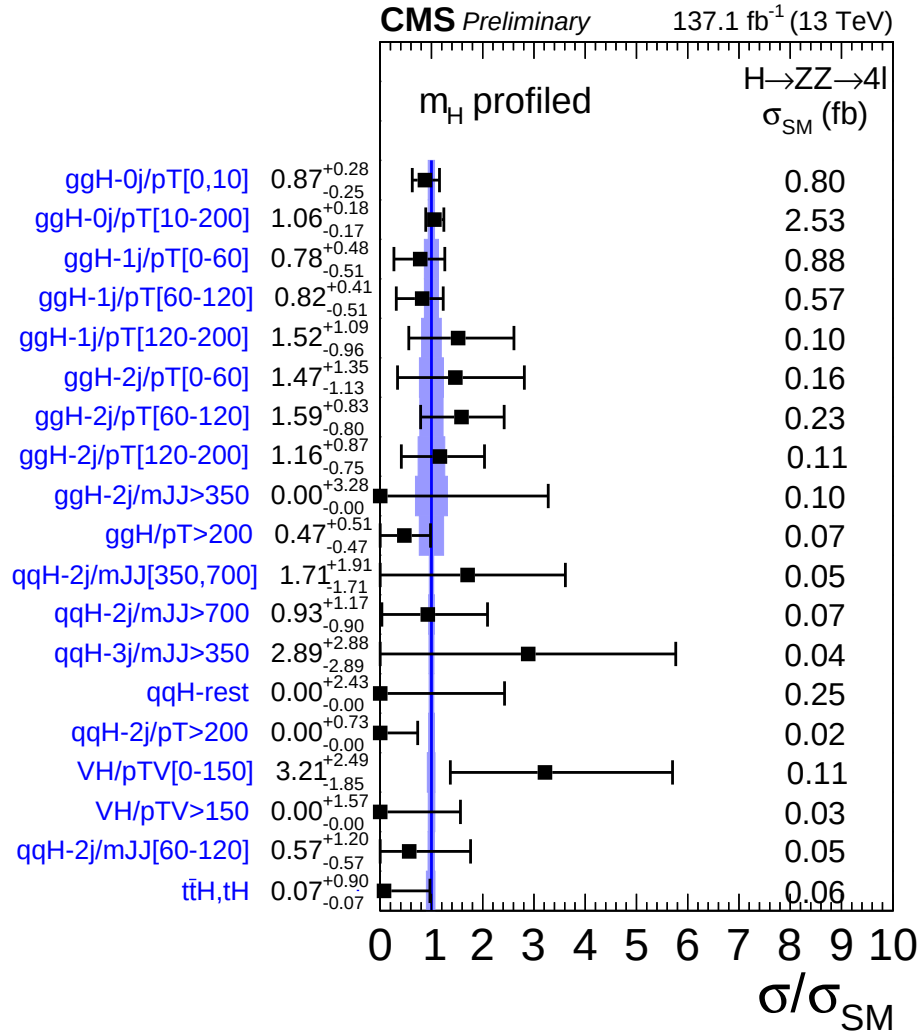


Figure 53: The ratios between measured cross sections and the SM prediction for stage1.1 bins, the m_H is profiled. The band around the vertical band shows the theoretical uncertainties on the SM cross section predictions for each stage 1.1 bin.

12.3 Differential Cross Sections

12.3.1 Fiducial Volume Definition

The fiducial volume is defined to match closely the reconstruction level selection and is very similar to the definition used in Refs. [18]. With respect to the Run 1 analysis, the leptons are defined as “dressed” leptons rather than Born level leptons. Leptons are dressed by adding the four-momenta of leptons within $\Delta R < 0.3$ to the bare leptons. The fiducial lepton isolation criteria is also updated to match the Run 2 reconstruction level isolation. Leptons are considered

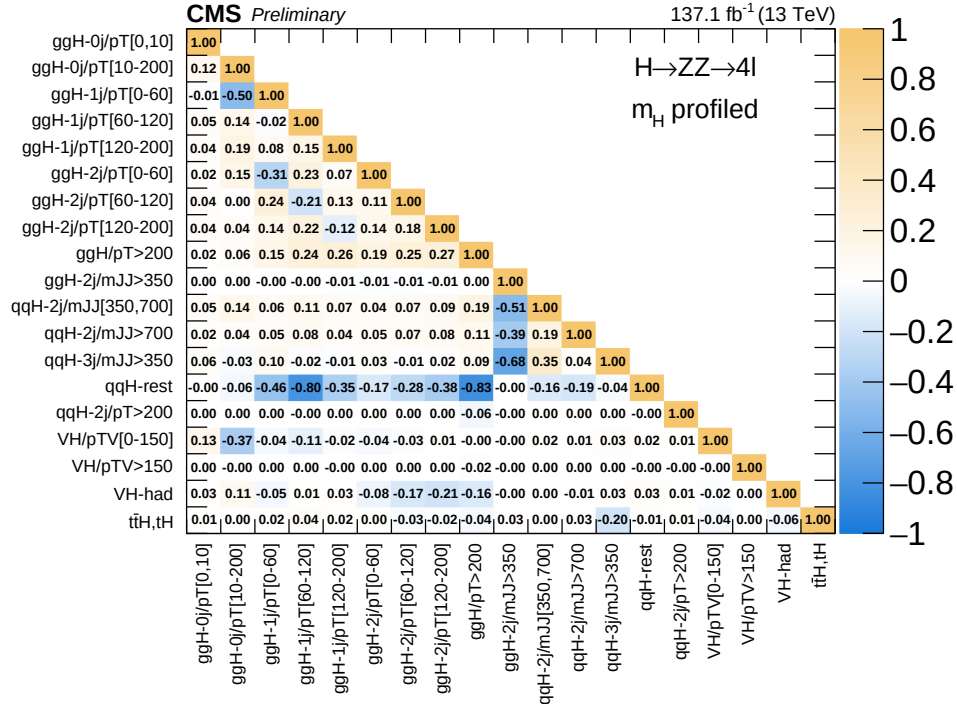


Figure 54: Covariance matrix of the fitted signal strength.

1010 isolated at generator level if the sum p_T of particles within a cone $\Delta R < 0.3$ is less than 0.35. The
 1011 fiducial volume definition can be seen in Table 26. The fiducial volume acceptance for various
 1012 SM production modes can be seen in Table 29.

Table 26: Summary of requirements and selections used in the definition of the fiducial phase space for the $H \rightarrow 4\ell$ cross section measurements.

Requirements for the $H \rightarrow 4\ell$ fiducial phase space	
Lepton kinematics and isolation	
leading lepton p_T	$p_T > 20$ GeV
next-to-leading lepton p_T	$p_T > 10$ GeV
additional electrons (muons) p_T	$p_T > 7(5)$ GeV
pseudorapidity of electrons (muons)	$ \eta < 2.5(2.4)$
p_T sum of all stable particles within $\Delta R < 0.3$ from lepton	less than $0.35 \cdot p_T$
Event topology	
existence of at least two SFOS lepton pairs, where leptons satisfy criteria above	
inv. mass of the Z_1 candidate	$40 \text{ GeV} < m(Z_1) < 120 \text{ GeV}$
inv. mass of the Z_2 candidate	$12 \text{ GeV} < m(Z_2) < 120 \text{ GeV}$
distance between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$
inv. mass of any opposite sign lepton pair	$m(\ell^+ \ell'^-) > 4 \text{ GeV}$
inv. mass of the selected four leptons	$105 \text{ GeV} < m_{4\ell} < 140 \text{ GeV}$
the selected four leptons must originate from the $H \rightarrow 4\ell$ decay	

1013 12.3.2 Measurement Strategy

1014 We measure the integrated and differential fiducial cross section for $pp \rightarrow H \rightarrow 4\ell$ by perform-
 1015 ing a maximum likelihood fit of the signal and background parameterisations to the observed

4 ℓ mass distribution, $N_{\text{obs}}(m_{4\ell})$, and the fiducial cross section (σ_{fid}) is directly extracted from the fit. The systematic uncertainties are included in the form of nuisance parameters and are effectively integrated out in the fit procedure. The results are obtained using an asymptotic approach [69] with a test statistic based on the profile likelihood ratio [70]. This procedure for the unfolding of the detector effects from the observed distributions is the same as in Refs. [18] and [71]. In the case of the differential cross section measurements, the finite efficiencies and resolution effects are encoded in a detector response matrix which describes how events migrate from a given observable bin at the fiducial level to a given bin at the reconstruction level. This matrix is diagonally dominant, with sizeable off-diagonal elements for observables involving jets.

Following the models for signal and background contributions described above, the number of expected events in each final state f and in each bin i of a considered observable is expressed as a function of $m_{4\ell}$ given by:

$$\begin{aligned} N_{\text{obs}}^{f,i}(m_{4\ell}) &= N_{\text{fid}}^{f,i}(m_{4\ell}) + N_{\text{nonfid}}^{f,i}(m_{4\ell}) + N_{\text{nonres}}^{f,i}(m_{4\ell}) + N_{\text{bkg}}^{f,i}(m_{4\ell}) \\ &= \epsilon_{i,j}^f \cdot \left(1 + f_{\text{nonfid}}^{f,i}\right) \cdot \sigma_{\text{fid}}^{f,j} \cdot \mathcal{L} \cdot \mathcal{P}_{\text{res}}(m_{4\ell}) \\ &\quad + N_{\text{nonres}}^{f,i} \cdot \mathcal{P}_{\text{nonres}}(m_{4\ell}) + N_{\text{bkg}}^{f,i} \cdot \mathcal{P}_{\text{bkg}}(m_{4\ell}), \end{aligned} \quad (15)$$

The parameter $\sigma_{\text{fid}}^{f,j}$ is the signal cross section in bin j of the fiducial phase space, and it is the parameter extracted from the measurement.

The shape of the resonant signal contribution, $\mathcal{P}_{\text{res}}(m_{4\ell})$, is described by a double-sided Crystal Ball function as described in Section 9.1.3 whose normalisation is proportional to the fiducial cross section. The shape of the non-resonant signal contribution, $\mathcal{P}_{\text{nonres}}(m_{4\ell})$, which arises from WH, ZH, and t $\bar{t}$ H production where one of the leptons from the Higgs boson decay is lost or not selected, is empirically modelled by a Landau distribution whose shape parameters are constrained in the fit to be within a range determined from simulation. This contribution is treated as a background and hereafter we will refer to this contribution as the “non-resonant signal” contribution.

The $\epsilon_{i,j}^f$ represents the detector response matrix that maps the number of expected events in a given observable bin j at the fiducial level to the number of expected events in the bin i at the reconstruction level. The $f_{\text{nonfid}}^{f,i}$ fraction describes the ratio of the non-fiducial and fiducial signal contribution in bin i at the reconstruction level. The efficiency is measured using signal simulation samples and corrected for residual differences between data and simulation. In the case of the integrated fiducial cross section measurement the efficiencies reduce to a single values.

An additional resonant contribution arises from events which are reconstructed but which do not originate from the fiducial phase space. These events are due to detector effects which cause differences between the quantities used for the fiducial phase space definition and the analogous quantities at the reconstruction level. This contribution is treated as background and is referred to as the “non-fiducial signal” contribution. The shape of these events is verified using simulation to be identical to the shape of the fiducial signal and its normalisation is fixed to be a fraction of the fiducial signal component. The value of this fraction, which we denote by f_{nonfid} , which has been determined from simulation for each of the studied signal models.

The variation between different models of the factor in the final column of Table 29, $(1 + f_{\text{nonfid}})\epsilon$, is directly related to the model dependence of the measurement. The model dependence is defined as the variation of the factor $(1 + f_{\text{nonfid}})\epsilon$ when the relative fraction of

each the production modes are varied within there experimental constraints. An increase in model dependence compared to Run 1 is observed when using the ZZ candidate selection at reconstruction level where the the candidate with the best $\mathcal{D}_{\text{bkg}}^{\text{kin}}$ discriminant value is chosen. Therefore the fiducial cross section measurement is performed with a different event selection algorithm than the other measurements, namely that the ZZ candidate selection is made using the same algorithm as in Run 1.

Table 27: Summary of different Standard Model signal models. The MC samples are from 2016 production.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
gg→H (POWHEG)	0.398	0.592 ± 0.001	0.049 ± 0.001	0.621 ± 0.001
VBF (POWHEG)	0.445	0.601 ± 0.002	0.038 ± 0.001	0.624 ± 0.002
WH (POWHEG+MINLO)	0.314	0.577 ± 0.002	0.068 ± 0.001	0.616 ± 0.002
ZH (POWHEG+MINLO)	0.342	0.592 ± 0.003	0.071 ± 0.002	0.634 ± 0.003
ttH (POWHEG)	0.311	0.572 ± 0.003	0.136 ± 0.003	0.650 ± 0.004

Table 28: Summary of different Standard Model signal models. The MC samples are from 2017 production.

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
gg→H (POWHEG)	0.404 ± 0.001	0.593 ± 0.001	0.054 ± 0.001	0.625 ± 0.001
VBF (POWHEG)	0.443 ± 0.001	0.605 ± 0.002	0.045 ± 0.001	0.632 ± 0.002
WH (POWHEG+MINLO)	0.329 ± 0.001	0.589 ± 0.002	0.080 ± 0.001	0.637 ± 0.002
ZH (POWHEG+MINLO)	0.341 ± 0.002	0.593 ± 0.003	0.083 ± 0.002	0.643 ± 0.004
ttH (POWHEG)	0.316 ± 0.002	0.597 ± 0.003	0.172 ± 0.004	0.700 ± 0.004

Table 29: Summary of different Standard Model signal models. The MC samples are from 2018 production. **FIXME: Scale factors not yet applied**

Signal process	$\mathcal{A}_{\text{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\text{nonfid}})\epsilon$
Individual Higgs boson production modes				
gg→H (POWHEG)	0.403 ± 0.001	0.630 ± 0.001	0.055 ± 0.001	0.664 ± 0.001
VBF (POWHEG)	0.443 ± 0.001	0.645 ± 0.002	0.043 ± 0.001	0.673 ± 0.002
WH (POWHEG+MINLO)	0.330 ± 0.001	0.632 ± 0.002	0.075 ± 0.001	0.679 ± 0.002
ZH (POWHEG+MINLO)	0.338 ± 0.002	0.638 ± 0.003	0.086 ± 0.003	0.693 ± 0.004
ttH (POWHEG)	0.314 ± 0.002	0.620 ± 0.003	0.184 ± 0.004	0.733 ± 0.004

Examples of the efficiency matrices for gluon fusion and VBF production can be seen in Fig. 55. The matrices for the p_{TH} and N(jets) observables are shown.

12.3.3 Measurement results

The result of the simultaneous fit to to the $m_{4\ell}$ spectrum is shown for each final state in Fig. 56. The fiducial cross section using this defintion is measured to be:

$$2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{sys.}) \text{ fb.} \quad (16)$$

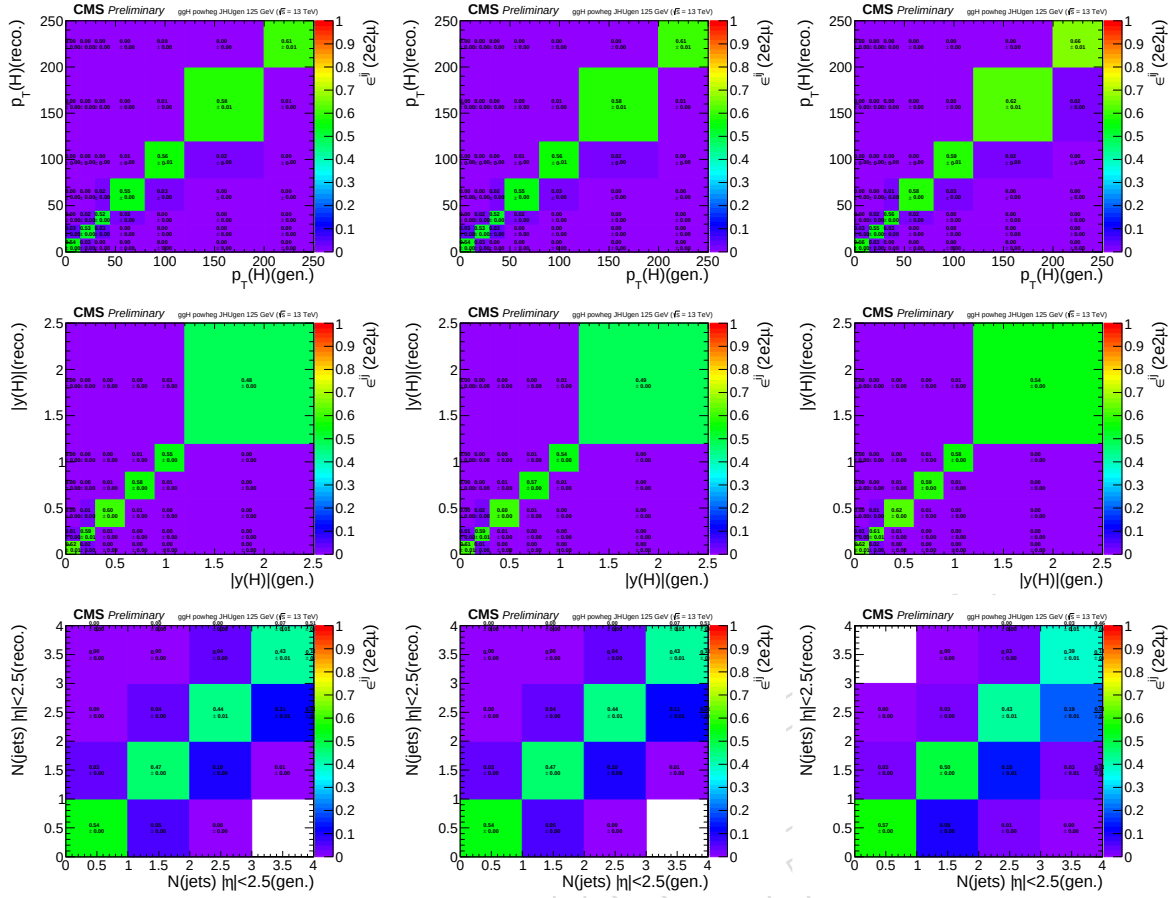


Figure 55: Efficiency matrices for the $p_{T(H)}$ (top) and $y_{H(H)}$ (middle) and $N(\text{jets})$ (bottom) observables for gluon fusion production modes in the $2e2\mu$ final state in 2016 (left) 2017 (middle) and 2018 (right).

This can be compared to the SM expectation of $\sigma_{\text{fid.}}^{\text{SM}} = 2.72 \pm 0.14 \text{ fb}$.

The integrated fiducial cross section as a function of center-of-mass energy is also shown in Fig. 60. The measured differential cross section results for $p_{T(H)}$, $|y_{H(H)}|$, $N(\text{jets})$, and $p_{T(\text{jet})}$ can also be seen in Fig. 60. The dominant sources of systematic uncertainty are the experimental uncertainties in the lepton identification efficiencies and luminosity measurement, and the theoretical sources of uncertainty are found to be subdominant. In order to assess the model dependence of the measurement, the unfolding procedure is repeated using different response matrices created by varying the relative fraction of each SM production mode within its experimental constraints. The uncertainty is determined to be negligible with respect to the experimental systematic uncertainties.

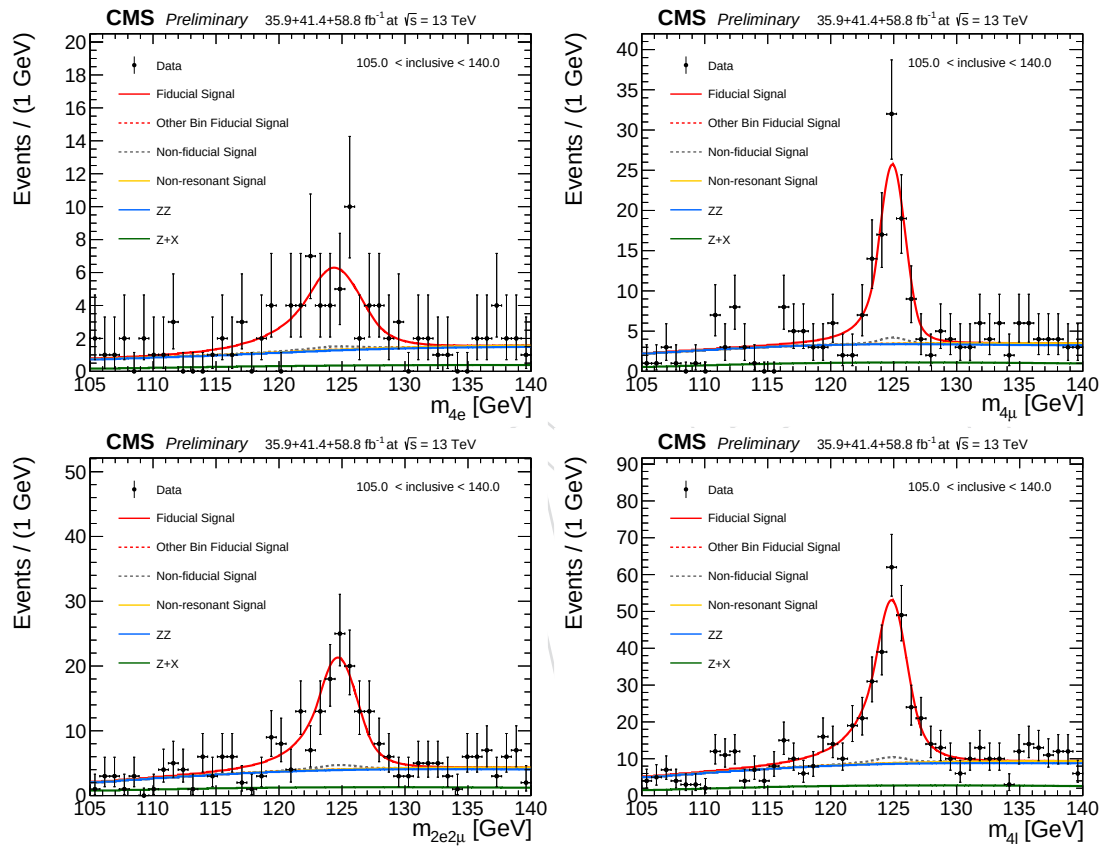


Figure 56: Result of simultaneous fit for the integrated fiducial cross section measurement in each final state. The results are shown for 2018 only.

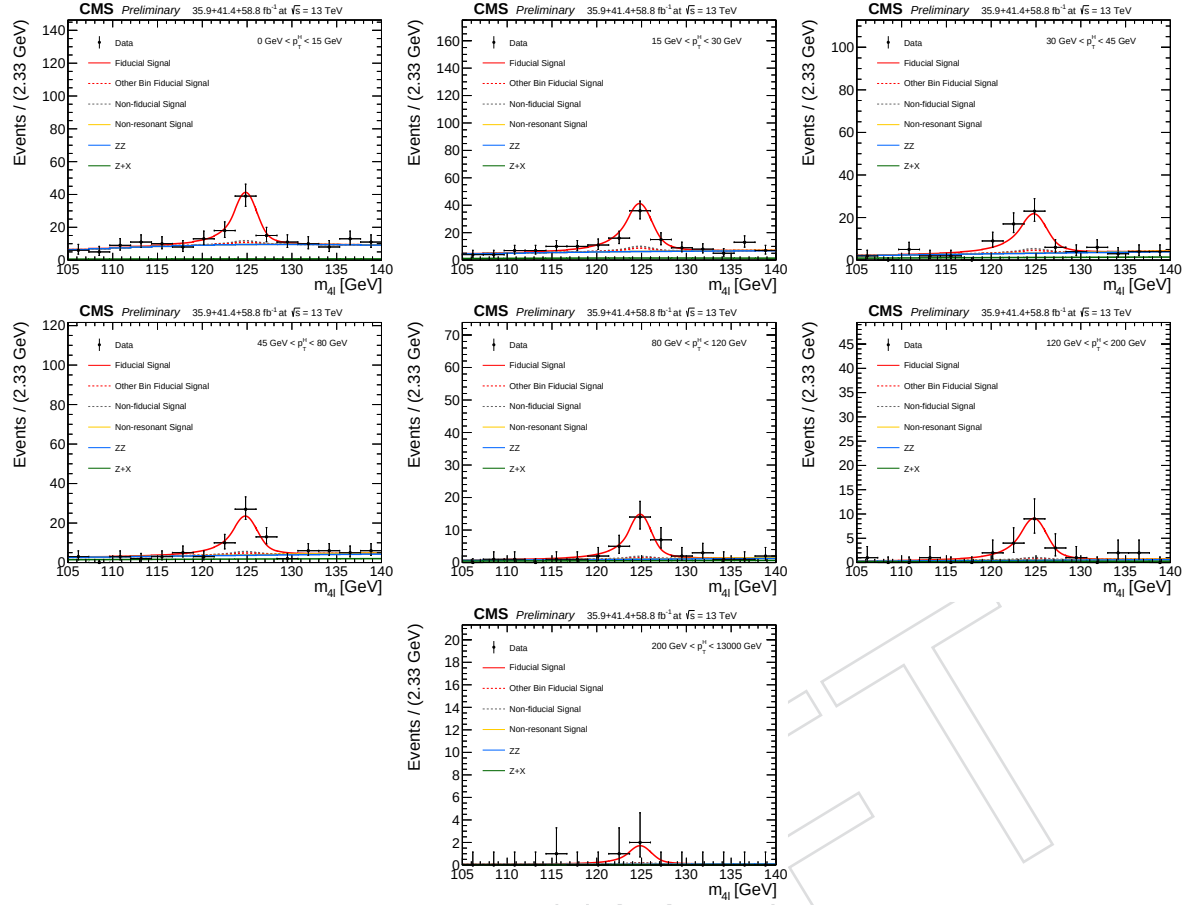


Figure 57: Result of simultaneous fit for the differential fiducial cross section measurement for $p_T(H)$ in each differential bin. The combined 4ℓ final state is shown. The results are shown for 2018 only.

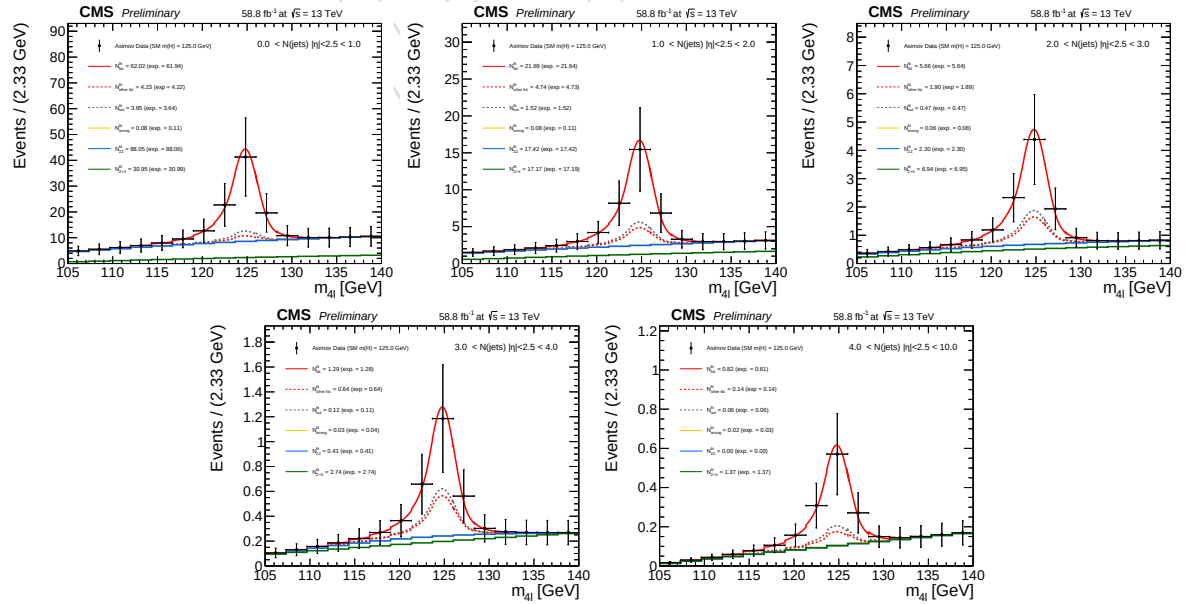


Figure 58: Result of simultaneous fit for the differential fiducial cross section measurement for $N(jets)$ in each differential bin. The combined 4ℓ final state is shown. The results are shown for 2018 only.

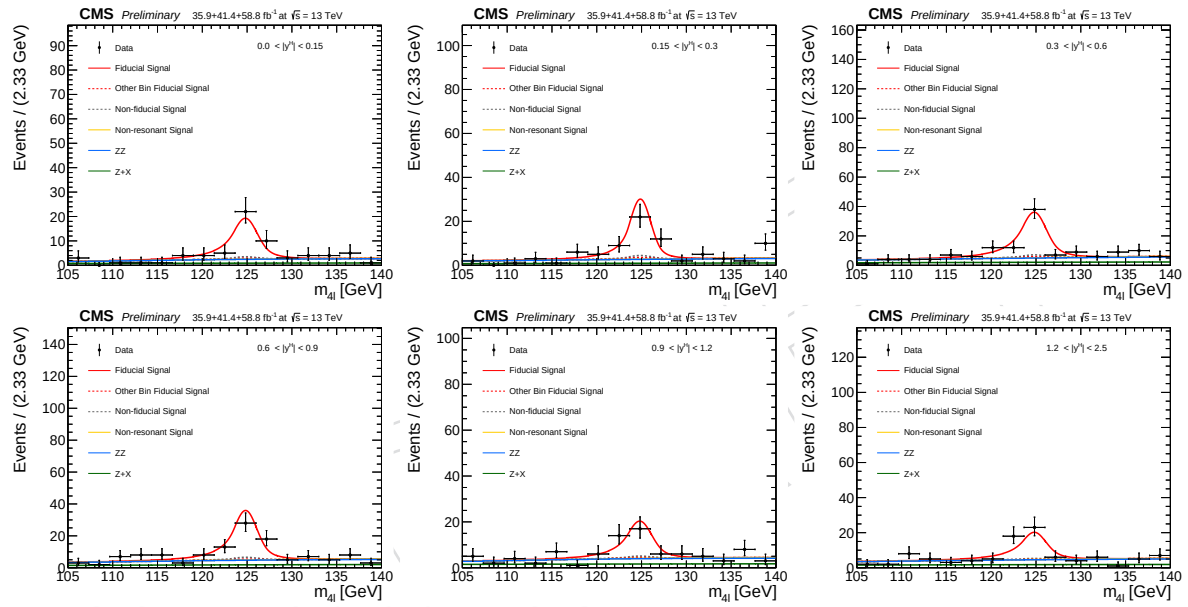


Figure 59: Result of simultaneous fit for the differential fiducial cross section measurement for $y(H)$ in each differential bin. The combined 4ℓ final state is shown. The results are shown for 2018 only.

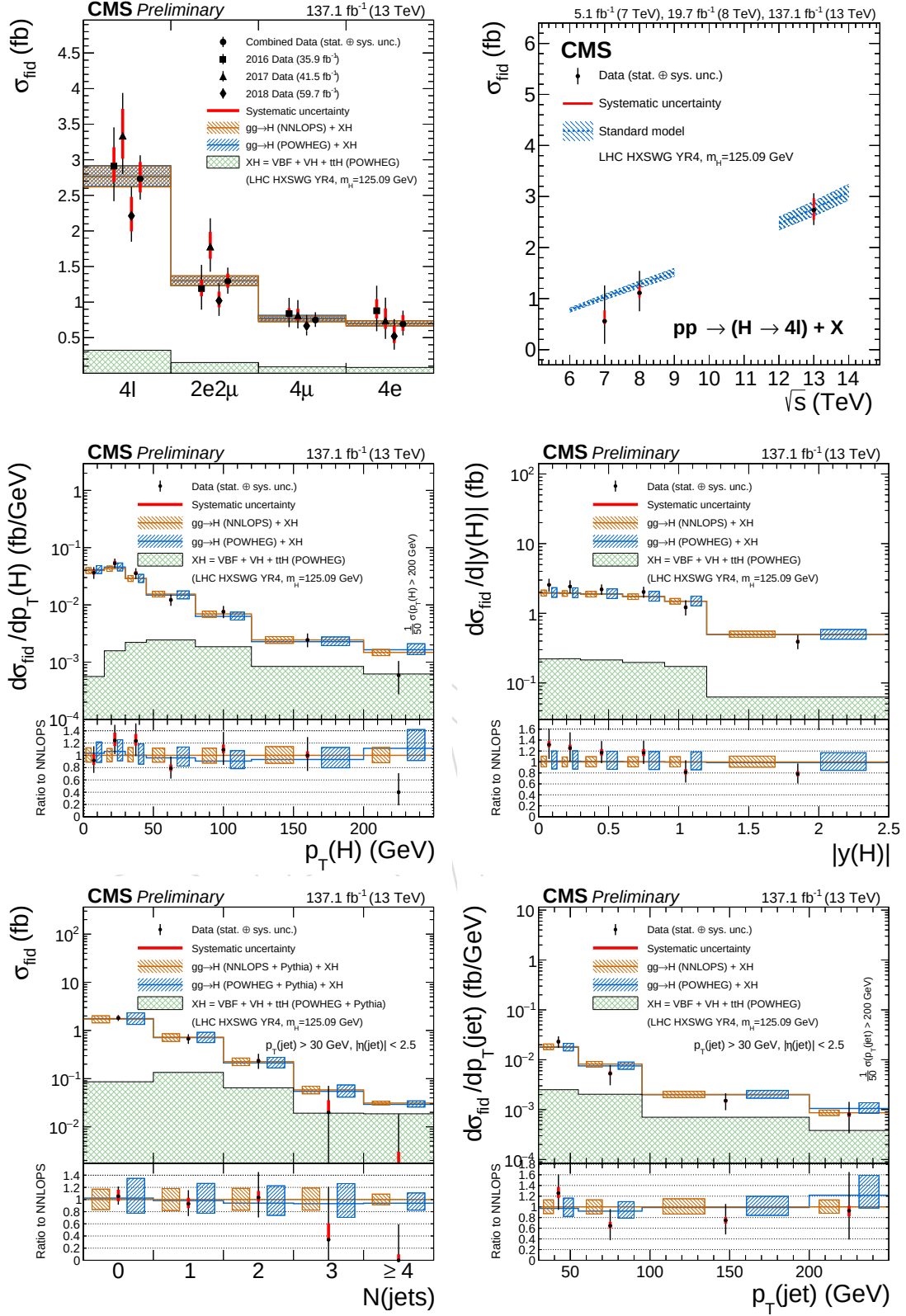


Figure 60: The measured inclusive fiducial cross section in different final states (top left). The measured fiducial cross section as a function of $\sqrt{s}$ (top right). The acceptance is calculated using POWHEG at $\sqrt{s}=13$ TeV and HRES [66, 72] at $\sqrt{s}=7$ and 8 TeV and the total gluon fusion cross section and uncertainty are taken from Ref. [73]. The fiducial volume for $\sqrt{s}=6-9$ TeV uses the lepton isolation definition from Ref. [18], while for $\sqrt{s}=12-14$ TeV the definition described in the text is used. The results of the differential cross section measurement for $p_T(H)$ (middle left), $|y(H)|$ (middle right) and $N(\text{jets})$ (bottom left), p_T of the leading jet (bottom right). The acceptance and theoretical uncertainties in the differential bins are calculated using POWHEG. The sub-dominant component of the signal (VBF + VH + ttH) is denoted as XH.

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